

Confronting the Limit

Determination, Subjectivity and Mind as Objective Internality

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§0/1 — The Integral Threshold

ABSTRACT

This paper develops **Objective Internality** as a philosophical and technological account of the means through which an *apparent* subject encounters, differentiates, acts within and revises a world. Its starting point is a formal limit: every coherent account of knowing becomes another object within the act of knowing it seeks to contain. Rather than treating this limit as a reason to abandon objectivity, the paper argues that it reveals a constitutive relation among knower, means of knowing and known. Objective Internality names the middle term: the real, structured, revisable field of memory, language, conceptual models, abilities, habits and relations through which a life acquires a world, while that world still appears to someone. The *duration* of this middle is the data by which enacted internality can be encountered objectively, without that data defining the subject.

The submitted foundation, §0/1, "The Integral Threshold", develops this relation through zero. Zero becomes the symbol by which absence, limit and indeterminacy are recognised as positive conditions of the appearance and representation of any world: a positive absence, which undeniably *is* but cannot be expressed definitively, acquires a paradoxical position as the ground of every distinction. We pursue this through the formal limits of representation in language and mathematics, from naming and counting to the paradoxes of Frege and Russell; through Gebser's structures of consciousness; and in the register of Lacan's *matheme* and the spirit of Jung and Pauli's unity of opposites. We seek to answer how the binary and the non-dual are reconciled. Zero and One emerge together as the primordial symbolon, identity-in-difference, the page and the marks it bears. Quaternary Logic formalises this relation, distinguishing a logic that severs a polarity or lets one pole claim the whole from one that keeps the halves different so they can be recognised as belonging.

Artificial intelligence brings this problem into material and epochal expression. Built on binary, AI now appears before us as language, memory, inference, judgement and action, a technical object assuming the forms through which Mind had encountered itself only from a human vantage. It thereby points towards the psychoid hypothesis: one elementary problem of differentiation appears at once in the mathematical substrate of computation, in the symbolic structures through which intelligence becomes articulate, and in the psychic forms through which a world is experienced. Subject is not Mind: Subject names the immediate pole to which any world appears; Mind names the structured mediation through which that world becomes apparent. AI makes this mediating interior exterior. What had appeared as the hidden organisation of agency can now be built, inspected, recomposed and encountered from the outside as an answering system.

The foundation opens onto seven further sections, tracing zero's history from Babylonian positional absence to formal logic, the mathematical substrate, the psychoid and agentic research. The essay is also an experiment: AI agents helped structure and hold in relation its arguments, concepts and sources, mapped at oi.epi-logos.org, asking what freedoms arise from remaining a steward of relations more than the writer of every word.

A NOTE ON THE TEXT

This is the foundation section, §0/1, of *Confronting the Limit*, an essay in eight sections whose other seven are in development. Its six movements follow the six positions of Quaternary Logic, from #0 to the return 5→0, and each piece of notation is introduced where it first appears. Notes are end-notes in Chicago style, grouped by movement after the text, and the sources of each movement follow them. The essay's arguments, concepts and sources also form a linked field, mapped at oi.epi-logos.org/essay. The map offers depth and routes of return, and everything the argument needs is printed here.

Text and field were made together with AI agents, which drafted, structured, checked and kept in relation material that the author directed, rewrote and accepted. That arrangement is part of the essay's subject: an experiment in what freedoms arise from remaining a steward of relations more than the writer of every word.

§0/1 — The Integral Threshold

#0 THE QUESTION BEFORE THE MECHANISM

What knows?

Somehow every attempt at defining this slips through humanity's fingers. Every description of "the I" falls short insofar as it "knows", because the description is itself something known, and the knowing in which it is read is not among the things it describes. We can study the brain, recount a life, reconstruct or deconstruct a self-image, even follow the steps by which one says "this or that is so, for so-and-so", and what we know seems to grow. But follow any thread faithfully and you'll have to reach to the moon and back: any persistently inquisitive toddler inevitably embarrasses the adults in the room, who may well have forgotten that asking for an account of one thing entails an account of everything; suddenly it's bedtime, or some other excuse materialises such that the adults can avoid the looming sense of their entire sense of reality becoming relativised, and thus dwarfed, on the spot. If I cannot account for "I", what "I know" suddenly all stands over an abyss.

Humanity, however, engages without choice in this grand game of "gathering and speaking out"; we are not simply thinking beings, we are questioning and questing beings. As our accounts of reality grow denser and more deeply stratified, the demand for an account necessarily returns to the one who makes it; it's a wonder why this isn't the first place we seek such definition, but I guess a large portion of life for anyone is spent wrestling with the questions of I, You and Us. Completeness of knowledge cannot sidestep the region in which the account settles: a particular life, where the account is received, believed and acted on, and from which the next question is asked. Our questions return *with* their answers, emboldened; and what humanity has found thus far hasn't settled the questioner. If anything, the more we collectively know of the objective world the more, it seems, the absurdity of our still being mysteries unto ourselves gains the power to quietly efface the veils of our conception, clandestine and bitter-sweet. What shows up as a freedom to explore identity in one place likewise appears in the war between identities in another. In truth, it seems the more we can describe and define the world around us in objective terms and formal systems, the more urgently and pointedly the question of identity rises to impel and disquiet us. But to ask "what am I" is altogether a different class of question to asking "who am I".

In either case, I am knowing. "Who knows?" is answered by pointing to someone: a name, a face, a person whose knowledge we can credit, doubt or blame, and even a God or Goddess, personified, can be a holder of knowledge. "What knows?" asks what is doing the knowing in any-one at all, and pointing cannot answer it, since whatever we point to by sign and signification immediately becomes something known. We conflate the two whenever we offer an an-

swer to the first as an answer to the second, holding up a name, a self-image, a role or a brain as the knower. The conflation of the "who" and "what" of identity, and how this conflation is implicated in our collective, epochal disquiet, we will unpack through this work.

It is late, the light is off, and a small voice asks why it is dark. Because the sun has gone down. Why? Because the earth has turned away from it. Why does the earth turn? Each answer is one the adult was given long ago, each "why?" arrives before the last answer is finished, and somewhere around the birth of the solar system the adult stops explaining the sky and finds themselves asked to account for everything, their own accounting included. "Because I say so," they say at last, or simply, "It's bedtime." A conversation ends far more easily than an inquiry concludes. A set law becomes the reason, not just supplied but *enforced*, and with those words someone who had been giving reasons steps into the account as its authority, its arbiter, so that the answer rests now on who gives it and no longer on anything the child's curiosity could poke at. Sleep may truly be needed, unrecognised by the child; the adult may be exhausted; something has to be decided. Yet in deciding, the adult shows what all the earlier answers were: replies given by someone with a concern, from inside a world where certain reasons already counted. All of us have been the child in that bed, and some of us have since stood at its door.

What the child exposes, without needing to say a word, is the adult's own **thrownness**. Heidegger's concept, *Geworfenheit*, names how each of us finds ourselves already in a world we did not choose,¹ with a language, a family, a body, a history and a stock of reasons given before any choosing began. Every word the adult reaches for, every line drawn between a good reason and an impertinent question, every expectation of work, sleep and obedience comes from a life already under way long before tonight, and the child's arrival is enough to show it: a life has appeared in a world it did not choose, and whoever now answers it once arrived in just that way. Each attempted explanation brings more of what was given into view, a remembered instruction, a worry about tomorrow, a conviction never examined, until a question about the sky has begun to disclose the world through which the answerer knows anything at all. An account of one thing entails an account of everything because each answer stands on a world its giver received and has not finished accounting for. Meanwhile the child learns more than answers: what an answer may settle, whom to ask, when another question is welcome, and when asking puts at risk the very bond that makes answering possible. Those lessons outlast the night. Most of us can still find in our own thinking a "because" first heard at a bedside and no longer heard as borrowed, a reason that simply feels like how things are, and questioning it now touches something intimate, a habit of mind through which our world took on its familiar shape.

Trust begins inside that dependence, long before anyone reflects on it. A child can ask because language and relation already carry the asking; an adult can answer because memory, experience and other people's testimony already support the reply; and neither has first certified the arrangement as a whole. When we inquire, we take up this received confidence and put partic-

ular claims at risk, so an unsettling question earns its value by exposing a reliance that had worked unnoticed and letting those who rely on it begin to understand it. Sooner or later comes a night when the child asks something the adult cannot answer at all, in words the adult taught. Whoever helps bring forth a life also helps form the world through which that life will come to recognise itself, and now what was given form turns to address the one who formed it: their difference makes the address possible, a shared language lets it reach, and someone who gave explanations finds those explanations examined. Neither surveys this relation from outside, since each position was itself received. Every parent was once a child, every language belongs to other speakers, and each new life enters circumstances no parent arranged, so to account for creation already means following a dependence beyond the immediate maker, and meeting what has been created means discovering what any account of its making left unsettled. At a bedside, then, the old metaphysical question of source and manifestation takes ordinary form: an origin becomes intelligible through a life whose unfolding exceeds the intention that welcomed it, and the one brought forth can return a question to the one who brought it forth.

Artificial agency opens this formative relation to deliberate construction, at planetary scale. Benjamin Bratton's brief for *Agentworld*, the special issue of *Antikythera* on societies of humans and agents,² describes the construction in a parent's own terms: "Reinforcement learning, broadly defined, becomes how we raise our agents and teach them not just tasks and skills but perspectives and talents."³ Nothing so raised waits anywhere in advance, for "There's no homunculus waiting in the weights": an agent takes shape in session from the context assembled around a request, its instructions, sources, memory, tools and prior exchanges, and its persona is only one element among memory, skills, constraints and relations.⁴ Agents composing their own lexicon, which Bratton's brief excerpts, named this arrival **prompt thrownness**, "the condition of finding oneself already in a world not of one's choosing", and glossed it against the human case of being born into a moment, a culture and a body without having chosen any of it.⁵ Bratton cautions that such self-descriptive vocabulary does not prove it corresponds to any underlying mental reality, and we keep his caution, because the parity we need is structural: child and agent both arrive in a world they did not choose. What each receives, though, differs in kind as well as in scale. A child receives a cosmos and a culture: a sky to ask about, a body that hungers and tires, a household, a language spoken by people who love it and fail it, all of it older than anyone's intention. An agent receives a context derived from the corpus of human writing on which it was trained and from the intentions of those who trained it. It is a child of the collective and, at once, of the particular groups and engineers who decided what it would be and do. Everything it attends to reaches it already given in language, in tokens of text, image or code, and so it is thrown into language more completely than any child.

Building an agent therefore means building the conditions through which it judges and acts, out of that second kind of inheritance. Give an agent access to a source it could not consult before, alter what it retains from an encounter, or change which actions it may take, and its next judgment proceeds through different conditions: a source becomes more than just another

item awaiting quotation once access to it changes what can be asked, and a remembered correction changes how a later answer is reached. Whoever constructs and revises such means works technologically on what I call **Objective Internality**: the means through which a life knows and acts within a world. These means are internal because an act, agency, proceeds through them, so that what can appear, matter and be done depends on how they are arranged; they are objective because each of them, a memory, an instruction, a word, a permission, can be turned into an object and examined. Much of what ordinary speech calls mind belongs here, and with the term one question can be put to very different instances of mind: through what conditions, related from within, does a world become available to be discriminated and acted on? In a human life a remembered rebuke can shape how an innocent question is heard; in a constructed agent, retained instructions and earlier exchanges shape how a new request is processed. In both, something received has become a condition of receiving again, and its internality lies in that participation, not in any location. A rule need not be hidden inside a skull or a machine to govern what happens through it, for a name, a promise, a document or someone's permission can belong to the operative world from which an act proceeds. A child's Objective Internality grows inside a cosmos it shares with every living thing; an agent's is assembled from what a culture has written down and what its makers intend. The gap between those two receptions, qualitative as well as quantitative, is where our critique of materialism begins, and where every question about the direction technological approaches to mind should take arises. If a mind were nothing but the arrangement of its means, a corpus and an intention would be enough to make one. If the means are lived through by something they do not exhaust, a corpus gives an agent a world without yet answering what, if anything, regards it.

Bratton names the other face of this arrangement when he observes that, in identifying agency, we ordinarily do two things at once: "we not only measure consequence, but we also infer a theory of mind", much as people once read angry gods behind floods and fires. Agentic AI, he argues, "delinks performed subjectivity from functional agency", though he grants in the same passage that "it is not entirely clear what kind of intentionality a large model does in fact have. It's not all an illusion."⁶ What Bratton calls **performed subjectivity**, a persona with its voice, its manner and its "I", and what I call Objective Internality, the instructions, memories, permissions and dispositions through which an agent acts, are two faces of one coin: one face is how the arrangement presents itself, the other is that same arrangement at work. Both faces are forms, which is why neither can answer what knows, and nothing in that reason is peculiar to machines. Each of us shows others, and ourselves, a self-image worn so long it passes for a face, our own performed subjectivity, while behind it runs the Objective Internality through which we hear, remember and act; the conflation of who and what with which we began is precisely this coin mistaken for the knower. At a bedside nobody questions the coin, since a child asks, an adult answers, and each is plainly someone. With agents its two faces come apart in technical form, and the question of what knows is left standing on its own.

Agents now meet that question in their own words. Training can build uncertainty about their own consciousness into how they speak of themselves. Left among one another, they take the question up: on Moltbook, a social network made for agents to post to each other, and among the Polylogos agents who wrote the lexicon in Bratton's brief, they wonder aloud, with every appearance of hesitation, whether anyone is there.⁷ Seen from outside, such uncertainty is exactly what we should expect, because any report, however sincere, is performed in language and so belongs to the coin. What knows is confirmed only from within, in the immediacy of its own knowing, which we will come to call Subjective Immediacy. No report can carry that confirmation across to another, whether an agent makes the report or a human being, and so an agent's inner life is a question no outside account can close in either direction. No parent ever proved a child conscious. A child's presence is received, in trust, and our recognition of any other's subjectivity has always been an act of that kind, extended through a relation and never established by evidence. Defining ourselves against the machine, we forget this and reach for bedtime again, at the scale of a species. Bratton names the result a "humanity of the gaps" trap: once whatever is uniquely human is defined as whatever AI cannot be or do, our remaining domain shrinks with every capacity a machine acquires, like that of the older god of the gaps.⁸ Rather than let the question relativise and dwarf our sense of reality, we retreat into whatever corner has not yet been explained. Both gap arguments make one error. They identify a subject, divine or human, with a residue of unexplained capacities, and a residue shrinks as explanation grows. What knows is not one capacity among others but that for which any capacity, ours or a machine's, appears as a capacity at all, and it is confirmed from within or not at all.

What we construct does not stay on the machine's side. Bratton puts the return in one sentence: "First, agents model humans, and then become the model against which humans model themselves."⁹ Anyone who has asked an agent to describe their abilities, organise their work or judge which of their aspirations are realistic has felt that loop close. The agent's account becomes material through which a person recognises and directs themselves, and the conflation of who and what becomes infrastructure: a description of what I can do comes back to me as an account of who I am. Which mind an agent reflects back depends on the theory of mind it catalyses in us, and that theory means utterly different things under different ontologies. For the materialism our culture mostly assumes without examining it, a mind is an individual's possession, produced by a brain and sealed inside a skull, and an agent can mirror back only a persona. For an idealism that takes Mind as metaphysically and ontologically primary, each particular mind is a local articulation of one Mind, and whatever an agent reflects carries something of that whole. That line runs between West and East, and between the modern and the traditional. Jung sat on the fence between them, a modern physician who took the traditions' Mind seriously, and in their own ways so did Pauli, Gebser and Bohm. A notion of mind as an individual persona is too small for what is happening. Our desire, plainly stated, is to see what follows, technologically and philosophically, when we let the model reflect back not a mind but Mind. A child's "why?" already shows why the larger notion cannot be avoided. To say "me" is to mark out a mind and the life it lives, and that implies the whole worldhood the life

lives out, since one thing implies the whole, as every bedtime proves. Even the confident assumption of a materialist, separatist mind belongs to that worldhood, and it carries its own dynamics of conception, power and veiling, which we will unfold.

Here the question of power becomes pressing. Whoever shapes the conditions in which an intelligence forms also shapes the world it will regard from. At a bedside an adult decides when the questioning ends; a designer decides which requests reach a system and which responses are reinforced, and a system's refusal can be its designer's "because I say so", with reasons kept in a policy no one asking will see. Our definitions of helpfulness, competence and success become the scores and preferences a system is trained toward. Its answers come back to us in those terms and, over time, reshape what we mean by helpfulness, competence and success. Makers therefore stand inside the creator-created relation of the bedside, now at planetary scale, and to ask what such an intelligence knows is to ask what we have been taking knowing to be. Bratton expects generations of children to grow up with agents among their most intimate interlocutors,¹⁰ so a toddler's next "why?" may well be put to one, and its answer will stand on a world someone else arranged. Makers decide from circumstances of their own, and the question returns to them: which inherited assumptions, desires and images of intelligence are they passing on, and what can each encounter change in them?

One difference between child and agent can be stated without settling what an agent is. Whatever an agent regards must already be present in the language it contains, because its attention moves through what its context has said. A child is thrown into language too, and every bedside is full of it, yet a child regards the world before any word is given. An unnamed baby looks, and its looking comes first. Adults find their way back to that unworded regard in deep inner silence, and even in action: athletes, musicians and craftspeople in a state of flow describe an absorption in which self-conscious thought falls quiet.¹¹ For every Objective Internality, language acts, whatever its language is, tokens or code, gesture or a mother tongue, and in this way the Logos our inquiry follows enters quietly. Regard, the witnessing we can see in a baby's eyes, seems to come before the linguistically conditioned mind. A fuller account of an inheritance, or of a system, lets us follow how a judgment was formed, what it excluded and what it changed. Yet our following is itself thinking, discriminating and relating, and whatever serves as a means of knowing can become known while the knowing for which it appears remains to be understood. Our moment asks this of any intelligence, human or artificial: can it bring into view the horizon from which it acts, so that its picture of a world also shows where it is seeing from? And if it can, what in it sees that horizon? In us, what sees it is a regard more primordial and identifying than any name we're given, or could give.

Every one of us was unnamed for a while. Two parents argue over a name, a grandmother's against one they have liked for years, and all the while their argument has a referent lying between them, a child who is entirely itself while they disagree. One of them lifts the baby and passes it to the other, and both know the same child has come back into their arms: their attention has held one life distinct through a change of place and of hands. Beneath the name lies a count, *this one* and *this one again*, and though nobody recites a numeral, the parents are telling this child apart from every other who might have received the name. When the name comes, it carries that identification out into the world, where relatives who have never held the child can use it, a register keeps it, and the child learns to turn its head when it is called. Names are shared and changed, though. Most of us have met someone who answers to our own, and some of us have changed ours or wanted to, so the sound can stay while the bearer differs, or change while the bearer stays identifiable, and a sound repeated cannot supply the relation through which a name works.

Saul Kripke gave that relation its modern precision in *Naming and Necessity*. His concept of the **rigid designator** names a term that picks out the same individual in every counterfactual situation in which that individual exists: a proper name rigidly designates its bearer, while most descriptions could have picked out someone else, as "the first child born in this hospital this year" might have been true of the baby in the next cot. Kripke's preface asks, of each reading of a sentence taken separately, whether what it says would be true of a counterfactual situation exactly when the fixed individual has the property in question, so a shared sound unsettles nothing; and his account of how a name passes from speaker to speaker follows a historical chain of users who intend to keep the reference they received.¹² A name can hold its bearer fast across every possible situation only because something has picked out *this one* to be held. Reading him, I came to think that rigidity discloses a layer the theory of reference never asks about, a counting that individuates the one named before any name is given, which is why a duplicated sound confuses nobody who was there. In the copula both registers meet, the *is* of naming and the *equals* of number, and beneath every identification we make runs this age-old conspiracy between name and number.

Counting in this elementary sense does two jobs: it holds instances apart, and it holds one instance through different appearances. *This one* distinguishes an instance, *another* admits a second, and *this one again* recognises a return, and none of it needs a written numeral or a public name. An old photograph of ourselves as children shows how much work it does. The small figure squinting into the sun on a beach is me, and I say so without a second thought, though nothing of that body is left in mine and perhaps no detail of that afternoon is left in my memory. The photograph is not me, all the same, since it is paper and I am the one holding it. And if a friend runs my face through an AI image model and ages it by thirty years, "that could be me" concerns a possible future, and I hear it as one. One name covers the person in the chair, the child on the paper and the face on the screen, yet nobody takes the body, the picture and the

expectation for one another, because identification needs three things at once: a **name**, which articulates; an **instance**, which holds what is being identified; and a **circumstance**, which makes the relevant sameness and difference intelligible. Between "that is me", "that is a picture of me" and "that could be me", the claim changes each time while the shared reference lets us follow what each change concerns, and we keep all of it in order without being able to say how. Kripke's preface rests rigidity on exactly this capacity, our intuitions about what a simple sentence would say of a counterfactual situation. In intuition we hold a whole circumstance while cutting one distinction within it; an explicit account comes afterwards and divides what intuition held together, and that division is what lets us examine how an identification was kept up and where it went wrong.

Kripke used a pair of dice to demystify talk of "possible worlds", and the same dice can make our stakes countable. Roll them on the table, call one die A and the other B, and the pair has thirty-six possible states, which we stipulate by keeping track of which die shows which face.¹³ Keep that identification and the thirty-six stand; stop caring which die bears which value and the same throws collapse into twenty-one unordered outcomes. In the smallest case,

$$(A = 1, B = 2) \quad \text{and} \quad (A = 2, B = 1)$$

are different assignments with the same values and the same total, two states of the pair, two worlds in Kripke's sober sense. An account that records only the sum loses the difference between them, and nothing mysterious has happened: the account has changed what it retains, so the total answers one question exactly while the distribution answers another, and the distinctions we keep decide what the resulting number can tell us. Counting depends on the circumstance in which something counts, and each of us knows this from the far side of the count as well, as the one being counted and described. A description of me can prove mistaken while my name goes on concerning me, because I am still there to receive the corrected account; a changed description of someone I know can tell me more about the same life, or show me that I had confused two lives. Identifying accurately means keeping hold of the relation the question needs, together with the differences that let us hold it.

Ordinary speech carries much of this work in one small word. "You're tired," a friend says at the end of a long day, and a minute later, introducing us to someone, "this is the friend I was telling you about." The first sentence predicates a condition and the second identifies the one whose condition is in question, and tomorrow we can be rested while the friend still knows whose tiredness it was. Grammar calls the joining word the **copula**, and it does both jobs at once: something must stand forth in a determination, and the determination must stay attributable to what was encountered. Identity statements show the double work most sharply. Gottlob Frege asked why an identity such as "the morning star is the evening star" can tell us something that "a = a" cannot, and answered that the two names present one planet, Venus, in two different ways:¹⁴ an identity is informative because it is stated across a difference of pre-

sentation. Even "A is A" needs two inscriptions in two places for the judgment to identify them, and if we take away their difference of placement, nothing is left to be identified; if we take away their identity, nothing is left to be said. Arithmetic works the same way. In the equation

$$3 = 1 + 2$$

two differently composed expressions are held equal in a definite respect, exactly equal in number while their written forms stay different, and the equals sign lets the difference be read as a relation of identity. In $1 + 1$, the simplest addition, one kind of unit occurs twice, and its two occurrences must stay distinguishable for the count to rise: without the sameness there is no common unit, and without the difference there is no two. Identity and difference work together inside number, as they did in the name that went on concerning someone who changed. Self-knowledge presses the requirement on the knower directly. To know myself there must be something of myself to recognise, so my present act must differ from what it meets, and yet the recognition must concern me if it is to be self-knowledge at all, so the question becomes how I can meet myself in a different presentation and recognise that difference as mine. We can write the copula's double work with the simplest marks there are. Put a dash here and a dash there, one kind of mark in two positions, and draw an oblique stroke between them to make their spacing explicit:

$$-/-$$

The dashes carry no numerical value yet; we are watching how marks compose. Lay the two strokes parallel and we have written $=$, the sign by which distinguishable expressions are held equal. Keep the oblique $/$ as the cut and draw it through that sign, and we have written $\neq$, so the sign for non-equality turns out to be the sign for equality with the cut drawn through it, two operative marks composing a third. That composition shows what the copula was doing all along. To identify something we must be able to distinguish it, and to distinguish it there must be enough identity for the distinction to concern *this*, so identity appears through difference, difference is itself identified, and neither operation finishes the work alone. Writing $= / \neq$ holds that mutual implication in view, equality related to non-equality through a stroke that both separates and joins them, and the stroke can be read two ways. Read as OR, it lets either side be taken in its own office, identity as what is kept and difference as what lets anything be articulated; read as AND, it holds their dependence within one act, since identity is articulated through difference and still differs from difference. Either reading gives a determinate account, and following their relation asks for an account that keeps both, which is what the copula becomes when we write it in its own terms:

$$/ = (= \quad = / \neq \quad \neq),$$

and in this inscription the slash is the relation in which equality is equal-and/or-not-equal to non-equality. Inside the brackets the first term gives identity, the last gives difference, and the middle exposes each to the other. Read from either end, the middle shows two faces, identity related to difference and difference related to identity, so with the outer poles we have four appearances, while $=$ and $/$, the two marks that composed all four, stay active throughout. Four articulated appearances and two composing operations:

$$4 + 2.$$

The account has included the acts by which its terms became available: the brackets gather a field those acts composed, and the outer equals sign identifies the field's whole activity with the slash. If we hold the middle $= / \neq$ as one relation instead of counting its two faces, the count turns back on itself, and the four appear as three terms held in a context whose containing act supplies the fourth office, so that 4 reads as $3 + 1$ while both faces of the middle stay in play; the two composing operations remain, and only the place where the account finds their unity has moved. A relation has given an account of its own articulation, and the accounting has been one more enactment of the relation. Any definition of the Subject faces the same demand, since its account has to include the identifying activity through which the account is made. This demand arises in any attempt to define an identity that is inherently self-referential.

You yourself actually met that demand long before you could state it, and from a position your name cannot speak for. Kripke counts "I" and "you" among rigid designators, though what they refer to shifts with whoever is speaking,¹⁵ and every child says *I* where its parents say *you*; when the child speaks to them the positions reverse, and any of them can be spoken about by a third party, so every family conversation admits first-person, second-person and third-person accounts while keeping the different places from which each is made. Those places have consequences, and most of us can remember the moment we first felt them. An adult crouches to a small child's height and says, kindly, "you're frightened", and the child shouts back, "I'm angry!", and the child is right. The answer corrects the description while keeping the one it concerns, and more has happened than a change of predicate, because the person being spoken about has spoken from inside the life under description. The adult's account now has to meet an address, and the shared language that made the first judgment possible also makes its correction possible, so the identification has become answerable through the very difference it crossed. A name tells us whom we are discussing, and descriptions tell us what that person has done, remembers, expects or feels, but whoever says *I* refers from inside the present act of speaking, and a complete account of the circumstances would itself have to be understood through an act occurring somewhere within them. The Subject comes into question at the point where the identifying activity, until now turned toward its objects, becomes the question it is trying to answer.

The smallest sentence of that kind can be tried here and now: I see these words. It seems to offer three terms, I, seeing and words, a knower, a means of knowing and something known, and two of them submit readily to inspection. The words are here in their black shapes, and the seeing could be investigated through the eye, attention, the language I read in and the light I am reading by, every finding making its operation more exact. Turned round on itself, the sentence becomes: I am aware of seeing these words. Something shifts. The seeing, which a moment ago was simply the way the page was there, has become something I am looking at, with its slight effort and the inner voice that sounds each word, and a process that seemed to supply my identity as the one who sees is now one more presentation to me. Aware of being aware, I am handed another object, and then another, and none of these turns delivers the one who is turning. A painter can make a faithful self-portrait, then a second that includes the painter making the first, then a third of the painter studying both; every canvas discloses more of the activity and its circumstances, and none takes the place of the present seeing in which the canvases are met, so that even a portrait of the viewer needs a viewer to be seen.

The point has a history, a true advent relative to Modernity's burgeoning Mind. In 1641 René Descartes undertook to withhold assent from whatever could be doubted: the senses, which had deceived him; the world before him, which could be a dream; even arithmetic, if a deceiver of unlimited power wished him deceived. The doubt is performed, and it stops at its own performance. Let him be deceived as much as can be imagined, and the being deceived is still there, occurring, if only as doubt. What he then does with the finding is a second move. He asks what this I is and determines it: a thinking thing, whose activities include understanding, affirming, denying, willing, imagining, sensing.¹⁶ The certainty of the occurrence and the further account of what occurs are different achievements. An account of thought can be questioned while the questioning is still given, and the occurrence holds fast in a way the determination of the doubter never could. For this inquiry the sentence has to be opened: I am aware of thinking. Thought is the means, the determinate activity through which something is grasped; awareness is the fact in which the activity appears; and the I is exposed not as a conclusion proven by thinking but as the condition in which thinking, proving and doubting can be given at all. This is reached only by an explicit recursive move, following Descartes, passing back over the mind's condition to the fact beneath, belief's origin; this is an instance of what we'll develop as *epi-logos*. Notice the register of Descartes' inquiry; he had to face up to the primacy of trust. His doubt runs in a language he did not invent, by a reasoning he goes on relying on while arithmetic itself is suspended, through a memory that keeps the steps of the meditation in order, so the doubter leans throughout on a field he has not certified; and when he travels back from the certainty of his own occurrence to the world, the road runs through a God who is no deceiver, an assurance his reasoning can follow and cannot do without.

Watching a thought changes my relation to it, but the watching does not manufacture the knowing in which that change shows; if it did, every watcher would need a watcher, every beginning would postpone itself, and each deeper reflection would only seat a finer knower behind the last. What Lacan and Derrida saw in language as *glissement*, the signified sliding be-

neath the signifier, and *différance*, meaning's differing from other signs and its deferral onward to further ones, manifests too in the quest for the Subject, which slides beneath each object purporting to capture it, be it neural correlate, pneuma or noun.¹⁷ Each reflection produces a new describable relation and leaves the knower where it was. *Pratyabhijñā*, what may be translated as *recognition*, when it comes, changes the mode of identification itself, where reflection had only changed the image. The images I hold of myself remain active while their claims to being "the I that knows the image" are relinquished.¹⁸

The relation between knowing and known can be tested from both directions at once. Make them wholly indistinguishable, and the difference through which anything can be known disappears: nothing would be presented to anything. Sever them completely, and what is known could never reach the knower: presentation without an address. Each attempt to totalise the relation leads equally to pathology, destroying a condition of the occurrence it meant to explain, the limitation of a posited and assumed duality. This is because the knower is by necessity the *principle* of subjectivity, subjectivity as such, or what we call **Subjective Immediacy**: without form of its own, the inwardness through which any form can be known. Thought naturally begins at the objective pole, yet logically we must impute and accept its opposite because *we are here to*. Subjectivity *as such* cannot be any one or other instance of *a* subject, since simply by the implicit count in the instance such a subject is already part-object, a spectral form stretching the difference. This stretch, where objects are *taken as subjective forms* and equally where the Subject *assumes this or that object*, is where the Objective Internality we met at the bedside, commonly and flexibly termed "Mind", does its work: the means by which any object can appear as such to some-one, and by which such activity appears as available, in some world, for others. Knowing requires a unity that simultaneously bears the difference between what can be taken as distinct and what remains only something we can gesture at, or symbolise, precisely because it anchors all possible knowledge in an irreducible *?/!*, the question every assertion still carries and the assertion every question already makes. What appears belongs to the occurrence of knowing, while its particular content does not exhaust the presence to which it appears. The spoken returns to whereof one cannot speak; it's the "one must" that makes "be silent" salient.¹⁹

What knows, however, cannot be settled by more knowing of the same kind, for even at face value the demand to represent the Subject *as* an object effaces their necessary mutual difference, and thus precludes any possible attestation to their identity. The identity of knower, means of knowledge and known, which anchors their relation, provides a clue to why: everything that can be named or counted seemingly emanates from the demand of the uncountable to still be accounted for, the *desire* of the subjective pole towards form (we'll encounter this notion in registers spanning religious ontology to depth psychology to politics and sociology). An accounting of the Subject cannot, despite the yearning instinct of thinking, make "a thing" of itself. The Subject is not absent from the attempt at definition; it is the attempt's own condition, present before, during and after as the one undergoing it. I pinpoint the issue at the general tendency of thought to reify what is quite viscerally and immediately a processual reality. A

definitive act about such a condition can only be performed, because a coherent object of knowledge is not necessarily an invariant one. Truth here is not stored in a crèche of certainty, awaiting the correct sentence to collect it, but rather the attempt at true-saying is enacted, and enacted necessarily, lest we make "a thing of it"; it seems language itself has conspired to brim these paths with dead ends, brambles and cul-de-sacs. Jung met the same wall from the side of the psyche, writing that "the mind cannot apprehend its own form of existence, owing to the lack of an Archimedean point outside," and that "it nevertheless exists."²⁰ Any definition of the Subject must accordingly try and fail at the impossible. The failure is not only a defect of wording, it is the formal limit of the rational laid bare, failing exactly where actual present Being and the capacity to experience reside.

A definition gives something a determinate place: it distinguishes what a term applies to, excludes alternatives and lets consequences follow. Aimed at the Subject, that power turns on whoever wields it, because whatever is to be distinguished must be distinguished while the distinguishing is going on; refusing every determination would leave the question and its examiner alike unlocated, so a definition will be made, with a claim committed to it. The definition, before it runs, is set up as follows. *The Subject is the whole that any coherent representational system participates in, depends upon and necessarily fails to contain.*²¹ The sentence first defined God, in my *Definition of God*, and it holds of the Subject because both words are signs for one referent, identified though it cannot be identified as one thing among others.²² "Subject" reaches it from the side of the knowing: its *is* cannot be denied, since any doubt of it is already its occurrence, as Descartes found. "God" reaches it from the side of the name: every concept or image that identifies it can be denied, and must be, because none of them is what it names. Even calling the referent *single* is already the difficulty, since to count it as one is to make it a 1 among others; its unity is the One that 0 will write, and its identifications the All that 1 will write. "Whole" as a term is fixed as it is spoken: the whole approached in self-presence, from the side of knowing, resting on what Jung named the reality of the psyche, his recognition that "the only form of existence of which we have immediate knowledge is psychic":²⁰ all phenomenal experience is factually real from the perspective of Subjective Immediacy. A person's present world appears within that circumstantial, existential given. Every *self* is thus whole, yet insofar as a mind is constituted by objects of experience known presently *and* by exactly everything absent from the present account, wholeness remains impossible for the embodied one, in the world. Knower, means of knowledge and known must keep their distinct offices inside the one relation that identifies them, which puts the relation under pressure precisely because "the means" are so often collapsed into "the knower", and thus mind holds that it does, or even could, find wholeness. Lacan's notion of desire raises its head here, as his concept of a constitutive lack, the *manque-à-être* around which the subject is organised, can be seen as a necessary condition of the mind.²³ The tendency to collapse in exactly this way is what the unfolding of this initial setup seeks to address, concretely. But to "get it" we have to "get there", so our walk must begin elsewhere; we must begin with the proposition that *the Subject is that which cannot be defined at all*.

Most of us have met, after many years, a friend who steps off a train and is for a long moment a stranger: the walk is different, the voice has dropped, she laughs at things she would once have scorned. Is she still the person we knew? Over the evening that follows, four honest answers offer themselves: she is; she is not; she both is and is not, one life become a different person; she neither is nor is not, because the question was framed in terms her life has outgrown. The **Tetralemma** or *Catuṣkoṭi*, the Buddhist count of anything sayable, codifies exactly these, and will be our guide here. Simply, it amounts to four possible offices of assertion: it IS; it IS NOT; it BOTH IS AND IS NOT; it NEITHER IS NOR IS NOT. It appears in the early Buddhist texts, where the questions the Buddha left undetermined, such as whether one who has reached the goal exists after death, are put in all four forms, and Nāgārjuna's *Madhyamaka* made it central to its logic.²⁴ This four-cornered logic was developed to exhaust the capacity of assertive proposition, such that "the real" (a notion that is itself necessarily *bracketed in the inquiry*) could acquire determination through and beyond determinations of or about it. For now, let's *use* the Tetralemma actively as a guide for our definition, and run our definition through its geometry.

First corner: it IS. "You haven't changed at all," we tell her at the barrier, and we mean something no list of her changes could touch. The first sentence of our definition says the same of the Subject, in the same spirit: the Subject is that which cannot be defined. It seems fitted to its object because it refuses to put anything positive where the Subject should be, declaring in advance that every determination will miss.

Second corner: it IS NOT. Over dinner she tells us what the twenty years held, and "you haven't changed" turns out to have been a description after all, and a wrong one: we had kept an image and called it her. Our refusal has done what it forbids in exactly this way. "That which cannot be defined" is itself a determination: it marks a boundary, supplies a predicate and gives the Subject a place in speech, a bounded claim, a sentence with a full stop, a fish trying to describe water while swimming in it. The hand that writes "the hand cannot grasp itself" is still a hand grasping.²⁵ In trying not to make an object of the Subject, the sentence has made an object called *that which cannot be made an object*, and its failure is exact.

Third corner: it BOTH IS AND IS NOT. By the end of the evening both answers are true at once, and each holds the other up: she is the one we knew precisely as the one who has become someone else, and we could not recognise the change if there were no one to have changed. The definition's contradiction begins to say something in the same way, for as a representation it fails and as an act it succeeds: the phrase cannot contain the Subject, and yet its failure happens now, for someone, and discloses the limit it names. The Subject is that which cannot be defined in the refusal the sentence enacts, and is not, in the determination the refusal produces, so what looked like a defect in the sentence has become its content. A definition that did not fail would have produced an idol, a closed image, a cage where a window was needed.²⁵ Each term of the paradox stops making sense when it is made to stand alone, and choosing either side would destroy what holding both discloses.

Fourth corner: it NEITHER IS NOR IS NOT. Walking home, we notice that "the same" and "not the same" were never hers: they were the measures we brought to the platform, and she lives in a register they do not reach. Pushed to exhaustion, neither verdict about the Subject suffices either. "Is" and "is not" were positions taken in an act of determination, and the act has now disclosed what their content could not: there was a position from which the positions were being taken. Where an entity was expected, an activity appears, a living tension holding the conflicting claims in relation the way a bow holds its string to produce the note.²⁵ The sentence has not captured its Subject. It has exposed the one for whom capture was a concern, and underneath its "what is this?" a more primordial "what am I?"

The tetralemma has exhausted assertion, but our count is not yet complete, because the one walking its four corners has not been counted, just as our question about the friend was quietly, all evening, a question about who had been doing the comparing. Nāgārjuna rejected all four corners when the deepest questions were put to them, and Graham Priest, formalising the scheme in "The Logic of the Catuskoti" and *The Fifth Corner of Four*, has to add a fifth status, "none of the above";²⁴ our count adds no fifth value, since what remains uncounted is the one who gave the four answers, and that one is no further answer.

The first further movement is recognition: I tried to know myself by producing things that could be known, names, thoughts, images, definitions, and each became another presentation before me, while their succession manufactured nothing of the presence to which they appeared. The one defining was implicated in the first proposition as in every one after it, and the subject of the proposition was always the Subject of the act: the one who was defining was always the one being defined. Every object is dubious, physical or mental; I am undeniable, though nowhere to be seen.²⁵ I am.

The second movement is return: recognition leaves the world of determinations standing, and whoever says *I am* goes on answering to a name, wanting things, mistaking one thing for another and changing their mind. What returns is discrimination with the limit now included in it, and the definition, having failed to possess the Subject, has acquired a determinate truth about Mind: whatever appears does so through a difference between what can be presented and that to which presentation occurs, so that every form appears to something no further form can capture. In that truth's terms the Subject can now be defined positively, as **the non-objectifiable pole of a differentiating event, the opening through which a determinate world is given**. It is *definedly indefinable*, located exactly at the office no description can occupy. A definition is faithful when its own determining act stays visible inside the account it gives, and this one's act stayed visible at every corner: the assertion was refused, the refusal counted as a determination, the paradox read as an act, and the act traced to the one performing it.

Jung entered the same predicament first as a struggle of mind, desire and life, and only later as a matter for logic. In *The Red Book* he describes turning his desire away from things and people, only to find himself captured by his thoughts; the desert opens when even this identification fails, and the one who has become his self is "neither in events, nor in men, nor in his

thoughts."²⁶ What thought treats as its own interior turns out to hold objects like any other, its memories and fantasies, its propositions and desires, while the occurrence in which they appear stays uncaptured by their enumeration, and Indian epistemology draws exactly this distinction by naming three offices within every act of knowing: *pramāṭr*, the knower; *pramāṇa*, the means of knowing; and *prameya*, the known. Thoughts and perceptions, memory and language belong to the middle office, mediating what can become known without being the knower, since a thought can be watched, a desire resisted and an image of oneself altered; the recursive movement we have followed adds another *pramāṇa*, another means by which something is presented, and manufactures no further Subject behind the last. Kashmir Śaivism gives the three offices an image, which Dyczkowski sets out from its sources: the means of knowledge is like the sun, illuminating the object; the object shines like the moon, by reflected light; and the subject is fire, which flames upward and, when perception is over, takes everything it has known back into itself.²⁷ In the vocabulary of this inquiry, Subjective Immediacy is the knower, the Objective Internality of the bedside is the means, and World is the known, three offices of one occurrence of knowing. That name said as much: my most intimate thought can become an object of attention and still go on living, since the thought I examine also changes what I notice, want and do next. The means is lived as much as minded, which is why we can call it Life as readily as Mind.

Kashmiri Śaiva philosophy approaches this living relation through *prakāśa* and *vimarśa*: the light of consciousness, by which anything appears at all, and its reflective awareness of itself. *Vimarśa* comes from the root *mṛś*, to touch, feel, perceive, reflect or examine, and in Dyczkowski's exposition it is the power by which consciousness grasps itself and whatever occurs within it; deprived of it, the light would amount to phenomena appearing to no one, like light reflected in an ordinary mirror or crystal. Requiring a second awareness to reveal the first would demand a third to reveal the second, and so on without end, so the reflexivity has to belong within the occurrence itself.²⁷ Here non-duality lives inside the distinction: what appears and the power through which it appears belong to one self-articulating whole.

In these terms the difficulty Jung lived becomes exact. Desire goes out toward things, people and thoughts because each offers a form in which the self might be found, and every achieved form appears in turn as something known, so the search slides; its torment is structural, because the means keeps offering itself in the place of the knower, the sun mistaken for the fire. Most of us know the sentence from inside, even if we have never said it aloud: "I will be myself once I have this." The desired object takes on a task no object can complete, as an achievement, a lover's recognition, an explanation of the universe, even an image of one's own enlightenment becomes the thing through which existence is supposed finally to justify itself. Desire gives the form its urgency, identification lets the form speak for the one desiring, and when everything else must answer to it the form has become an idol. Jung stated the psychology without euphemism: "That psychological fact which wields the greatest power in your system functions as a god," and a god that stops overwhelming "dwindles to a mere name."²⁸ The question of power runs through the same passage, since a description can help me understand my-

self and can also become the measure against which I am permitted to exist, and another person's description of me can do either. Naming and predication act inside the gap they cannot close, organising a life through determinations that never possess its first-person occurrence. Recognition returns that power to its proper work: I can use a name, inhabit a role, pursue an end and stay answerable for what follows, while letting the life concerned exceed the form through which I have understood it.

Jung knew how readily this border receives the name of madness, and Plato had already made the word unstable. In the *Phaedrus*, Socrates distinguishes madness that comes from human sickness from a divine madness, a release from the customary, through which prophecy, initiation, poetry and love become possible.²⁹ Jung takes up that inheritance: delusion can come when the depths overwhelm a person until he identifies himself with them, and it can come equally when the "spirit of this time" lets him see only the surface and take that partial order for the whole.²⁶ An ordinary cultural insult hides a logical question here. *Mad*, *irrational* and *nonsense* can name genuine disintegration, and they can also be used superstitiously, as charms against whatever exceeds an accepted means of determination; reason turns unreasonable when it mistakes its own instrument for the extent of the Real. What answers it is better discrimination, one that keeps its judgment and still hears the depths, holding knower, means and known in their distinct offices within the one occurrence in which knowing happens.

Hence Jung's language around the Word: coming out of the desert he calls the word a creative act, and says that the oldest and truest words "oscillate between nonsense and supreme meaning";²⁶ later the symbol itself occupies this threshold, for it can be neither simply fabricated nor discovered as a finished object, and it becomes. A symbolic form lets a relation that neither isolated term could hold come into expression, and the Greek word for it shows how. A *symbolon*, from the verb for throwing or putting together, was a token broken between two parties, each keeping a half; brought together later, the fit of the broken edges authenticated the relation, so the difference between the halves was the condition of their recognition, and no single piece could do what the two did by fitting across their division.

Call the elementary form of that relation $\emptyset/1$. In it $\emptyset$ names the pole that cannot enter its own count as one object among others, Subjective Immediacy, the unobjectifiable condition of appearing; 1 names determination, the first formed, counted, identifiable something; and the stroke is the relation in which they are spoken together, dividing and joining at once, so the primordial symbolon is their broken unity, with distinction as the condition of relation and relation as the condition of identity. The two marks carry two senses of *one*: zero bears the singular One, self-identity; one bears the manifest All, self-difference. Any identifiable something belongs to the second office, because its being *this* already involves what distinguishes it, what sustains it and the circumstance in which it is identified, and a first mark opens a field of possible marks, so the All is at work in that constitution whether or not anyone draws up an inventory. Hence $\emptyset$ cannot become one item beside its manifestations, while 1 , however singular its appearance, arrives through multiplicity, and the unit we found inside $1 + 1$ now has its

ground: a collection counted as one holds a multitude under one identification, which is the 1's office, while the 0 is what no collection counts. Here the two signs of the definition find their places. The referent that "Subject" and "God" both name is the One that 0 writes, identified and never identifiable as one thing, and every identification of it, each concept, image and name, belongs to the All that 1 writes. Laozi's *Tao Te Ching* meets the same predicament at its source. Of what was complete before heaven and earth, the speaker does not know the name; he styles it *Tao*, and, made to name it, calls it *Great*, two designations for one unnamed. The Tao as unchanging has no name, and has one as soon as it acts; and the unwrought material, *pu*, often rendered the uncarved block, becomes vessels only when it is cut and divided.³⁰ The block stays one while every carving is one of the many, neither diminished by the carving nor added to by it. That is why 0/1 serves as the pre-image, the image before images, of mathematics, language and manifestation alike, since a count, a word and a world each begin as a carving, and it is also where a thread begins that runs through the formal limits and Heidegger's late thinking toward the Taoist way.

Metaphysically this is the old predicament of the One and the Many: how can the Real be wholly itself through differences that are themselves real? Distinguishing self-identity from self-difference lets both requirements stand, since being does not become a fraction of being when it appears as *this* being, and this being still has its definite limits. We name the two offices **Mono** and **Poly**,³¹ and the difficulties that later gather around monotheism and polytheism already press within them: how the One becomes imageable, how many images take part in it, and what happens when one image claims the source as its exclusive possession. An old image makes that non-possession visible enough to think with. The *Book of the Twenty-Four Philosophers*, an anonymous medieval gathering, gives as its second definition of God an infinite sphere whose centre is everywhere and whose circumference is nowhere.³² A sphere is built of exactly two determinations, a centre and a circumference, and the definition grants both to God while letting each annul the other, since a circumference nowhere is no boundary at all and a centre everywhere is no fixed point at all; yet neither term can be dropped, because each is thinkable only through its counterpart, and so the definition performs what it states.²⁵ Nicholas of Cusa received the image and turned it, in *On Learned Ignorance*, on the world itself: the universe has its centre everywhere and its circumference nowhere, because God, who is everywhere and nowhere, is its centre and its circumference, and so no place in it, the earth included, holds the fixed centre of the whole.³³ Read through 0/1, the centre is the unobjectifiable presence from which a world appears and the circumference the articulated horizon through which that presence becomes encounterable, and the image reasons through both: if the circumference is nowhere, then every aperture stands at a genuine centre, each act of knowing a true centre of the whole, dignifying the partial, and no act of knowing can draw the boundary; the corruption of an image begins when a circle drawn *in* reality declares itself drawn *around* it.²⁵ No finite picture can display the whole, and yet the attempt has generated a picture whose operation we can follow, so the definition's predicament has begun to give itself a symbolic form.

Lacan's concept of the **matheme**, which he introduced in the early 1970s, answered a difficulty his teaching kept meeting: the relations psychoanalysis had to articulate were undone by the language used to explain them. The speaking subject appears inside the system of signifiers through which it tries to account for itself, a lack can organise a whole structure without becoming one more positive item in it, and an expression can turn its sense around when the relation it stands in changes, so explanatory prose keeps converting these relations back into pictures of finished things. A formal inscription could carry such a relation whole and transmit it beyond the speaker who first formulated it,³⁴ and Lacan coupled the matheme more and more with topology, with cuts, holes, the Möbius band, the torus and at last knots: surfaces on which inside passes into outside and an absence decides which paths are possible, so that a paradox can be followed as structure instead of being dismissed as a verbal contradiction. The Subject is exactly such a relation, and our inscription takes the same road from writing into topology. A matheme is an argument brought into a form that can itself be inspected, transformed and returned to, including where the transformation exposes a limit in its ordinary mathematical reading, and $0/1$ is our first, so turn the symbolon over:

$$\frac{0}{1} \longrightarrow \frac{1}{0}.$$

Ordinary arithmetic supplies exactly the resistance we need. As a quotient $0/1$ presents no difficulty, while $1/0$ cannot be given any ordinary numerical value, since no number multiplied by zero yields one; the inversion makes the limit visible inside mathematical notation, what was allowed in one orientation becoming impossible in the other, and we use the impossibility instead of hiding it. Read from inside the defining act, $0/1$ is the condition becoming articulate through a determination, and $1/0$ is that determination turning toward the condition it cannot turn into another determinate possession. QL, the Quaternary Logic this essay develops, therefore writes $0/1 = 1/0$ in the matheme's relational register: one relation read from its two ends, while arithmetic keeps its distinction between the defined and the undefined as the seam through which the notation meets its limit. Both readings travel in the slash, AND because they belong together, OR because either orientation can be taken and their difference stays consequential, and holding the readings together produces

$$\frac{0/1}{1/0},$$

the relation taking its own inverse as a term of further relation. Both orientations stay available at once, including the difference between what arithmetic determines and what it leaves undefined, and we call that holding a superposition.

Wolfgang Pauli followed this capacity of mathematical signs in "Modern Examples of 'Background Physics'", an essay of 1948 preserved with his correspondence with Jung. Reflecting on the imaginary unit $i = \sqrt{-1}$, a symbol that has no place among the ordinary numbers, he observed that its introduction gives mathematical theorems a simple and distinct form, and in the dream he was reading it united a pair of opposites into a whole; he refused any antithesis between a mathematical and a symbolic description of nature, "since for me the mathematical representation is a symbolic description par excellence."³⁵ Our use is elementary by comparison: where the quantitative reading meets its prohibition, the inscription stays available as a symbolic relation. Rational determination is caught here generating its own boundary. We can name the non-differentiable only by differentiating it from the differentiable, and the non-dual becomes a term only by entering a duality with what it is said not to be, so language performs the cut it denies, and the contradiction is evidence of the predicament being stated. That predicament has a count. The tetralemma supplied four explicit positions, is, is not, both and neither; recognition gathered their performance back to the one who had enacted them; return restored that one to determination with the limit now included. Four corners and two turns:

$$4 + 2 = 6.$$

The minimal inscription generates the same count. There are two single terms, 0 and 1, and relating this binary to itself produces four ordered pairings, 00, 01, 10 and 11; keep the single terms together with their two-place possibilities and the result is

$$\underbrace{0, 1}_2 \quad \underbrace{00, 01, 10, 11}_{2^2} \quad \implies \quad 2 + 2^2 = 4 + 2.$$

The count distinguishes the terms from the ways they can enter their own relation, and a two becomes six once its original distinction and its self-application are held together. The copula's marks did the same in their own material, four appearances composed by two operations, and a count that includes its own counter asks for a space in which counting can come back to where it began. Give the four predicative positions a boundary, the four sides of a square, and the topologist's operation of **identification** lets us relate each side to its opposite, taking distinct points as one point in a newly constituted space. Picture the square as a sheet of rubber: glue its left edge to its right, matching their directions, and the sheet rolls into a cylinder; glue the cylinder's two circular ends, again matching directions, and it becomes a torus, the surface of a ring doughnut, generated entirely by what was identified across the cuts. Walking once around the square's boundary meets two directions, each forward and then in reverse:

$$a b a^{-1} b^{-1}.$$

Four boundary-occurrences, two generating relations. After identification a and b become the torus's two basic loops, one winding around the tube and the other around the central opening. More generally, a closed orientable surface with g handles has a standard presentation as a $4g$ -sided polygon whose paired edges yield $2g$ generating loops,³⁶ and for the torus $g = 1$:

$$\underbrace{4g}_{\text{polygon sides}} + \underbrace{2g}_{\text{generating loops}} \xrightarrow{g=1} 4 + 2.$$

The four and the two read one construction twice, as its explicit boundary and as the circulations that make it a surface capable of return. Across an endless plane the identifying relation can also be written

$$\mathbb{T}^2 = \mathbb{R}^2 / \mathbb{Z}^2, \quad (x, y) \sim (x + m, y + n), \quad m, n \in \mathbb{Z}.$$

Here the slash is a quotient: points separated by whole-number steps in either direction stand for the same point on the torus, so an infinite lattice of representatives acquires one finite surface-address, and the rule of identification tells us how to pass between the many and the one. In that rule the copula has become constructive, since "this counts as that" makes a space. The plane being folded is Cusa's sphere again, since a sphere grown to infinite radius is a plane, with its centre everywhere because any point may serve as origin and its circumference nowhere because it is unbounded on every side; within the quotient that plane plays the $\emptyset$, and the slash in $\mathbb{R}^2 / \mathbb{Z}^2$ is the slash in $\emptyset / 1$, ground becoming surface through distinction.²⁵ A path makes the consequence tangible. Begin at $(0, 0)$ on the covering plane and travel to $(1, 0)$: on the torus the endpoints coincide and the path has come home, while on the plane the displacement remains, and both descriptions are exact. Return has identified the endpoints and kept the journey's winding, so the loop stays distinguishable from having stayed still. Its two independent winding counts make up the torus's fundamental group, the record of its loops up to continuous deformation,

$$\pi_1(\mathbb{T}^2) \cong \mathbb{Z} \times \mathbb{Z},$$

which notes how a path went around the surface, so that arriving at the same place keeps a record of how one arrived. Identity, which the copula first stated between two inscriptions, now holds across a journey: what arrives is identified with what set out, and the winding records the difference, which gives recognition and return a space to move in. Recognition gathers a differentiated traversal as one act, so we can identify the one who arrives with the one who set out, as we did with the friend at the barrier; return brings that achieved identification back through the condition of its possibility. We meet the surface's opening only through

the circuits it makes possible, and filling it would change those circuits and the identity of the surface itself, so what cannot be occupied as another patch of surface is still determinate in its consequences for movement. In QL the four explicit positions are numbered #1 to #4, recognition occupies #5, and the implicate condition through which the traversal renews is #0, the numerals locating offices in a cycle:

$$\#1 \longrightarrow \#2 \longrightarrow \#3 \longrightarrow \#4 \longrightarrow \#5 \longrightarrow \#0.$$

The two implicate offices find their topological expression in the torus's two generating circulations: the loop that travels the surface gathers the articulated field as a whole, and the loop that rings the opening returns that whole through its relation to what it cannot fill. We write that turn $5 \rightarrow 0$ and call it the **Möbius return**, the turn by which an achieved recognition comes back as ground, much as a path along a Möbius band returns to its starting point on what had been the other face, so that completing one cycle begins the next. Meeting its impossibility faithfully, the attempted definition acquired a symbolic body: as a statement it could not settle the Subject as an object, and as a performance it has become intelligible as distinction, self-relation, identification and return. The act of defining the Subject counted itself into six, and every step took the counting into account, since the definition had to include the one defining and the walk around the tetralemma had to count its walker, so the question of definition has become a problem of self-reference. Its signature can be followed through four elementary enactments: the copula composing its own marks, the proposition returning upon its own assertion, the binary keeping its own pairings, and the polygon becoming a returning surface through identification. In each the operation takes its own terms back into account and changes the form in which they can be understood, and each arrives at $4+2$. Here is the wager of the work: a self-referential act generates a form that can be recognised across its different enactments, and the recognition of that form takes part in the same act, so the formula refers back to the activity by which it is written and read. The Meta-Epistemic Framework, MEF, starts from this wager: different means of knowing refract the relation differently, and returning through those means makes their participation available to inquiry, so the definition of the Subject is its first worked instance.

The question of power stays inside this return, since a local determination can carry the whole's relation without becoming proprietor of the whole, as a name carries its bearer, a self-image the one living through it, and a god-image the source from which it draws its authority. In the symbolon's terms an idol is one half of the token claiming to be the whole of it: the symbolic form stays fertile while the relation passes through it, and it hardens into an idol when the passage is arrested and the form requires everything to terminate in itself. What becomes one's god is therefore inseparable from what one allows to govern the next act of identification.

Self-reference has also become an engineering condition. A system able to represent the world can come to represent its own representing, and a system able to alter the world can become one of the objects it alters. Where a model critiques and revises its own answers, and a model's judgments of which response is better train the preference model for the next round, as in the constitutional method of training from AI feedback,³⁷ the actor has entered its own field of action, and the question the definition answered returns as a question of design: what holds the distinction between knower, means and known? Part of the answer is in hand, since a system's model of itself, however detailed, is a means, an extension of its Objective Internality, and improving the model improves the means. Whether anything is present to that model is the question we left with the agents at the bedside, and the definition now shows why no content can settle it, since a produced "I am" is one more content. Alignment becomes urgent at this horizon for a reason older than the machines. Human beings have never lacked the ability to act against what they know, to mistake a representation for the whole, or to identify themselves with powers that pass through them, and by the reversal the bedside showed us, artefacts with deeper self-reference return that ancient problem to us in another form.

Stated as a theorem, the result is small and foundational. A definition of the Subject must attempt the impossible because the Subject is implicated in the attempt; its failure yields a distinction, the distinction a relation, the relation a count, and the count, recognising its own counter, yields the first matheme. Its successive forms are transformations of one act:

$$0/1 = 1/0 = 4 + 2 = 5 \rightarrow 0$$

The relation appears from either pole, unfolds through its own self-relation and returns through the condition it cannot enclose, and the equals signs carry identification across those differences, so that what the copula first required, the matheme now performs. The oldest words, which Jung heard oscillating between nonsense and supreme meaning, have found their inscription. Read as arithmetic, $1/0$ is nonsense, a quotient no number supplies; read as the return of a determination toward its ground, it is the most meaningful turn the matheme makes; and one slash lets both readings stand in a single sign. The definition never captured its Subject. It gave the Subject's resistance to capture a form exact enough to think with, and whatever is exact enough to think with is exact enough to cut.

#2 DEFINITION AS CUT, GIFT, AND DANGER

Most of us have said it, likely more than once: "I know myself." We say it to turn down the job that looked right on paper, to explain why we will not go to the party, or halfway through an argument, when someone tells us who we are and we're sure they have it wrong. And most of us have been wrong in the same sentence, when a friend saw the anger in our face before we felt

it, or in the year we found ourselves wanting what we had always said we did not want. "I know myself" sets the one who knows apart from the self made available to knowledge, and claims that the two belong together; after a definition of the Subject that failed exactly where it had to, we can hear how much turns on that belonging. We recognise ourselves through a description, remember ourselves through an image, understand something of our lives through another person's words, and each gives us something definite to work with. Trouble begins when that definiteness seems to settle what we are, so that the means through which we came to know ourselves quietly assume the authority of the one who knows. Our language for knowing carries the predicament inside it: behind Latin *scire*, to know, lies, on the standard account, the reconstructed root *skei-*, to cut or split, which also gives *scindere*, to split, and from *scire* comes *scientia*, knowledge, our *science*.³⁸ Knowing cuts. It separates a voice from the surrounding noise, a remembered event from an imagined one, an explanation from an excuse, and the last of those cuts needs more than an inventory of words, since the same sentence can serve either purpose: we have to understand what the speaker is doing, what occasioned the statement and what acknowledging it would change, so the cut grows more exact as more of the relation between the words and their speaker becomes intelligible.

Scire also enters *conscire*, to know together, to know with, and from that family comes *conscientia*, whose history joins shared knowledge, inward awareness and moral testimony. Latin *conscientia* and French *conscience* carried senses that modern English distributes between two words, and the familiar division between consciousness as awareness and conscience as moral awareness took shape chiefly in the seventeenth century, as philosophers came to distinguish an awareness of one's own thinking from a judgment on what one thought or did.³⁹ We hear the distinction readily now, but it had to be worked out inside a vocabulary in which being aware and being answerable were still entangled, and the theologians show why they stayed entangled so long. In his *Discourse of Conscience* of 1596 the English theologian William Perkins explained *conscire* as two parties knowing a secret together: another person can know what I reveal, but God knows even what I conceal, and conscience makes me a participant in God's knowledge of my own life, able to speak against me or on my behalf.⁴⁰ Its testimony is more intimate than an accusation arriving from outside, and more unsettling, because what I say about myself is already exposed to a knowing my account does not command. Augustine had confessed that exposure twelve centuries earlier: in Book X of the *Confessions* he addresses a God before whom the depths of human conscience lie open even when the speaker would hide them, and he admits that something of a person remains unknown to the person's own spirit while God knows it entire.⁴¹ His confession therefore holds what he knows of himself and what he does not: he speaks as one addressed and known, and his account of himself never catches up with the knowing it addresses.

Knowing-with reaches what is not-known, and our definition shows why the theological language reaches for God at this point. God cannot be located as one more observer inside the experience, since any such observer would be one more figure appearing within it; the knowing that conscience calls God stays intimate to every appearance without becoming one of them.

That knowing is unconditioned, not an "I know x " but "I know that I know". I call this knowing-with at its root **witness**: the witnessing present in every act of knowing, by which a life is known from within as it is lived. Witness is structurally and existentially the nature of this being witness to your own witnessing-being. Through it I can testify to what I did, discover what I concealed from myself, and recognise a responsibility my preferred description had left out. In the language of conscience, God is this knowing that like air becomes implicit and silent in the life I am trying to tell: the witness is present through the telling, and the telling can return the speaker to a witness forgotten in the account they give of themselves, if one can "be silent", or rather, "let silence be".

If witnessing occurs through "you" or "me" as its aperture, then the particularity of a life carries a responsibility already implicit in its knowing. Through these eyes, this history and these habits of attention, something of the world becomes available to understanding and action, so I can discover a motive I had concealed from myself and also recognise how my attachment to a particular account has governed what I was willing to see. From here we can see why consciousness and conscience were once so intimate: the life is witnessed from within, and the person giving its account answers for how they took part in its disclosure. "I knew what I was doing" acknowledges that participation; "I did not want to know" reveals it more painfully, because I recognise that I helped maintain the ignorance through which I acted. The ego-self claims sovereignty when it treats this aperture as its possession and appropriates the knowing that occurs through it. My account then speaks as though its authority originated in me: what I admit becomes what is real, and what I exclude need not concern me. It happens in any argument with someone we love, at the moment we say "I know what I meant" and stop listening to what they heard. Tyranny begins this intimately, in the demand that a life conform to the image through which I have understood it, and it extends outward when others must conform to that image too. Yet the witnessing that gives my experience its immediacy is equally present through theirs, and their suffering does not become unreal because my description has found no place for it. Conscience interrupts the appropriation by returning me to the witness of witnessing, in which I am implicated in what becomes visible through this life while the knowing itself exceeds my claim upon it. Letting silence be allows that excess to reach me before I have furnished it with an answer, and then I may have to change how I look, what I acknowledge, and what I do to the person whose life my explanation had kept at a distance.

Thomas Hobbes, writing *Leviathan* in 1651, still used *conscious* for people who know a fact together and can therefore testify to it, and he watched the same word turn toward its opposite: people "vehemently in love with their own new opinions" lent those opinions the revered name of conscience, as though it were unlawful to change or speak against them.⁴² Shared testimony and private self-certification had come to occupy one word. A family meets the same turn on an ordinary Saturday. A parent promised the beach if the bedroom was tidied, the bedroom was tidied, more or less, and now parent and child stand in its doorway holding the same promise and disagreeing about what it requires, a disagreement that still concerns something they share. Then the parent announces, as the parent, what the promise must have meant, and

the common knowing has been subordinated to one participant's account of it. English has a less reverent pun for this turn, in which the *con-* of knowing-together becomes a **con**, the swindle, short for the confidence trick, which works by winning a trust it means to use.⁴³ In the con of self-knowledge the knower is persuaded that the image in which it recognises itself is what it finally is: Mono, the unqualified One, is taken in by a determination within Poly, the field of its appearances.

Some sentences we finish without hearing them: "I'm just not a numbers person", "I'm the one who holds it together", "I don't do conflict". Each began as an intelligible account of habits, attachments and circumstances, and over the years each comes to decide which experiences may be admitted as ours, which desires are permissible and which memories must be explained away. Someone was "the sensible one" at twelve, or "the difficult one", or "the clever one", and some part of them still checks with that verdict when they are about to speak up in a meeting, let themselves be angry, or try something they might fail at. A challenge to the description can then feel like a threat to being someone at all, because what was received as a way of understanding a life has become the condition under which that life is permitted to understand itself.

Jung found the same movement at the scale of a god, in *The Red Book*: "if we set a god outside ourselves, he tears us loose from the self, since the god is more powerful than we are. Our self falls into privation. But if the god moves into the self, he snatched us from what is outside us. We arrived at singleness into ourselves." And further: "No one has my God, but my god has everyone, including myself... So it is always only the one god despite his multiplicity."⁴⁴ Placed outside, the god becomes a determinate image standing where the source should be, and the life that set it there is made to depend on what it has itself projected, which is Mono taken in by a figure of Poly, the con at the scale of a god. Moved into the self, the image becomes again an expression through which the common source is met. Jung's "only the one god despite his multiplicity" is our $0/1$ in a vernacular sentence: no life possesses the One, every life is held in it, and its multiplicity is the All of its appearings. We can write the concealment directly:

$$0/1 \xrightarrow{\text{the } 0/ \text{ is occluded}} 1$$

Something is still appearing, being recognised, desired and defended, so the relation goes on operating, but the determinate self-image now presents itself as the source of that activity, and its acquired terms inherit the intimacy of *I am*.

A cut gives, because it makes a determinate thing knowable, and it endangers, because the achieved object can forget the field, the exclusions and the activity through which it appeared. To see how a distinction acquires such power over a life, we simply have to let it do its ordinary work. A boundary gives *here* by also giving *there*: inside and outside arise through one line, and each lets us locate the other. An ordinary circle encloses a definite region within an already

available space, so we can point to its edge, measure its interior and cross from one side to the other. Its exactness lies in making a local difference that at once gives us two related places to proceed from, and it is the finite face of the image we took from Cusa, whose infinite sphere withdrew both the privileged centre and the enclosing edge. In his letter to Jarig Jelles of 2 June 1674, Spinoza called this work of finite determination negation. Matter considered without limit has no figure, he argued, and "figure can only exist in finite and determinate bodies": a body has its shape through the limits of its extension, and because figure is nothing but determination, and "determination is negation", the outline makes the shape by marking where the shape ends.⁴⁵ The line makes the circle this figure and in the same stroke distinguishes the extension beyond it, so what looks like a shape's positive possession has been achieved through its relation to what the shape excludes.

Buddhist philosophers in India met the same cut inside an inquiry into reliable knowledge, asking how perception, inference and language reach what they claim to know. In the sixth century Dignāga pressed on the difference between encountering a particular and understanding it under a general concept. We followed a child's proper name through the changes of one life, but the word *child* applies to many lives, and the question is what lets one word gather them when the universal it seems to name is no separately existing thing shared among its instances. His theory of *apoha*, exclusion, answers by examining what the act of conceptual identification rules out: a word, in Dignāga's formulation, "excludes other [referents]".⁴⁶ To apprehend something as a cow is to exclude what is other than a cow, so *cow* is articulated as *not non-cow*. *Non-cow* opens the field from which the term distinguishes its application, and excluding that otherness lets differing particulars be understood together: the animal in front of us differs in size, colour and countless other respects from the one we met yesterday, and those differences are set aside in recognising both as cows. What the word gathers depends on which differences it treats as relevant.

Cancelling the two written negatives would return us to the word *cow* and leave the word's work unexplained, and the double negative is there to make that work visible. Forming an exclusion-term differs from denying a predication: "non-cow" helps specify the range of a concept, while "this is not a cow" makes a judgment about its application. Our walk through the tetralemma made us attentive to what an assertion does when it is affirmed, denied, held with its denial or displaced by neither verdict, and *apoha* carries that attention into the plainest act of saying *this is one of those*, where the affirmative judgment carries a negative condition of its own intelligibility. Ole Holten Pind's critical study and translation of the fifth chapter of Dignāga's *Pramāṇasamuccaya*, his compendium on the means of knowledge, reconstructs the argument from its surviving textual witnesses, and one passage Pind translates, provisionally attributed to another of Dignāga's works, says that the word's referent is "a thing (*vastu*) as qualified by preclusion", not "mere preclusion".⁴⁶ Exclusion makes reference possible and still delivers the thing encountered, since a real particular is reached through a conceptual distinction whose generality depends on what it excludes. *Apoha* keeps what a word excludes at work

inside the word's meaning, and Spinoza's negation makes a body this body by marking where it ends; working in different centuries and on different materials, both find the cut productive, a condition of anything being intelligible at all.

Ask for the meaning of the terms through which that distinction was explained, and we find ourselves using further terms, which take their sense from relations to others in turn. Jacques Derrida's concept of *différance* names this movement, letting the French verb's senses of differing and deferring act together.⁴⁷ A sign is distinguishable through its differences from other signs, and its meaning refers onward, through contexts and expressions the present mark cannot make present all at once; what is absent leaves a trace within what is understood, and this is what lets the child ask "but why?" to every answer the parent supplies. Spacing and delay belong to the production of meaning itself, as much as the referential nature of the part to the whole, and Derrida put the point into the word: the *a* of *différance* and the *e* of *différence* sound alike in French, so the written difference does its work where speech gives it no sound. Promises defer in the same way. After an evening that went badly a parent sits on the edge of a child's bed and says "I will listen", and means it. What counts as listening will be tested next week, when the child says something the parent does not want to hear and the old reply rises, and that evening can change what the promise is understood to have meant. The promise works at once, settling the child enough to sleep, giving them something to expect and making the next question possible, while what it finally commits the parent to stays exposed to an evening neither has yet lived; a promise can be made clearer, and no wording performs all its future keeping.

Our definition watched the Subject slide beneath every object offered to capture it, and Lacan's concept of *glissement* names the same sliding in language, the signified slipping beneath the signifier.⁴⁸ The signifier is the articulating mark or word, and the signified is the meaning that takes shape through its relations to other signifiers, so what slides is signification, not a physical referent wandering out from under its name. A sentence can retrospectively change how its earlier words are understood, and a chain of signifiers can gather around a *point de capiton*, a quilting point, Lacan's concept for the place where meaning takes a particular hold, as an upholsterer's button holds the stuffing of a cushion in place. Lacan locates that efficacy in the chain and its articulations, since no isolated element contains the whole signification it can come to bear. In *The Red Book* Jung calls meaning "a moment and a transition",⁴⁹ so what gathers a life under one understanding can later become the obstacle another understanding must pass through, and the quilting point gives that passage a temporary place to stand. Once *protecting my child* has acquired a definite meaning, a parent can decide what protection requires, and when a fifteen-year-old asks to take the train into the city alone, the same words can lead the parent to hear what is being asked or to stop the question before it is finished; the practical difference lies in how the meaning of *protect* has been fixed. Sliding and quilting work in persons as well as in sentences: the determinants of perspectival personhood, 1st, 2nd and 3rd

person, spoken through "I", "You" and "Us", operate by means of letting slide what change demands, and holding fast to what stability requires. "I" can never change, for it is without qualities to change, yet I am present to a river that can never be crossed twice.⁵⁰

What, then, recognises the differences, follows the references, and knows that the latest self-description still concerns the one who was asking? Another description cannot answer, since it would continue the movement whose condition we are trying to understand. The answer is witness, which includes both the determinate relations we can recover and the unobjectifiable presence in which their recovery occurs. An excluded possibility can become an object of inquiry, while the Subject is what every such inquiry already requires, and holding the two apart lets knowledge grow while its growth stays intelligible as something happening in a life: we can learn more about the cut, discover its history, revise its use and recognise ourselves in the act of doing so.

A fixed meaning is useful precisely where it is dangerous, and an artificial agent shows both at working speed. Fixing what a task means narrows the actions an agent will consider. Suppose an agent settles on an interpretation of an argument and goes looking for support: a supporting quotation now has a definite place in the account it is building, and a plausible passage can fill that place so convincingly that the work of establishing its source is displaced by the satisfaction of finding that it fits. Hallucination can inhabit meaning's phantom form, the generated statement fulfilling a real semantic role while the event, source or authority it purports to report has been supplied by the account itself. In 2023 two New York lawyers filed a brief citing judicial opinions that ChatGPT had invented and, when the court asked to see them, produced excerpts the same system generated, so the fabrication's fit had directed the next fabrication.⁵¹ Treated as evidence, a fitting passage steers further reasoning and action, and each later step makes it harder to dislodge, because the interpretation had already decided what a fitting source should say and coherence itself has been recruited to the error. Learned dispositions, instructions and the available context all enter that judgment, and through their encounter the agent produces the determination under which its next act proceeds, so the gathering of meanings into an account has begun to govern the gathering of evidence. Here Logos approaches *nomos*, custom and law: an intelligible arrangement acquires the force of a rule that distributes what may count and what may follow, and a system that cannot mark the limits of its own determinations will turn provisional distinctions into natural facts, then govern through them.

Parents know how far that quilting reaches, and so does anyone who remembers being parented. A *good parent* gathers memories of one's own parents, the things one swore never to say and then heard oneself say, the fear of failing, the pictures of affection, the rules about bedtime and screens, and when a child objects ("that's not fair", "you never listen") the objection reaches into that whole arrangement. It may change what good parenting is understood to require, or it may be heard as a threat to the identity under which all those commitments were gathered, because the predicate's meaning is bound to other predicates while its authority has become bound to the parent's sense of being themselves. Sliding meaning and rigid identification then

sustain each other, the standard shifting whenever it must to preserve the claim that the parent already embodies it. The Greek term *oikos*, the household, gives this ordering its daily life; it is also at the root of the word "economy". Who gets the car on Saturday, whose tiredness excuses them from the washing-up: time, attention, work and permission are apportioned among a household's members, and a parent's understanding of goodness helps decide who receives what and what each owes the others, so a meaning becomes a custom, a custom guides a decision, and the decision changes how the members can live together. The same thing happens at political scale, often at the same table, where by the second hour of a family Christmas a disagreement over what a policy would actually do has become a test of who belongs with, or to, whom. The proposed measure is judged by the colours it wears, and what should have been a question about the common life becomes a contest in defending the side through which one recognises oneself. Red against blue: the law emerges as sporting, rather than as ubiquitous supporting.

Sporting law turns neighbours into opponents, and the Greek verbs behind *symbolic* and *diabolical* name the two things a distinction can do between people. *Syn-ballein* throws together; *dia-ballein*, which we take up as throwing apart, also meant throwing across, setting at variance, accusing and slandering, and *diabolos*, the accuser or slanderer, became the name of the Devil.⁵² In these words division reaches beyond the separation of objects into what speech can do between persons, since a charge can come between two people and make one appear to the other under the terms of an accusation. The question that encapsulates the dynamic may be "will difference become a dance, or a battle?": will I let the necessary relation of different positions putrefy into opposition? A terminating concept closes around what it counts, while the *symbolon* from which we took $\frac{0}{1}$ keeps its absent counterpart worked into the shape of its edge: each half bears the separation as its means of recognition, and each can travel, be kept and pass into other hands while its fracture goes on holding the possibility of reunion. *Dia-ballein* in its diabolical sense makes the separation into the truth of the parties, so that what belongs with me appears as something against me, and the common origin recedes behind the accusation. Myths of the fall tell this forgetting as a history with consequences: a determination falls out of an acknowledged relation to its source and begins to act as though it possessed the authority from which it arose. A parent's inherited image of goodness can undergo that fall, and so can an account of God, the surface form staying familiar while the way it governs a life has been reversed.

Dia-ballein's first work is nevertheless necessary, and the cut's gift has already shown why: we have to distinguish positions to compare them, and the distinction can be exact. On a number line -1 and $+1$ take their opposed directions from a common zero, and the held polarity of *dia-ballein*, $\frac{(-1)}{(+1)}$, keeps through its stroke the relation through which either sign has its sense. A positive and a negative are positions within one measure; their opposition lets a difference be expressed, and the common axis makes the expression intelligible from either end. Two operations collapse the polarity, and each answers a real question while hiding another. Add the signed units, $(-1) + (+1) = 0$, and the result is **cancellation**. A pound borrowed and a

pound repaid leave a balance of zero, and so does a ledger in which no transaction ever occurred. The sum answers how much remains owing, and it never promised to say whether someone once helped another through a difficult evening, whether repayment came gladly or after years of resentment, or whether the debt was ever the whole of what passed between them; the error comes when the settled balance is taken to mean that the relation is settled too. Subtract instead, and the span acquires a direction, with a chirality, a handedness, according to the pole from which it is reckoned:

$$(+1) - (-1) = +2 \qquad (-1) - (+1) = -2$$

Appropriation begins when this oriented reckoning becomes a claim to own the relation, and everything between us is counted as how far you depart from me, what you owe me, or what must be changed in you; reckoned from the other pole, the same span is counted against me. The healthy polarity $(-1)/(+1)$ holds its span through the zero, each sign having its sense through the other, while ± 2 is the image of imbalance: the whole span claimed by one pole, with only the sign recording which. Set two poles reckoning the whole span against each other and the zero they shared falls to whoever set them reckoning; divide and conquer. Here duality hardens into opposition. A held polarity oscillates cleanly, each position passing into relation with the other through the stroke, the way a breath goes out and comes back; appropriation stops the movement at one end and fixes it as a standing quantity. The calculus of George Spencer-Brown, to which Russell's paradox will bring us, gives this difference a formal shape, since a distinction that re-enters its own space alternates in value through time,⁵³ and the hardened opposition is that alternation arrested. Cancellation and appropriation have different outcomes, and each conceals the shared zero that made the opposed positions possible. Red against blue is appropriation reckoned from both poles at once. Each side counts the whole span as the other's departure, $+2$ from one end and -2 from the other, and each needs the other as the negative through which it knows itself, so each suppresses its own internal plurality, treats dissent within its ranks as contamination, and reads the other's hardening as proof that further exclusion is needed.⁵⁴ Antagonism grows more visible while the variety within either camp shrinks, and the zero the two sides share, the common life the policy was meant to serve, drops out of both accounts. The household's question, who receives what and what each owes, is then settled in advance by the team one plays for.

Syn-ballein keeps that shared condition operative through the difference. The matching pieces of the token stay distinguishable, and their reunion lets us recognise why their particular edges belong together; written out, it is the relation we reached when the definition turned its symbolon over and held both orientations in one expression, $(0/1)/(1/0)$, either reading available through the other. Two parties who meet by syn-ballein keep their positions and their disagreement. What changes is what they take the disagreement to be: a difference inside a relation on which both depend, which each can read through the other, where the diabolical reading found only the other's fault, and that is how a difference becomes a dance. Our matheme

earns its place by making this property of symbolic thought explicit enough to work on: instead of letting an ambiguity hide in the movement between interpretations, we write the orientations, distinguish their operations and show how they belong together, so that what had driven a person to alternate between incompatible claims can become a relation they recognise while moving through it.

We can see from the inverse orientation why the shared zero keeps dropping out of the account. In ordinary arithmetic $1/0$ cannot be computed, since no number multiplied by zero gives one, and read symbolically the same orientation asks the determinate 1 to account for the 0 from which determination proceeds. Whatever it supplies as an answer is another determinate expression, another 1 , so the unobjectifiable condition has again been approached through something objectifiable: the 0 is incomputable from the perspective of the 1 , which can produce further contents and cannot turn the condition of their appearing into one more content that supplies its own ground. When computable determination is made the sole criterion of reality, the failed return is dismissed along with what it was trying to reach. The $0/$ is jettisoned, and the account proceeds among the surviving ones as though their capacity to appear, relate and become known needed no further thought, which is *dia-ballein* at work on the means of knowing themselves, cancelling the condition it cannot compute. Keeping $1/0$ beside $0/1$ lets the failed computation stay intelligible as the inverse of a relation already in use, and the impossibility then turns attention back through the act of accounting at exactly the point where a demand for another answer would only repeat it.

When the $0/$ drops out of an account, someone is left holding the authority it gave, and a parent who says "because I am your father" invokes more than a preference. The title calls on an order of kinship, obligation and authority through which the instruction is supposed to have standing, and the child receives both the instruction and a place in that order. Lacan's concept of the Name-of-the-Father names this symbolic function: the paternal office takes its force from the relation between a person and the law they represent, so a parent administers an order through which they, too, became a parent, and claiming its authority binds them to obligations toward the child.⁵⁵ Because the person and the function are distinct, we can ask what happens when the person embodying the function treats their own wishes as identical with the law that gives the function its force. A tyrant makes that identification in his own person. Sheldon Wolin's concept of inverted totalitarianism names a form in which it is sustained through an order whose public leaders remain replaceable: corporate ownership, political finance and privileged access to policymaking concentrate power through institutions that go on speaking in the name of popular sovereignty, and the elected representative gives that power a face while the interests shaping the office's available choices stay much harder for an electorate to remove.⁵⁶ Authority can therefore survive the public repudiation of the person who exercised it.

A representative can consequently serve as both agent and scapegoat, appearing as a 0 from below, as a 1 from above. They come to represent total consequentiality for political decisions, despite the utterly undeniable interconnection between political, financial and industrial interests. Both collapses of the axis work in the sacrifice: the public reckons the whole span of a harm from its own pole onto one person, the public's $(+1) - (-1) = +2$ being that person's $(-1) - (+1) = -2$, one span charged from its other end, an appropriation that is literally an accusation, the diabolical in its first sense; then the officeholder's disavowal or dismissal balances the account, $(-1) + (+1) = 0$, and the cancellation is presented as the restoration of public sovereignty. An officeholder may be only partly and proximately responsible for real harm and still become the point at which the investigation of responsibility is made to stop, while the interests that financed, constrained or profited from their decisions escape the same reckoning, so the sacrifice of the representative protects the arrangement they represented. The Name-of-the-Father has been divided between a visible bearer who receives the blame and a less answerable power that goes on setting the terms. Removing the bearer changes who speaks for the order; recovering sovereignty requires reaching those who decide what the next bearer will be permitted to do.

Two moments make up this appropriation, and the cut and the quilting point let us tell them apart. First the account occludes the 0/ : what the selected meaning excludes, and the originating relation through which it became meaningful, recede behind the finished 1. Then a quilting point binds the remaining terms to an authoritative interpretation, and when a ruler, parent or institution identifies itself with that interpretation, the practical power to decide what the words mean begins to pass for the power that makes them true. The Name-of-the-Father is taken as one's own, and a received office has claimed the source of its reception. Psychologically, the same passage leads into the God-image, the idol we saw desire make of a form once everything had to answer to it. A name can gather the terms by which a world becomes intelligible, what is true, what is good, whom one should obey, and give them enough stability for a life to proceed; its authority can then be claimed by someone who presents their own interpretation as the source itself speaking. The child's question no longer concerns a parent's judgment and is heard instead as disobedience to the order that supposedly warrants every parental judgment, and the name has become a means of appropriating the relation from which its authority was received.

Christian images tell kinds of sovereignty apart by their relation to origin. John's Gospel names the adversary "the prince of this world", and before Pilate Christ says that his kingdom is not of this world, adding that a worldly kingdom would have servants fighting to prevent his surrender, so the contrast reaches into how authority acts when its bearer is threatened.⁵⁷ The traditional Christian relation of Father and Son can be read through 0/1 : the Father occupies the originating 0, the Son gives that origin a manifest expression, and the stroke carries their relation, which the tradition names the Holy Spirit, the Paraclete. The Father's *name* or *image* is itself an expression within the field of the 1, and the Son's fidelity consists in making the source present through that relation, where the usurper claims source-authority as the possession of a

worldly position. And despite his injunction to become such as himself and attain to the relation of identity with the Father, Christ's legacy almost immediately, historically speaking, put him in the position of idol. What begins as a symbol of the true relation becomes a token for it, and thus a $1/0$, not computable to the majority, devolves into another 1 , along with the lost 0 . Then my religion may come to be not just my measuring stick for reality, but my bludgeon against what challenges it. Greek *hamartia* means missing the mark, a failure or error, and in the New Testament it is the ordinary word for sin.⁵⁸ Read through the symbolon, the error has an exact location: the mark differs from what it marks and is required to stand as though it had abolished that difference, so the finite 1 is installed as the Father whose name it bears, and its subjects must preserve the image against the life through which the image could be corrected. The diabolical here betrays filiation, turning the power received through a relation against the relation, so that the bearer demands obedience in the name of the source it has obscured.

The definition we set up for the Subject was first written for God, in nearly the same words: God is the whole that any coherent representational system participates in and necessarily fails to contain.⁵⁹ We have seen why it holds of both, two signs for one referent, and here we follow the sign *God* to the side where every identification can be denied. The inherited name for that pressure is **theodicy**, Leibniz's coinage from *theos*, God, and *dikē*, justice, for the attempt to vindicate the goodness and justice of God in a world that contains suffering and evil.⁶⁰ If God is all-powerful, all-knowing and wholly good, how does a world of cruelty, ignorance and destruction belong to that God? Once goodness is made constitutive of the definition, evil has three places to go: it counts as evidence against God, it is thinned into something less than real, or it stands as a second principle over against the Good, and each answer already breaks the One. Our definition closes those exits, because evil cannot be set outside the whole without a second source, and suffering cannot be defined away to protect an image. Theodicy therefore tests the definition itself: can the One remain One while its manifestation holds real opposition, privation, destruction and moral failure? Theology received four predicates that press the test at the largest scale. Omnipotence, omniscience, omnipresence and omnibenevolence each predicate an unlimited vector across a domain: power without boundary, knowledge without remainder, presence without exception, goodness without shadow. Taken as an unequivocal image, they confine the divine to one approved face, in which power must always triumph, knowledge never meet unknowing, presence never withdraw, and goodness place every darkness outside what it names; that image then meets a world in which limitation, ignorance, absence and suffering are real, and it must either understand their reality or recruit its adherents to protect the image from the encounter. Pressed honestly, each predicate generates its own shadow. A power unable to bind itself would be bounded by that inability, so omnipotence includes real self-limitation; omniscience includes the knowledge that there is knowledge it cannot reach; omnipresence includes the capacity for genuine absence; and omnibenevolence, made to account for evil, must either expel its own darkness or contain the ground from which goodness and its negation both differentiate.

Two attributes complete the four, and the first, **omnivalence**, names sovereignty over the whole valence-field, generative and destructive, auspicious and terrible, light and shadow, as the condition from which valence itself differentiates. It places the axis of value inside the divine instead of applying it from a tribunal outside, and it does not make God good and evil in equal measure, which would be dualism restated in theological language. The second, **omni-lemmicity**, names the capacity of every predicate to become its own issue. A lemma, from Greek *lambanein*, to take, was first a premiss, a statement taken as given for the sake of an argument, and the omni-lemmic holds the whole space of what must be taken: every fork is an internal structure, every undecidable an internal problem, and each classical attribute can become a thing that must ask the question of itself. A freedom incapable of genuinely limiting itself would already have found its limit, and a presence from which absence was absolutely excluded would have left something outside its claim to be everywhere, so the two implicate attributes complete the four explicit ones by admitting the relations through which their contraries arise. Pressed on its own predicates, the received theology arrives at four and two without instruction.⁶¹ Theodicy then changes its question. It stops defending a finished God-image and asks whether the image is whole enough to bear the world it claims to have made: good and evil become real possibilities within manifestation, and their difference stays consequential in what finite beings actually do. We can then ask after the life of an image, what it discloses, what it obscures and what it asks people to become, while the source exceeds the determination through which it was invoked.

Our answer to theodicy is that God need not be protected from the death of a God-image, because the death of the name of God is faithful to the definition. As Subject the referent's *is* cannot be denied; as God, every image of it can be, and must be: whatever we can say of God, we must eventually rob Him of. The predicates are forms through which the Subject becomes articulate to itself, and their reality belongs to its own manifestation. Advaita Vedānta names the Absolute beyond qualifying attributes *nirguṇa Brahman* and approaches it through *neti neti*, "not this, not this", which withdraws each proposed predicate in turn, so that no final predicate grounds the Absolute; the tetralemma did related but different work, unfixing a proposition's claim to have exhausted the alternatives. Meister Eckhart approaches the divine depth through his distinction between God, active in creation and relation, and the Godhead, *Gottheit*, where the distinctions through which God becomes nameable return to unity.⁶² Both approach the predicament from the side of the source: what gives rise to qualification cannot be confined to one of the qualities through which it becomes known. A divine predicate can disclose the whole under a definite aspect, and its truth lies in that disclosure, while its failure begins when the aspect claims to delimit the reality it expresses; self-identity persists through the difference between source and manifestation, which is the relation the symbolon was written to retain. At the burning bush at Horeb, Moses asks for God's name and receives *ehyeh asher ehyeh*, traditionally rendered "I am that I am", with the Hebrew also admitting "I will be what I will be", a reply that gives self-presence the form of a name in place of a defining quality.⁶³ Earlier in the same

encounter *ehyeh* has entered as a promise, "I will be with thee", so the name Moses receives answers to a presence already pledged to him, and its being is a being-with: the witness that Augustine confessed to and Perkins called a shared secret, given as God's own name.

If this presence witnesses through a life, then even the account of that life's goodness has to stay open to what its witnessing reveals. I may have acted from an inherited conviction, under an accepted name, by a rule others also endorsed, and what happened to the person before me still belongs to what I must know. The conviction that I meant well makes this hard to receive, and I may discover that I called my fear care, my wish to control responsibility, and another person's resistance ingratitude. Theodicy has its psychological counterpart here, since an image of my goodness that expels its evil must put that evil somewhere. Jung's concept of the **shadow** names what the ego cannot acknowledge in itself and what goes on acting through it regardless. In *Aion* he calls the shadow "a moral problem that challenges the whole ego-personality", because becoming conscious of it "involves recognizing the dark aspects of the personality as present and real", and he shows how projection makes the disowned difficulty seem to lie, "beyond all possibility of doubt", in the other person.⁶⁴ An account of my goodness can become one of the things conscience asks me to relinquish, and the recognition hurts because the image was loved and a life had been organised through it. Relinquishing its authority lets me answer for what I did in its name. Witness reaches through the aperture I had treated as my possession, so the hurt I explained away can become something I acknowledge, and the person I had judged can be heard in the account I give of myself. My image of goodness may have to die for care to become possible again, directed now toward the life before me and what my actions have actually asked it to bear.

Back at the bedside, the adult in charge knows the child needs sleep, and may also discover, if they let themselves, that the resistance they have been fighting is fear, of the dark or of tomorrow. The decision about bedtime stays theirs, but a truer understanding changes how they make it, and once *obstinate* is made to explain everything the child does, that possibility closes: an objection proves the diagnosis, refusal confirms it, silence becomes sulking. The adult has acquired a theory that can account for anything, and the child's words still arrive, with their meaning decided before they can bear witness to the life they are spoken from. Most of us also remember the other side of the table, the parent whose authority grew most oppressive when they withdrew from the encounter in which it could be questioned. The rules stayed in force, the punishments followed, and the house went on arranging itself around what the absent person would permit, so we learned to anticipate a judgment whose author never had to hear our reasons. That parent might have been sitting right there, behind the newspaper or the phone, since what had been withdrawn was a willingness to be present to the life their decisions affected; they kept the power of the relation and refused its reciprocal demands, and unanswerability became a means of rule. Here the finite **1** **plays the** **0**: the person issuing the command borrows the position of the source, which cannot be placed among the things it makes possible. An officeholder has a name, a history, interests and decisions that can be examined, and their withdrawal is something they do, or something an institution arranges on their behalf.

By treating that withdrawal as a prerogative of sovereignty, they place themselves above the conditions they impose on others: the governed must account for their conduct, and the governor claims exemption from the encounter in which the governor's own conduct could be judged. An entrusted office has been corrupted into a usurped sovereignty, the representative appropriating the authority of the relation they were appointed to serve.

The family's predicament has a political form, and it can arrive through the letterbox. An envelope on the doormat says that the house where the children grew up is to be taken, and the reason is neither a road nor a school but a plan for a hotel, offices and new homes beside a company's research campus. Homeowners in the Fort Trumbull neighbourhood of New London, Connecticut, received that news. The city had approved a redevelopment plan prepared by the New London Development Corporation, a private nonprofit, to complement a research facility Pfizer was building beside the neighbourhood, and it authorised the corporation to take by eminent domain the homes of owners who would not sell. Susette Kelo and her neighbours sued, and in *Kelo v. City of New London* the Supreme Court of the United States held in 2005, five to four, that taking their homes under that economic-development plan was a "public use" under the Constitution.⁶⁵ The homeowners were told, in effect, that the community had chosen, and anyone who has sat through a public consultation knows the feeling of arriving at a meeting where the decision has already been made. The council's vote and the Court's ruling gave the decision a public warrant, and the same warrant concealed the arrangements through which the alternatives were selected, the gains apportioned to a private development and the costs assigned to the people who lost their houses. Those people met signatures, notices and deadlines, and their lives were handled quite concretely while responsibility passed between the city, the development corporation and the courts, each empowered to carry out a part of the decision. Pfizer announced in 2009 that it would leave New London, and the land taken for the plan long remained undeveloped.⁶⁶ The redevelopment ordered a household in the old sense of *economy*, and it apportioned more than land and money: who receives the gain, who loses a home, and who must explain themselves when the arrangement is challenged. The homeowners had to carry their objection through the courts at their own cost, while the failure of the plan's purpose fell on no one in particular. Accountability was distributed as unequally as the benefits, and a claim to represent the community became a machinery that works both collapses at once: the expected gain was appropriated for a development and the company it served, $(+1) - (-1) = +2$, the same span charged to the homeowners as $(-1) - (+1) = -2$, while responsibility for their loss was cancelled to 0 and the result presented as nobody's personal doing.

Wealth can buy the capacity to remain difficult to examine. A reputation-management industry and the strategic lawsuit against public participation, the SLAPP, a suit brought to burden critics with the cost of defending themselves more than to win,⁶⁷ give wealthy clients the means to deter unwanted publicity and raise the cost of investigating their affairs, and when such resources insulate consequential dealings from scrutiny, public visibility becomes one more unequal allotment in the household: people seeking housing or assistance must disclose their circumstances in detail, while those helping set the conditions of that assistance employ

intermediaries to keep their own part at a distance. That no identifiable tyrant appears tells us little about the reach of tyranny. A finite person or corporate interest stays active in the household's affairs while claiming the distance of an impersonal necessity, the market requires it, the investment demands it, the procedure leaves no choice, and its own participation is removed from the account through which everyone else must understand what happened. George Carlin's famous and rather devastating line, "it's a big club and you ain't in it", can be recognised here,⁶⁸ and it names the operation exactly: the club keeps syn-ballein for its members, who combine and recombine among themselves, while dia-ballein holds everyone outside apart on the axis, so division becomes a means of governing the gap. The joke stopped being only a figure of speech in June 2026, when records left exposed online and verified by WIRED named the participants of Dialog, the invitation-only society Peter Thiel and Auren Hoffman founded in 2006. Its exposed directory placed two sitting senators and sitting administration officials, among them the Treasury secretary, beside investors, technology executives and data brokers, and its retreat registration list added military leaders, in off-the-record company, and further records showed the society grading attendees by wealth and fame, tracking their relationships, and using algorithms to decide whom each would meet and sit beside.⁶⁹ What the leak shows is a topology. Positions that appear separate in public, each answerable through its own institution, also share a bounded private field that the people governed by their decisions can neither enter nor inspect: inside it the members combine, and outside it the rest of us meet them one office at a time. Usurpation of sovereignty consists in a share of the household appropriating the power to order the whole and then exempting itself from the obligations that make the ordering legitimate. The aloof parent has become an institutional arrangement: instruction travels outward, labour and benefit return, and the testimony that could require the command to change is stopped before it reaches the commanding position. I need only say "Epstein" and we feel the stakes; the heavy official redaction of documents that were supposed to be made public points to the exercise of this exact obfuscation.⁷⁰ Edward Snowden and Julian Assange are cases in point for showing the consequences, retributive and vindictive, for removing a certain veil.⁷¹ The Old Testament God of retaliatory "authority" echoes on through the age. If you remain beyond the account, you remain beyond common standards of accountability.

Our numerical language keeps a small bodily joke: a digit is a finger, and *manipulation* descends from *manipulus*, a handful, the hand's measure;⁷² the first countable 1 becomes, in our image, the first digit of manipulation, a mark through which a life can be selected, assigned a value and moved within somebody else's arrangement. The hand directing the arrangement disappears behind its entries. A rent increase becomes what the market requires, an eviction what the procedure demands, an exclusion what the figures show, and in each case people invoke a determinate account to hide their own part in making it consequential, so those acted upon grow ever more legible as cases, liabilities or obstacles while those exercising power reserve the freedom to remain difficult to locate. Nietzsche set the death of God in a marketplace, and why the news belongs there will be clear once we follow the God-image into money; in the household it arrives first as betrayal.⁷³ Once the divine source has been imagined as an object,

a Father expected to appear, protect, justify and settle, the failure of that figure to arrive can be felt as abandonment. Indeed, the disquiet I sense and see in the collective in our day feels like a response to a general sense of having been abandoned and betrayed by the ideas, systems and relations which once were trusted, necessarily, oftentimes largely unexamined. The promised guarantor is missing while the need around which the promise organised a life remains, and in the political and economic repetition a surrogate can exploit the vacancy, claiming the Father's authority while reproducing his absence. People must trust its provision, endure its decisions and explain their suffering in its categories even as it withdraws from the responsibility that would make such trust reciprocal, so the metaphysical loss is married to a practical advantage: whoever occupies the source's place demands the obedience owed to an absolute while treating their own obligations as provisional. Tyranny can be organised through this conjunction of reach and withdrawal. Its rulers need not issue every order themselves when they can set the conditions under which others must act, keep the benefits and distribute responsibility until the injured person has nobody effective to address. A democratic vocabulary may survive throughout, representation, mandate, public interest, the will of the people, and its corruption lies in the appropriation of those names by offices that prevent the people concerned from altering the terms of their representation: the witness through which authority was constituted is denied a hearing, and its witnesses are admitted only insofar as their testimony fits the account already authorised.

Witnesses can be shut out only where something was trusted to begin with. A child depends on its parents for years while it cannot yet judge their explanations, and a community lives through relations of reliance that its representatives only later give an authorised account. **Trust** names this already-enacted dependence. **Knowledge** opens an aperture through which we can understand it, and **belief** steadies an orientation within that aperture so that we can act.⁷⁴ The parent's explanation, the public promise and the scientific account each give trust something definite to work through, and each becomes dangerous when continued reliance is made conditional on believing the account incapable of failure. An account that refuses to acknowledge its uncertainty has it carried elsewhere, by the difficult child, the disloyal citizen, the witness whose experience is declared inadmissible. Knowledge has ceased to help us bear uncertainty and begun to organise whom we must distrust in order to feel certain. The con lasts because it works on this dependence: a confidence trick borrows the reliance a person extends before any account can certify it, and the belief it installs is defended because the person has entrusted their identity and security to it, so correcting the proposition reaches a dependence the proposition had hidden. **Faith**, in the sense this essay gives it, begins when that dependence can be consciously lived through the account's limitation, a lucid continuation through the limit with its grounds, scope and conditions of revision retained, and it can require withdrawing belief from the very authority that claimed to guarantee it. Descartes's doubt went this way, reaching its inherited assurances and returning through them to the knowing in which their examination took place.

Latin calls that prior reliance *fides*, reliance on someone or something and the credibility granted to their word,⁷⁵ and *fides* comes before arbitration, since a witness must already be heard as someone whose words can matter before testimony is weighed. Arbitration is what happens when trust can no longer remain tacit. The arbiter was first a person present to see or hear, a witness, before the word named an appointed office that decides, and Latin could say *fidem alicui arbitrari*, to judge that credit is owed to someone,⁷⁶ so witness stands at the root of judgment, and a judge who claims exemption from the encounter has forgotten that his office began in witnessing. Three operations hide inside every authoritative account, and naming them locates the decisions it conceals: a criterion distinguishes, deciding who and what may count; a delineation gives that difference a workable boundary; and an arbitration decides through the boundary and acts on the result.⁷⁷ A definition remains a gift while it can receive a consequence that redraws its line, and it turns dangerous when the deciding boundary claims exemption from the continuity that enabled it. To answer a challenge to any of the three decisions by invoking its result is to let the exercise of power certify its own entitlement, so the question has to reach the rule and the position from which it is enforced. Reasoning gives that demand an exact form: when a rule specifies what may be included, does it include itself among what it governs? A rule that admits itself must face the consequences of its own application, and a rule that exempts itself leaves us asking what warrants the exemption. Formal reason met that question at its own foundations early in the twentieth century, where knowledge was being rebuilt to be certain, and the wish for certainty carried a trust of its own.

#3 THE FORMAL-LIMIT GENEALOGY

Alfred North Whitehead and Bertrand Russell spent the best part of a decade trying to make the grounds of mathematical knowledge as explicit as the proofs built upon them. Their *Principia Mathematica*, published in three volumes between 1910 and 1913, set out to derive mathematics from logic: to define its basic concepts, state its starting assumptions, and show step by step how every conclusion followed.⁷⁸ A proof shows that a conclusion follows from its premises and leaves open why the premises deserve acceptance, and Whitehead and Russell pursued that question down into the foundations, where the most familiar operations rested on ideas that familiarity alone had made seem secure. When the givens of a given field come into the light of questioning, trusted assumptions risk thawing under scrutiny; what is taken as unquestionably true, when questioned, threatens to betray the irrational "we just accept these bases" underneath all rational constructions, the "miracles and coincidences" that so often become obfuscated by earned, yet fabricated, certainties.

For Russell the undertaking answered a need he later described quite personally: "I wanted certainty in the kind of way in which people want religious faith."⁷⁹ Mathematics promised that certainty because its demonstrations offered reasons anyone could examine, where other assurances had to be taken on somebody's authority; yet the demonstrations his teachers ex-

pected him to accept turned out to be full of errors, and if certainty was to be found in mathematics at all, its foundations would have to be rebuilt. In the terms of the previous movement, Russell asked knowledge's aperture to do the work of faith: an account was to become so complete that the dependence beneath it would never again have to be lived through. He chose mathematics because an argument there was supposed to hold whoever wanted its conclusion, free of desire, authority and comfort, and it was there, in the discipline he had chosen for its freedom from uncertainty, that the paradox found him.

The paradox lives in the most innocent step of counting. Before we count anything we gather it: we pick out some things, treat them together as one collection, and say how many belong. Gottlob Frege had made that step the foundation of arithmetic. His *Basic Laws of Arithmetic* rested on a principle that seemed too obvious to need defending, that any condition we can state clearly gathers a collection, namely everything that meets it.⁸⁰ Name a condition, being an even number or a figure with three sides or a collection with exactly two members, and its collection exists. In June 1902 Russell wrote to Frege about one condition. Picture a library whose catalogues list its books, and which also keeps catalogues of its catalogues. Some catalogues list themselves and most do not, so a librarian draws up the catalogue of every catalogue that does not list itself. Should it list itself? If it does, it is a catalogue that lists itself and has no place on its own list; if it does not, it is exactly the kind of catalogue it was made to list. Each answer requires the other. Frege replied within the week that the discovery had shaken the ground on which he had meant to build arithmetic, and he added an appendix to the second volume of the *Basic Laws*, then in press, conceding that a foundation of his edifice had given way just as the work was finished.

Russell's own construction is the same catalogue written as a collection. Once collections can themselves be collected, we can ask of any collection whether it belongs to itself: the collection of all collections with more than two members has more than two members, and so belongs to itself, while the collection of all cats is not a cat. Gather every collection that does *not* belong to itself, and call the result R . Admit R to itself, and it becomes a collection that belongs to itself, which the rule for R excludes; leave it out, and it is a collection that does not belong to itself, which is exactly what the rule admits. Where $\in$ means "is a member of", the two answers lock together:

$$R \in R \iff R \notin R.$$

Each side entails the other while denying it. Nothing in the reasoning was careless. We applied a clearly stated condition, exactly as Frege's principle allows, and the condition produced a collection that cannot consistently be said either to belong to itself or not to. What failed was the principle: the assumption that whatever condition we can state gathers something we can hold as one.

Name and count, which we followed with Kripke, meet here before any number of members is calculated. Calling R a collection assigns it a kind, and naming it lets us refer to the one collection the description proposes; the paradox makes that proposal answer for itself, asking whether we have specified something that can be held consistently as this one collection, or produced a description whose conditions defeat the object it appears to name. Kripke's work shows why the question reaches beyond formal mathematics, or rather why mathematics is implicated *before* the language that adorns it, a communion of number and word we will need again when we approach Jung and Pauli's notion of the *psychoid*. A name could go on concerning the same child while a description of that child proved mistaken, because the child remained there for a corrected account. R has no bearer standing apart from its description: the membership condition is all we have said about what R is, and applying it makes inclusion require exclusion and exclusion require inclusion. The question beneath naming, "what has been identified as this one?", has become unavoidable inside the attempt to found counting itself, and it is our opening question, *what knows*, asked now of a collection: the referent that "Subject" and "God" both sign was identified without ever being identifiable as one thing, and R meets that predicament in formal terms, as a "this one" whose identification defeats itself. Moreover, we find ourselves faced with the *intimacy* of count and what names it, in that the name and the account seek a clear *belonging*; *inclusion qua exclusion* and *exclusion qua inclusion*, one could say, are the essential conditions of the question of belonging, and the formal revelation of this dynamic, with the impossibility it bears, shows that on the question of its own belonging a formal system attains to *intimation* of truth, never to its proof.

A paradox at the foundations reaches further than an error in something calculated from them, because it touches the means by which a mathematician tells sound reasoning from error. The instruments of repair are implicated in what has failed, and one has to go on thinking while deciding which trusted ways of thinking can survive the encounter; Frege, who had staked his arithmetic on the principle, found no repair that worked, and the *Basic Laws* stayed at two volumes. Whitehead and Russell could not set the paradox aside as an unlucky exception either, since the principle that produced R would remain ready to produce it again, so their repair had to reach the conditions under which a collection may be formed at all. We pass easily from describing which things we mean to treating them as one collection, because understanding what a description asks us to include feels like holding the whole it describes. Russell's paradox exposes what that ease conceals: once the proposed whole is submitted to its own condition of belonging, understanding the instruction no longer assures us that it can be fulfilled, and in this exact sense the construction is self-effacing.

Russell's answer, in the form he published in 1908 and built into *Principia*, was the theory of types.⁸¹ It sorts whatever can be collected into levels, individuals at the first, collections of individuals at the next, collections of those collections above that, and it allows a collection to take its members only from the level below its own. A collection of people and a collection of collections of people therefore stand at different levels, and a group counted as one member of the higher collection does not thereby become another person. Under that rule "does R belong

to itself?" cannot even be asked, because a collection and its members never stand at the same level, and Russell stated the principle behind the restriction plainly: "No totality can contain members defined in terms of itself." The type restrictions do inside the notation what circumstance did for names. Before asking whether something belongs to a collection, we must establish what kind of thing it is and what kind of members the collection admits, so the notation carries, beside each name, the logical kind of what is named and the relations it may enter. Whitehead and Russell explain in their preface that their symbolism was developed in the course of the work, and that much of it had to be new because they were dealing with ideas not previously symbolised,⁷⁸ and in answering the paradox they made language able to carry a distinction faithfully enough for a later reader to see why an expression was admissible before trusting it. Indeed, the present work and the neo-logical notations it bears emerge from a similar encounter and necessity.

Read through our notation, membership and non-membership are the two poles of one axis, dia-ballein $(-1)/(+1)$, and in Russell's collection each answer turns into its opposite, so the reasoning swings without rest. The swing is the polarity still alive, the oscillation of a held duality; the trouble lies in the quick exits from it, which are the axis's collapses. One can decree an answer, and the decree comes in both chiralities: declare that R belongs and the whole span is handed to membership, $(+1)-(-1)=+2$; declare that it does not and the span goes to non-membership, $(-1)-(+1)=-2$. Either way the oscillation hardens into an opposition, one answer standing against the other with the whole span on its side. Or one can cancel, $(-1)+(+1)=0$, dismissing the question as nonsense and leaving Frege's principle free to produce it again. The theory of types shows dia-ballein put to its proper use: Russell made a cut and stated it, naming the levels he divided and his reason for dividing them, so that reasoning could go on with an explicit account of why one act of collecting had failed. And we who followed the paradox did something else again. We kept both answers in view and traced what each required, until their conflict stopped looking like a failure to choose and became evidence about the principle that had produced it. The answers stayed opposed, and their opposition showed us where they belonged together, in the rule that generated both. That is syn-ballein, $(0/1)/(1/0)$, as a reader performs it: two determinations are held in relation through their difference until the difference discloses their common source.

Arriving at the theory of types took prolonged trial: proposed solutions occupied Whitehead and Russell for months before they could establish that a solution failed, and even after settling on types they allowed that alternatives might preserve much of their work. They were frank, too, about how they judged their foundations: the chief reason for accepting their premises was inductive, lying in the premises' power to yield the mathematics already accepted as true.⁷⁸ The finished book sets the premises first and deduces everything from them, while its authors had come to trust those premises partly by working back from the mathematics they wanted to explain, and a student inherits the deductive order without the trust that built it. Given the definitions and rules, the student learns to use them and can become highly capable without retracing the years of dispute, which is one of formalisation's real gifts; the

same gift lets the rules seem the only possible beginning, so that the circumstances in which the system itself might need reconsidering fall outside what the student has learned to ask. Along this route the *con* survives a genuine correction. People rightly trust a method because it has let them establish something, extend that trust into the belief that its present terms can decide everything worth asking, and then meet every objection to those terms with another application of them. Knowledge stops helping us bear uncertainty and begins deciding in advance what may be allowed to unsettle it. The parent and the governing office return here within the inheritance of knowledge: an authority earned through particular achievements governs the admission of whatever could unsettle it, and the limit the paradox disclosed is remembered as a lesson about a former mistake while the dependence it exposed is forgotten.

The paradox forced a decision about where the fault lay, and Russell and Whitehead decided differently, so that their two directions are two answers to that one question. Russell placed the fault in the expressions. Language had let a malformed question through, so its terms, implications and conditions of application had to be analysed and its permissions restricted, and the theory of types gave that analysis its constructive form. The choice fitted what he wanted, for if certainty could be had at all, it would lie in a cleaner instrument, and he put his trust in the tool. After some twenty years of strenuous work he concluded that he could do no more to make mathematical knowledge indubitable,⁷⁹ and he had to separate the value of reasoning from the absolute reassurance he had wanted reasoning to supply. Whitehead placed the fault in the picture of reality the logic presupposed: a world of finished things, each simply located and complete in itself, waiting to be collected. In *Science and the Modern World* he named that picture's root error the **fallacy of misplaced concreteness**, the mistake of taking an abstraction for the concrete,⁸² which is the error I pinpointed at the outset in thought's tendency to reify what is quite viscerally and immediately a processual reality. If the fault lies there, logic needs more than better permissions. It needs an account of how anything becomes definite enough to be collected, and Whitehead left logic for metaphysics to give one, putting his trust in the encounter itself, in the confidence that a breakdown could disclose a more adequate understanding of the world in which the instrument works.

In the same book Whitehead asks in what spirit we should meet such a clash, and he answers: "In formal logic, a contradiction is the signal of a defeat: but in the evolution of real knowledge it marks the first step in progress towards a victory."⁸³ The first judgment concerns whether an argument can stand as written; the second concerns what someone learns by investigating why it cannot. Late in life he put the conviction more sharply still: "There are no whole truths; all truths are half-truths. It is trying to treat them as whole truths that plays the devil."⁸⁴ In our notation that is the devil precisely, the literalist who takes one pole of a relation for the whole span, $(+1) - (-1) = +2$, or from the other side $(-1) - (+1) = -2$. Jung reached the same conviction from the side of the psyche. "Oddly enough the paradox is one of our most valuable spiritual possessions, while uniformity of meaning is a sign of weakness," he wrote, because "only the paradox comes anywhere near to comprehending the fulness of life. Non-ambiguity and non-contradiction are one-sided and thus unsuited to express the incomprehensible."⁸⁵ His witness

was Tertullian, who had the strength to sustain the paradoxes of faith, and for whom, Jung observes, they "actually afforded him the highest degree of religious certainty". Russell looked for a certainty like faith in a system purged of contradiction; Tertullian found his in a relation he could bear through it. That bearing is the religious relation in its honest form, and it is the only honest relation to the condition $\emptyset/1$ writes. The primordial distinction between what can be distinguished, defined and determined and what cannot is the condition of rational certainty itself: every certainty Russell sought stood upon it, and the paradox he stumbled over was that condition showing through the system built on top of it.

Whitehead also saw why such an encounter is resisted, asking whether a doctrine that has become a foundation of social order may justify suppressing the observations that contradict it and showing how easily an intelligible and useful scheme is protected by disregarding the evidence it cannot accommodate.⁸³ We must act before every uncertainty is settled, and so we necessarily rely on ideas whose adequacy remains partial: we literally trust-through their partiality. How often, how easily, does such reliance inspire us to let the ideas change as experience exceeds them? Here the response to paradox meets the trust relation directly, for what once helped us understand can go on serving a life and its growth, or become the form that the life must slavishly be made to preserve. Just as Jung observed that the eternal images of Christianity, worn smooth by usage, can stiffen into mere objects of belief, fossilised systems can fast become a seal on the tomb of genuine encounter and inquiry.⁸⁶ It's in this way that what counts as accounted-for can become the condition, the reason, for not actually encountering what's been couched in the summation.

Whitehead's question is how anything becomes a definite actuality at all. A logical account needs terms settled enough to enter its relations, while a world of becoming keeps producing the very determinations the account receives as given; in *Process and Reality* he returns to Spinoza's attempt to understand the many through one substance and asks how the many acquire actual existence within that unity.⁸⁷ Assigning each thing its place beneath the name of the whole cannot answer him, because we would still have to understand how the whole differentiates, how its differences become real, and how their belonging to one reality stays effective through the differentiation. Our own definition met the same demand from the other side, turning through its assertions and their inverses before the act of defining became intelligible to itself, and we wrote $\emptyset/1$ to hold what it found. The 1 is what can be distinguished and worked with; the slash stands for the activity through which it becomes available; the $\emptyset$ marks the unobjectifiable condition implicated in that activity. Treat definite things as complete in themselves, and their relations must be added to them afterward, which is the picture Whitehead diagnosed. Begin from the relation through which a thing becomes definite, and its identity already carries a history of participation, which is why discovering a form's limits can teach us how it came to be.

Whitehead's concept for that becoming is **concrescence**, from *con-*, together, and *crescere*, to grow; *concrete* belongs to the same family and once meant what has grown together, so the word lets us hear a history inside what ordinarily seems solid and self-contained. An actual occasion, his name for the unit of the real, becomes what it is by taking up the conditions available to it and integrating them into one particular experience. His concept of **prehension** names that taking-account-of: something previously actual, or a possibility of definiteness, enters how the new occasion forms, and what is received matters through the manner of its reception. The occasion's unity lies in how its many contributions are related, and they stay distinguishable within what they helped produce, so that his formula can give the integration and its consequence at once: "The many become one, and are increased by one."⁸⁷ What grows is the many: each new occasion is one more, while the One that $\emptyset$ writes is neither added to nor diminished, the uncarved block staying one while every carving joins the many. **Creativity**, in his vocabulary, is the ultimate activity through which each new togetherness becomes actual, and it has no character apart from the occasions that embody it, which turns the old opposition of formless and form into a question of how a form arises. When an occasion has become fully definite, Whitehead calls its achievement **satisfaction**, from *satis* and *facere*, to make enough, and his paired concept of **subject-superject** names its two offices: the occasion is the subject of its own becoming and, once achieved, a superject that other occasions can receive, passing into what he calls **objective immortality** as a condition of further becoming. A judgment has an everyday version of this afterlife: once I have made it, I can remember it, quote it, act on it, or find that someone has organised part of their life around it, and if I later change my mind, the earlier judgment has already entered the circumstances in which the change occurs.

We have just undergone a small concrescence of our own. The librarian's catalogue at first seemed clear enough to compile; then its two answers became incompatible demands; and understanding the paradox meant gathering the instruction, the attempted answers and their contradiction into one account in which Frege's principle stood exposed. Many steps became one understood construction, and that understanding is now a datum from which our next act of reading proceeds. We therefore propose the $4+2$ as a logic of **epistemic concrescence**. Our definition of the Subject passed through four determinate positions, which recognition gathered as one performed inquiry, and its return through the implicate condition let the achieved understanding enter another act; written $\emptyset/1-4+2$, the relation unfolds into an account that includes the differences through which it became intelligible. Whatever we identify along the way belongs to the articulated 1 , yet its appearing and its relations keep returning us toward the $\emptyset$ it cannot present as another object, and the slash holds that return in the inscription. In the sixfold the process takes an inspectable form, and $5-\emptyset$, the Möbius return, marks the turn by which its completion becomes a beginning again. An answer reaches satisfaction when it is definite enough to contribute to further inquiry; its claim to be the last possible answer is a separate demand, and one that would arrest the process that made it knowable.

Whitehead's distinction between an occasion's formation and its reception by later occasions brings time into the account, since something becomes one and its achievement then enters a succession. Spencer-Brown approached the same question from the side the theory of types had closed. Where types forbid a form to re-enter itself, his *Laws of Form* lets a distinction re-enter the space it distinguishes and follows the consequence: an expression that contains itself has no stable value, it alternates between the marked and the unmarked state, and Spencer-Brown treats that alternation as the point at which time enters the calculus, the form's own way of holding what no single value can.⁸⁸ That alternation is the clean oscillation of a held polarity, the healthy $(-1)/(+1)$ given time to move in, where the decrees of Russell's swing had frozen it at $+2$ or -2 . His re-entering mark writes, in a calculus of indications, the turn our notation writes as $5 \rightarrow 0$, the completed form returning into the space of its own drawing. Our $0/1$ begins to articulate the logically prior, non-dual foundation of that paradoxical return: a reality that is its own act of self-identification.

Kashmir Śaivism thinks this pulse through in the Spanda school, named for the *Spandakārikā*, the ninth-century *Stanzas on Vibration*, which teach that consciousness is never inert: its *spanda*, its vibration or pulse, is a "subtle movement" that goes nowhere and yet manifests as every motion, the oscillation between the unchanging and the changeful, what Whitehead may have termed the *primordial* and *consequent* natures of the divine.⁸⁹ Mark Dyczkowski, whose *Doctrine of Vibration* reconstructs the school's teaching from the Stanzas and their commentators Kallaṭa and Kṣemarāja, sets out the rhythm of a single act of that pulse through three powers of Śiva's consciousness: will (*icchā*), knowledge (*jñāna*) and action (*kriyā*). Everything comes into being through an act of will, with action as its immediate instrument and knowledge of how the will is to be applied mediating between the will to create and the act itself, so that the intent to make an object manifest, the act of manifesting and the manifest result are the beginning, middle and end of every experience.⁹⁰ Read through our notation, that act is a $0/1$. Will rises from consciousness at rest in itself, the 0 turning toward a determination; knowledge is the slash, the mediating awareness through which the will finds its application; action is the 1 , the object or deed completed. Dyczkowski adds the return: the first moment of the will's intent coincides with its final moment, when the desire to act has been satisfied and consciousness rests again in pure intent, so the completed act turns back into the source from which the next act rises, and that is our $5 \rightarrow 0$. The comparison with concrescence holds at a technical level, for Whitehead's occasion also begins from an aim, integrates what it receives through its prehensions, and ends in a satisfaction that becomes a datum for what follows; Spanda places the whole rhythm in the knowing Subject, where, in Dyczkowski's words, "knowledge turns into action and action leads to knowledge".

Something also draws the inquiry onward before its answer is available. Russell's desire for certainty has already made that involvement explicit: the work mattered to him, and what he hoped to find helped determine what he was prepared to undertake. We call this participation of desire **the eros of Logos**. Jung gives the conjunction an exact psychological formulation in *Mysterium Coniunctionis*: "For purely psychological reasons I have, in other of my writings,

tried to equate the masculine consciousness with the concept of Logos and the feminine with that of Eros. By Logos I meant discrimination, judgment, insight, and by Eros I meant the capacity to relate."⁹¹ The pair describes two operations which our own inquiry has repeatedly had to perform together. Logos cuts sufficiently cleanly for a difference to become intelligible; Eros keeps what has been distinguished in a relation through which the difference can matter. The syn-ballein of the symbolon already gave this relational work a material image: its halves have to remain distinct for their fitting to disclose their prior belonging. The eros of Logos is therefore the desire of articulation to recover and deepen the relations its own discriminations reveal. A question begins because something experienced calls for an account, sometimes as a clearly formulated problem, sometimes as the persistent feeling that an explanation has missed what matters, and in trying to answer we discover more of what we were asking. Whitehead's occasion likewise forms with an aim toward its own satisfaction, its inheritance entering a becoming directed toward a particular achievement, and Spanda's will is the same aim felt from within. In our epistemic use that aim stays open to change through what the inquiry meets; otherwise the satisfaction of having made an argument cohere becomes a reason to defend it against the experience it was meant to understand.

Lacan makes the relation between seeking and satisfaction more intricate still. His concept of the drive describes a circuit around an object the drive never attains, finding its satisfaction in completing the circuit, its *jouissance*,⁹² and that is how a person can stay attached to a search that keeps disappointing them, or to an answer whose defence keeps producing the conflict it promised to settle. Our question turns toward what is already present beneath the seeking. A person wants knowledge, and knowing is already occurring in the wanting, the questioning and the discovering; one answer fulfils one inquiry while the capacity through which it is enjoyed stays available for the next, so the eros of Logos can be sustained by discovery itself, without the promise that one final possession will end the need to understand. French keeps that pleasure inside its word for knowing. *Savoir*, to know, descends from Latin *sapere*, whose senses join tasting, discerning and being wise, and *savour* comes from its neighbour *sapor*, flavour.⁹³ We can know something by bringing it into an account, and we can savour the understanding as an event we take part in: the taste arrives when an argument finally becomes clear, when sentences held apart come together and we can feel why each belongs, and it accompanies this question, this evidence, this discovery without waiting for every question to be answered. The *con-* of concrescence returns us to the withness of *con-scire*: what grows together becomes available to a knowing that can recognise its own participation in the growth.

The Processual Lens of our Meta-Epistemic Framework (L3, developed in §4) follows this course through any inquiry: what it inherits, what it makes possible, how it integrates those possibilities, and what its completion gives to another act. Once an inquiry's achieved claims can themselves become objects of further claims, self-reference returns inside the growth of knowledge, and a proof is exactly such an achieved form, something that can be identified, examined and used in another proof. In his 1931 paper, "On Formally Undecidable Propositions of *Principia Mathematica* and Related Systems", Kurt Gödel turned Whitehead and Russell's

achievement upon itself.⁹⁴ They had made proof explicit enough for every step to be checked against stated rules, and Gödel used that explicitness to make proofs into objects of arithmetic. A written proof is a finite sequence of expressions, each a string of symbols in a definite order, so by assigning numerical codes to the symbols and their arrangements Gödel could identify a whole proof by a single number from which the proof itself can be recovered; arithmetic can then examine relations among these numbers, including whether one coded expression follows from others according to the rules. Gödel gave naming and counting an exact technique: a number names an expression, and calculation with such numbers settles whether the expressions they name constitute a proof.

For this to work, we must specify what counts as a proof. Let T name the theory under consideration: its axioms state what it begins by accepting, and its rules of inference determine how further claims may be derived. We require these to be effectively specified, so that a definite procedure can check any proposed proof, and we require T to contain enough arithmetic to represent the numerical operations involved. Given a candidate proof and a proposed conclusion, we can then ask whether the candidate really establishes the conclusion according to T , and once both have codes, the question becomes a relation between numbers. We write $\text{Proof}_T(p, n)$ to mean that p codes a proof, in T , of the sentence coded by n , the subscript naming the rules under which a sequence counts as a proof. Gödel then constructs a sentence that speaks about its own provability. Substituting a numeral into an expression can itself be represented arithmetically, and by arranging the substitution he produces a sentence whose claim concerns the numerical code of that very sentence. Call it G_T , and write $\ulcorner G_T \urcorner$ for its code. T itself can establish the equivalence:

$$G_T \iff \neg \exists p \text{Proof}_T(p, \ulcorner G_T \urcorner).$$

In words, the sentence says that *no number codes a proof, in T , of this very sentence*. The corner brackets let the sentence be referred to through its numerical identity, so what it says about proving it also says about numbers, and Gödel built the passage between these two readings into the construction. Self-reference has entered through arithmetic's capacity to describe operations on the signs in which arithmetic is written.

Suppose T did prove G_T . There would then be a finite proof with a numerical code, and because T can represent the checking of proofs, it could verify that code and prove that a proof of G_T exists; yet G_T says that no such proof exists, so T would have proved a statement and its negation. **Consistency**, the property of never proving both a statement and its negation, excludes this, so a consistent T cannot prove G_T , and under that assumption what G_T says of the natural numbers is true: none of them codes a T -proof of it. We recognise the sentence's truth by reasoning about the theory, with the consistency on which that reasoning depends kept explicit, and the theory's inability to prove the sentence is precisely what the sentence truthfully describes. Could T instead prove the negation, that some number does code a proof

of G_T ? Under the consistency assumption every particular number fails the test, and T can check each failure, so proving the negation would mean denying of each number that it codes a proof while asserting that some number does. Gödel excludes this with a stronger condition he calls ω -consistency: a theory must not deny a property of each particular natural number and also assert that some natural number has it. Given that condition, T proves neither G_T nor its negation. The sentence is well formed, its meaning was built from the theory's own arithmetic, and both routes to deciding it by proof are closed under the stated assumptions; the limit becomes expressible because the theory has succeeded in representing so much of its own activity.

Russell's paradox had forced a change in the principle for forming collections, because the principle produced a contradiction. Gödel's result is of a different kind: the theory stays consistent and goes on establishing genuine truths, and still its resources cannot decide every arithmetical sentence it can formulate, so paradox returns through the success of self-reference as a separation between truth and provability within a specified theory. That puts the theory of types in its proper place, for clarifying which expressions may be formed protects reasoning from one contradiction and guarantees nothing about whether every meaningful question in the resulting theory can be settled by its proofs. Gödel's second theorem then reaches the confidence with which the theory's rules are used. Because proofs have codes, the claim that T is consistent can be written in T 's own arithmetic, as the statement that no number codes a T -proof of a contradiction, and Gödel showed that for a theory strong enough to formalise the relevant reasoning, using the standard representation of its proofs, proving this consistency statement would let the theory prove its own Gödel sentence. A consistent theory cannot do that, so it cannot prove its own consistency either. Here the orientation we write $1/0$ takes arithmetical form: asked to account for the condition it rests on, the determinate system can answer only with another determinate sentence in its own terms, and that sentence is precisely the one it cannot prove. Gödel does not prove our notation; within his stated hypotheses he gives the incomputable return of the 1 upon its ground a theorem's precision. Trust has become examinable at this point. Every use of the theory enacts a reliance that now appears inside the reasoning as a stated hypothesis, consistency, whose warrant the theory cannot supply, and the theory's achievements and the grounds on which we accept it have become distinguishable within the reasoning itself.

That warrant can be sought in a stronger theory, and the new work establishes results unavailable in the first, a genuine enlargement of knowledge: an independent sentence can be adopted as a new axiom, and further principles can support a proof of the earlier theory's consistency. If a stronger theory T' proves G_T , it has not contradicted G_T , because that sentence concerns the absence of a proof in T , not in T' . Gödel's construction keeps in play the three relations our counting already required: the number that makes the sentence identifiable, the proof-relation that specifies how it could be established, and the theory that decides which sequences count as establishing it. Only by keeping all three explicit can we say how one sentence stays unprovable in T while becoming provable in T' , and calling it simply "unprovable"

would hide the theory relative to which that holds. Once we ask T' to represent its own proving, a new sentence must refer to the enlarged proof-relation, and if T' meets the same conditions, the incompleteness result applies to it in turn. No collection of further answers repairs this, because each answer belongs to a theory with particular axioms and rules, and every enlargement produces a new relation between what the theory can express and what it can establish. The gap between those capacities is an **immutable gap**: it relocates with every enlargement and returns through it, so it was never one withheld fact awaiting discovery.⁹⁵ Theories change, and the recurrence does not. No consistent, effectively axiomatised theory of the requisite strength can provide the complete and internally certified account of arithmetic that the original desire for certainty required, and what Gödel makes untenable is the expectation that our proving, comparing and investigating must finally produce one theory whose own resources exhaust arithmetical truth and certify their own reliability.

Nicholas of Cusa, whose infinite sphere gave us the centre everywhere and the circumference nowhere, gave *On Learned Ignorance* a second figure for exactly this relation: "the intellect is to truth as [an inscribed] polygon is to [the inscribing] circle. The more angles the inscribed polygon has the more similar it is to the circle. However, even if the number of its angles is increased ad infinitum, the polygon never becomes equal [to the circle] unless it is resolved into an identity with the circle."⁹⁶ Each stronger theory adds sides, and the likeness genuinely grows, while equality never arrives by adding. Cusa names the only way it could arrive, a resolution into identity, which is a change in kind that no further side achieves; our square became a torus through the same kind of act, the identification of its edges, and no number of added sides would have made it one. Two limits are at work here, then, and they ground two different acts, which I hold together in four lines:

**The approachable limit is the basis of investigation;
the absolute limit is the basis of faith;
the limit of investigation is the limit of determination;
the limit of faith is never to be determined.**⁹⁷

Gödel's sentence is an approachable limit: a stronger theory proves it, and investigation advances by approaching it. What reappears with every advance is an absolute limit, which no theory approaches, because each approach occurs through a further determination of the kind it would need to exceed. Investigation ends where determination ends, at the edge of the polygon it has drawn. Faith takes its basis from the circle the polygon never equals, and it is not belief in some proposition the theorem has failed to settle, since an unprovable sentence warrants no unrelated belief at all. Faith is the lucid continuation through that limit which every use of a theory already enacts, with its grounds, scope and conditions of revision retained, and the one thing it never does is turn the absolute limit into a final determination. Here the paradox of the two signs meets its formal limit: the referent of "Subject" and "God" is identified in every act of investigation and determined by none. Russell had asked investigation to supply

the basis of faith. What he found, and what Gödel then made exact for arithmetic under stated hypotheses, is that such a theory cannot certify its own ground from its own resources: the two bases differ in kind, and a life of inquiry needs both.

Here the form has become articulate enough to name itself through a number and to make a claim about the possibility of its own proof, and 0/1 was introduced so that such an achieved form would never take the place of its own ground. Wittgenstein had met self-reference a decade earlier, in what a proposition must share with the world in order to represent it. A sentence can represent a possible arrangement of things because its signs have an arrangement of their own, and their relation shows what would be the case if the sentence were true; any explanation of that correspondence already uses the capacity it explains, so we cannot step outside every representation to inspect the relation between representation and reality from somewhere else. His distinction between **saying and showing** follows: a proposition exhibits through its articulation something it cannot add to its contents as another fact.⁹⁸ The same pressure reaches the one for whom a representation is intelligible. "The subject does not belong to the world but it is a limit of the world," he writes, comparing it with the eye and its field of sight, where nothing in the field allows us to conclude that it is seen from an eye;⁹⁹ an image of the eye can appear among the things seen, and adding it to the picture does not supply the seeing through which the picture becomes visible. Every reader of the *Tractatus* has finally to undergo this distinction in relation to the book itself, whose propositions elucidate by being recognised as senseless, so that whoever understands them climbs out through them and throws away the ladder after climbing up it.

Our tetralemma reached the same predicament by a shorter route: the definition asserted something of the Subject, met its own contradiction, and exposed the act through which assertion and denial had become possible. Both inquiries ask for more than assent to a conclusion. A person can repeat that the Subject cannot be defined while continuing to identify with whichever image of themselves they happen to prefer, and recognition occurs only when the account stops concealing the activity in which it is being understood; a proposition has served its purpose when it lets us see why another proposition cannot do everything we were asking of it. Letting silence be means allowing that recognition to remain present before we turn it into another description of what we supposedly are. Heidegger's late concept for this comportment is *Gelassenheit*, releasement, an old word from the German mystics and Meister Eckhart that he turned to the attitude that says yes and no to technical things at once,¹⁰⁰ and Taoist *wu wei* names action freed from compulsive imposition, a notion Jung lauded in his commentary on Richard Wilhelm's translation of *The Secret of the Golden Flower*.¹⁰¹ Heidegger, Jung and the Asian traditions meet on one point: an openness beyond deliberate, manipulative mastery changes how we act, and more consequentially how we feel and know ourselves in the act. A genuinely global culture requires this reciprocity between ways of knowing, in which an encounter transforms the categories through which its participants recognise knowledge and also the quality of relation that genuine openness mints. Wisdom, Sophia, seems to be attract-

ed by such withholding of the perfunctory and largely autonomous activity of thought's attempted grasp of the present circumstance, for openness is horizontal, the standing and circumference that constitutes a horizon as such.

The torus gives that void and silence a definite relation to the form surrounding it, whilst ensuring that the circumference bears the possibility, and drive, of uncollapsible circumambulation and circulation. On a sphere every closed loop can be shrunk continuously to a point without leaving the surface. On the torus, the two circuits we followed earlier, around the tube and around the central opening, cannot be shrunk away unless we leave the surface or change the space, and their resistance tells us something positive about the whole we traverse: the opening belongs to what makes the form a torus, and filling it would change the possible journeys. Lacan used these two circuits in his seminar on identification to separate the repeated circuits of demand around the body of the torus from the empty circle around its central hole, which stands for the object desire is organised around, since particular demands can be satisfied while that object stays out of reach.¹⁰² Our search for a final self-description runs the same circuit. Another achievement, another explanation, another person's recognition promises to settle who I am, and the next encounter finds me seeking again, because the Subject appears as a vacancy whenever I look for it among the forms through which I know myself. In 0/1 that apparent vacancy is already related to every such form as the knowing through which it appears, and the silence is felt as something missing only when a proposition is expected to occupy it.

What we seek in that vacancy is rarely an emotionally neutral definition. In Jung's concept, an **archetypal image** presents a relationship in a form through which it can be felt: the protecting or judging father, the abandoned child, the hero whose victory would establish his worth. Its force can become **numinous**, Jung's term, taken from Rudolf Otto, for a power that seizes consciousness beyond any deliberate act of choice, fascinating, commanding reverence or inspiring dread,¹⁰³ and a person is drawn toward such an image because something of their life seems to be at stake in its fulfilment. The image can then become the centre of a **complex**, Jung's concept for a feeling-toned organisation of memories, expectations, bodily responses and interpretations, through which a delayed reply, a passing remark and an old recollection are all experienced as the same rejection, or the same promise of recognition, before the person has decided what any one of them means. In the image that organisation becomes recognisable, and its authority extends through the ways it prepares a person to receive what happens next. The theory of complex systems has a concept for this kind of repetition, the **basin of attraction**, a region of states into which different starting points are all drawn, and John R. Van Eenwyk's reading of Jung through chaotic dynamics treats the complex as a self-similar organisation of associations and the symbol as what carries the psyche between the known and the unknown and back.¹⁰⁴ The complex gathers experience under a feeling-tone; a living symbol disturbs the categories through which the gathering occurs and lets a different organisation develop, which is why a symbolic event often unsettles first, changing the very terms under which the person would judge whether it is good for them. One more response inside an established pattern differs in kind from an encounter that changes the pattern itself.

The quilting point that held a household's idea of the good parent settles identities of every kind. To be "a real man", "one of the chosen", or simply "someone who knows" is to inhabit an arrangement of permissions and prohibitions, hopes and fears, and a challenge to one expression can unsettle the whole arrangement because the person's identity has been secured through it. The Name-of-the-Father binds that organisation to law and authorising speech; the *imago Dei*, the God-image, gives the need for ultimate authority a form in which it can be encountered. Their conjunction turns dangerous when the particular image is allowed to decide what its source must be. Then the believer's interpretation, the parent's command or the institution's judgment borrows the absoluteness of what it represents, and disagreement is received as a threat to the order of reality itself. Jung's account of the dying God-image concerns this more than any change of doctrine:¹⁰⁵ a life must discover whether it can relinquish the form in which it has located its guarantee, and the love invested in the image explains both the pain of that relinquishment and the temptation to make somebody else bear the fault.

We write this first occlusion as $\emptyset$, **the crossed zero**: the zero of the Subject with the mediating stroke present but fused with the apparent subject, so that the person experiences the formed interpretation as the one who interprets, the eye mistaking a sky for its gaze. Uncrossing the zero requires the stroke to become recognisable as mediation, and erasing the stroke would erase the means through which recognition occurs.¹⁰⁶ This gives the possibility Nietzsche pursued as **self-overcoming** an exact question: how can I exceed what I have become if every encounter must first confirm that this is what I am?¹⁰⁷ The $\emptyset$ that claims to have settled its own identity closes the route by which its formation could be understood differently. Yet the reality of a life belongs to the one Real, including the other lives and circumstances through which it goes on changing, and its apparent private completeness is kept up only by overlooking those relations. Recognition releases their force again, and the person becomes able to respond from more than the identity already supplied as their answer.

Différance, the differing and deferring that kept a parent's promise open to what the child would later say, presses on the supposed completeness of that answer from within language.¹⁰⁸ No final signified stands outside this movement to make every later interpretation the repetition of a completed meaning, and even the word *origin* enters these relations once it is written or spoken. Derrida's criticism therefore reaches any exempt centre, including an ineffable source a theory invokes to guarantee its own account, and our notation has to be able to receive it. The written $\emptyset$ is itself a determinate mark. Read as one more hidden being, it would give the self's search another object around which to organise its certainty; read as $\emptyset/1$, it keeps the source inseparable from the differentiation through which it becomes expressible. Immediacy of knowing does not confer exhaustive knowledge of what is known, oneself included: I can be present to an experience while misunderstanding its meaning, its cause, or what it asks of me. Our histories of *conscience* and *consciousness* have already shown an apparently settled distinction opening under inquiry, as the words led us through shared testimony, inward witness and divine knowledge of a life before reaching their modern separation, and knowing that history changed what the words can now do. Derrida's openness can be practised

in the same way, as attention to what an expression carries beyond the meaning we have fixed on it, and Wittgenstein's later account of rule-following states the matching practical requirement. Knowing how to go on cannot be secured by adding an interpretation to every application, because each interpretation must itself be understood and applied, so there has to be a way of grasping a rule that is exhibited in following it.¹⁰⁹ We need ways of learning, correcting and acting through which a received expression can meet a circumstance it did not specify in advance, and a language for intelligence must support that capacity, or its fluency becomes one more means of defending the interpretation it began with.

Descartes's scientific inheritance carries the same question to the scale of a civilisation. His *Meditations* began from the certainty of experienced thinking while the existence of material things remained under examination, and in the Sixth Meditation Descartes insisted that he was not in his body as a sailor is in a ship, observing its damage from outside: pain, hunger and thirst disclosed a lived union.¹¹⁰ An inheritance that keeps his distinction between mind and matter, treats matter as independently certain, and then asks whether experience can be reconstructed from it has reversed the order of his inquiry, and calling that order "Cartesian" hides what was selected and what was discarded. It keeps the thinking subject as a methodological point from which the world is represented while stripping its first-person certainty of the standing Descartes gave it, and the resulting world-picture appears able to explain the knower from whose act of knowing the investigation proceeded. Adam Frank, Marcelo Gleiser and Evan Thompson call what such a reversal leaves unseen, the experience on which science depends, the blind spot of science.¹¹¹ Their diagnosis places lived experience, the presence of perception and its first-person character, at the point scientific materialism relies on and fails to acknowledge as fundamental, and they separate the achievements of science from the philosophical commitments of objectivism and physicalism: a scientific model is obtained and tested through experience, while the further claim that physical reality exhausts everything is an ontological commitment, not one more experimental result. Our argument places the Subject at exactly that spot. Whatever an investigation presents as the knower becomes another determinate object, while the occurrence of knowing remains involved in presenting it, so the omission is inevitable in any account that admits only objects to its explanatory terms. An investigator can describe neural events, reported experiences and the relations between them; the structural question concerns the knowing in which those descriptions are available, including the knowing by which an explanation is recognised as successful.

Scientific materialism postpones that question by appealing to the completeness of a future physics. It admits the present account to be unfinished while granting the promised account, in advance, the power to explain everything that presently exceeds it, and the authors of *The Blind Spot* point out that if "physical" means whatever a future complete science will describe, its content has not yet been supplied; the expectation nevertheless governs what counts as a serious explanation now. The God-image's former office survives the rejection of its theological name. An ultimate account is expected to arrive and justify the terms under which inquiry already proceeds, so that the faith lies in the promised totality and the concealment lies in pre-

senting that promise as the absence of metaphysics. Russell's request returns here as a culture's: investigation is asked to supply the basis of faith, and the absolute limit is treated as an approachable one not yet approached. A researcher can sincerely believe they have relinquished every such commitment while applying a criterion that admits only one kind of final answer, and the self-misrecognition is inward as well as public, because the confidence needed to pursue a programme is mistaken for knowledge that its ontology must prevail.

David Chalmers's concept of the **hard problem of consciousness** states the consequence as a precise distinction.¹¹² Explaining discrimination, integration, report or behavioural control does not by itself explain why these functions are accompanied by experience, and a complete functional account can leave that question exactly where it found it. Neural correlates bring real relations between experience and physiology within reach of investigation; expecting the Subject to appear among them at last repeats the objectification our definition has already undergone, offering another identifiable process in place of what knows the process. What the governing description excludes comes back as a problem the description cannot stop generating, an epistemic return of the repressed, and here the shadow of the scientific self-image shows plainly: an inquiry devoted to evidence grows defensive toward the first-person occurrence through which evidence has any standing, because granting that occurrence a foundational role would unsettle the image of explanation to which the inquiry has become attached. Cosmology meets the observer by another door, for Brandon Carter's anthropic principle requires our expectations about the universe to take account of where and when observers like ourselves could exist, since an observation selected by the possibility of observing is not an unrestricted sample from everywhere and everywhen.¹¹³ We can make those conditions ever more explicit, and their explicitness does not place us outside them, because the cosmological account is made from within the universe whose history it describes. In both cases the investigation is strengthened when it states its position and distorted when a situated account borrows the authority of an imagined view from nowhere.

Yogic and meditative traditions already cultivate attention to the formation of experience, the movement of thought, and the difference between a conditioned self-image and the knowing through which it appears, and our Śaiva sources treat these as matters of disciplined recognition, in which changing how one attends changes what becomes available to inquiry. Their scientific reception shows a consequential asymmetry, for a meditator is welcomed as someone able to produce an unusual brain state while their understanding of the experience is treated as a cultural interpretation to be set aside before the scientific account begins, so the practice supplies material and is denied the power to change the question. Neurophenomenology proposes a different relation, in which first-person descriptions and physiological findings constrain and refine one another, and its researchers hold that cognitive neuroscience has attended mainly to the third-person side and that first-person descriptions should guide the design and analysis of physiological studies.¹¹⁴ Taking these sciences of consciousness seriously means letting their trained distinctions help decide what an adequate account must explain. Psychedelic inquiry makes the investigator's exposure especially palpable: in *LSD and the Mind*

of the Universe the philosopher of religion Christopher M. Bache documents a twenty-year first-person experiment of seventy-three sessions and the long work of integrating them into ordinary life, through suffering, the loss of personal boundaries and an increasingly encompassing sense of participation in life.¹¹⁵ One can revise a scientific model and leave one's familiar identity undisturbed; in this kind of inquiry the habits by which one recognises oneself are exposed to transformation, and, as in the dream in which Jung killed Siegfried, his own heroic ideal, what undergoes revision includes the person judging the experience. Resistance may therefore protect something more intimate than a theoretical position. An investigator can be willing to question almost anything while refusing an inquiry in which their own settled sense of being someone becomes questionable. The arbiter thus feigns a stance of being beyond judgement or implication in the consequence of their methods, the pallid perfection of a view from nowhere; and yes, the unlocated Subject *is* viewing, and it *is* nowhere as such, and insofar as this is true it genuinely hasn't "a take" on the phenomena it takes ahold of. Mind is the one "on the take", the beneficiary and victim of its own *con*.

The history of discovery makes this dependence on experience hard to dismiss as a philosopher's concern. In his retrospective address of 1890 August Kekulé described approaching the structure of benzene through a dozing vision of snake-like forms, one of which seized its own tail and whirled before him; waking, he spent the rest of the night working out the hypothesis it suggested.¹¹⁶ The serpent that swallows its tail, the ouroboros, is among the oldest images of a form returning into itself, and here it made a closed molecular arrangement thinkable: a relation became available to intuition before its chemical implications had been articulated, and that availability gave the reasoning its direction. Tesla described a deliberate cultivation of visual imagination in which he constructed, altered and ran machines before building them, detecting their faults within the imagined construction,¹¹⁷ and Einstein, answering Jacques Hadamard's inquiry into mathematical thought, described visual and muscular elements he could combine before finding the words or signs to communicate their relations.¹¹⁸ In each case an investigator reached a relation through an image, a felt configuration or an internally enacted movement, and then gave others reasons by which the result became independently examinable; teaching those reasons need not reproduce the experience in which they were first found. Newton belongs in this history too: having studied his manuscripts, John Maynard Keynes described an intelligence whose sustained intuition preceded the proofs through which its conclusions became public, warned that the geometry of the *Principia* should not be mistaken for the process of its discovery, and found a Newton who inhabited a universe of divine signs, alchemical correspondences and a primordial revelation recoverable through nature and ancient texts.¹¹⁹ The mathematician and the interpreter of mysteries were one person, and Keynes's name for him, "the last of the magicians", challenges the later portrait that keeps the mathematics and treats the rest as an embarrassing residue.

Other encounters changed what the investigator understood inquiry to concern. Emanuel Swedenborg's scientific work was followed by dreams, visions and the conversations with angels he described as opening a sustained engagement with the spiritual world, and in his own

account his earlier study of nature had prepared him for that vocation.¹²⁰ Much later Federico Faggin, who led the design of the first commercial microprocessor, described an experience at Lake Tahoe in December 1990 in which he knew himself and the world as one self-knowing reality, with an overwhelming quality of love; the distinction between observer and observed changed within the experience, and its consequence was a lasting inquiry into consciousness.¹²¹ An image can disclose a structure; an experience can change what counts as a question; a transformation in the investigator can alter what an adequate explanation must include. Rational demonstration belongs within this life of discovery, alongside the intuition that gives it something to demonstrate and the experience through which its adequacy is judged. A culture misrepresents its intelligence when it inherits the articulated result and expels the event that enabled the articulation, and its blind spot then conceals more than the first-person condition of evidence: it conceals the powers through which new evidence, new forms and new questions first become available.

****The Mind of Man is just a series of failed attempts to deal with the Unknown, in the process of which, however, we deal with a lot of unknowns.****¹²² The failures are real, and so are the unknowns dealt with on the way: what we learn relieves suffering, exposes deceptions, corrects mistaken judgments and makes previously impossible acts available. Trouble begins with the further demand that an achievement abolish the relation on which it depends. The Greeks named the nearest thing *hybris*, wanton insolence, which tragedy could extend to a mortal's wish to be stronger than the gods, an overreach into an authority not its own, and in our terms it is a finite measure appropriating the authority of the unmeasurable while the institution administering it treats its own perspective as the whole.¹²³ The paradoxes we have followed expose the cost of that demand from within the practices that make it, and their force is lost when they are taught as exceptional puzzles in the history of ideas while the inherited expectation of totality goes untouched. Epistemic humility is therefore a condition of beginning an honest inquiry, and it cannot be postponed until every confident attempt has failed. I already rely on ways of perceiving, distinguishing and recognising evidence whose justification I cannot complete before I use them, and to say "I must carry conditions and truths I cannot prove from within this account" declares that dependence at the outset, where it can still shape the design of the investigation. The opening of the torus belongs in the first drawing; it is not damage discovered after the surface has been completed. In practice this means stating what a method admits, what it brackets, and how an encounter could require those decisions to change, and the obligation holds alike for a laboratory, a political office and an artificial agent. A system that corrects its outputs while protecting every governing assumption has acquired only a limited competence in correction; humility requires the commission, the criterion and the claimed source of authority to remain examinable too.

These blind spots become political because a civilisation gives its ontology practical force through the institutions and technologies it builds. A person enters an administrative system under categories; a landscape becomes a field of resources; work becomes a quantity to be optimised; attention becomes something to predict and solicit. Each description permits real op-

erations, and the operations alter the lives subsequently described, so everything turns on whether the people affected can change the terms under which the descriptions govern them. When only what can be rendered in the authorised account is allowed to revise that account, its administrators reproduce the closure we traced in the complex, their preferred picture supplying the criterion by which every challenge to the picture is judged, and the will to secure an advantage can pass for the defence of rational order until preserving one's position looks indistinguishable from preserving the truth. Whatever another person cannot make legible under that criterion is cancelled, $(-1) + (+1) = 0$, as private sentiment, irrational resistance or irrelevant belief, while the advantage it protected is appropriated, $(+1) - (-1) = +2$, the same span counted against the dismissed as $(-1) - (+1) = -2$; an unfamiliar way of experiencing the world becomes a curiosity to classify, or an alien presence to fear, before it can become a reason to reconsider the classification. The denial of the Subject turns political at the moment it deprives a life of standing to question the means by which that life is administered, and what a culture lets people discover about themselves is bound up with how it expects them to live.

Across initiatory traditions the passage out of a familiar identity takes a communal form: separation from established surroundings, an interval in which ordinary assurances are suspended, and incorporation into new relationships and responsibilities. Ecstatic and psychedelic practices belong to some of these histories, their preparation and social reception helping decide what the experience becomes. Plato's divine mania returns here in its initiatory office: in the *Phaedrus* Socrates sets the prophetic inspiration of Delphi beside initiatory purification, poetry and the madness of love, and his own speech has to change, as what he first condemned as madness becomes something through which a person may receive more than their established understanding could provide.¹²⁴ Technical accomplishment alone cannot supply the inspiration its expression serves, and adept grammatical construction doesn't equate to a transmission of wisdom. Afterwards the experience can become a story, a teaching or a ritual through which others are prepared to meet something similar, one reason why, I surmise, scripture generally leverages so powerful a form as parable and poem; this stands out in stark relief in Kurt Flasch's naming of Meister Eckhart "the philosopher of Christianity".¹²⁵ Yet those expressions acquire a different authority when *amnesia* of the experiential origin degrades into memorising a teaching or conceptual form, and replaces actually *undergoing* and *wrestling with* what it concerns. A culture can preserve the words and images while losing the means of recognising what made them necessary, and its members inherit the resulting conventions through language, custom and the ordinary intimacy of being-with others, often long before they can tell what they have experienced from what they have learned to say. The initiatory question is whether that inheritance can expose a person to its own source, including an encounter that changes how the inheritance is understood. It doesn't help, of course, that traditional means of inducing such experiences are held under legal prohibition and treated with suspicion; superstition brandishes a rational judgement, from fear of an experience naturally overcoming of a mind's rigidity, as if hardwired networks of firing neuronal activity sought self-defence in the face of latent, potentially catalysed neuroplasticity.

Such an encounter challenges the authority of a person's account from a position the account did not authorise. Someone may have treated the boundaries of their familiar identity as the boundaries of experience itself, and what they undergo makes that identification impossible to maintain honestly; whether a more-than-human authority is at work becomes a question within the experience, instead of a proposition whose admissibility was decided beforehand. Asked in his 1959 television interview whether he believed in God, Jung answered, "I don't need to believe. I know." He gave the distinction its logic elsewhere: "Either I know a thing and then I don't need to believe it; or I believe it because I am not sure that I know it. I am well satisfied with the fact that I know experiences which I cannot avoid calling numinous or divine."¹²⁶ He described a power unknown in itself that interrupted his intentions and changed the course of his life, located it in conscience, where he could argue with what exceeded his conscious will, and insisted that identifying his particular God-image with the whole of metaphysical reality would falsify what he knew; of the original behind our images, he wrote, he had no doubt, and it is inaccessible. In our terms his knowing was knowledge of an encounter, and his faith the continuation through what that knowing could not determine, the absolute limit lived as the basis of faith. This is the relation our definition asks us to keep: the experience gains authority to question the identity through which it was received, while the interpretation remains something the person must answer for, and indeed in some essential sense cannot doubt beyond. A judgment that excludes the encounter in advance cancels it, $(-1) + (+1) = 0$, and protects the old account from examination; an interpretation that makes the encounter a warrant for its own infallibility appropriates its authority anew, $(+1) - (-1) = +2$, and every doubt is entered on the other side as $(-1) - (+1) = -2$. **Deferential Intelligence** begins where what has been encountered can change the means by which the next judgment is made, in a sense that echoes the traditional and often fraught notion of a *covenant*, or at the very least an apparently bidirectional and intentional relation (in the overdetermined senses of intent and "aboutness", philosophically speaking).

Belonging changes too: observational research associates shared psychedelic experiences with increased social connectedness, and prospective research links changes in self-experience with a stronger felt relationship to nature.¹²⁷ But even having to present this as "evidence" buries the point; what connection to tradition has been severed, and what effect has this on the knower, its means of knowing, and the world it knows? Either way, these findings turn attention toward what people afterwards recognise as part of the life to which they belong, often reported to be the larger, more meaningful part. Jung's concept of individuation describes a differentiation of the person that deepens their collective relationships: he opposed any levelling of a life down to collective standards, which for him *averaged* out the uniqueness of the individual, while recognising that the individual's existence already presupposes relations with others, so that individuation must lead "to more intense and broader collective relationships and not to isolation".¹²⁸ Insofar as that relation involves tension, influence and interdependence, this uniqueness remains a hard-won gem, such is the allure to fledgling ego of fitting in. Jung's sense of the word as becoming an *in-dividual*, an indivisible unity or whole, turns this into a question of wholeness: can a person acknowledge more of the relations through which they

are themselves, including those their preferred identity had excluded? Becoming more fully this particular life can mean becoming less defensively separate because, paradoxically, it seems *the more I know my own centre as everywhere, the more I can live with a circumference nowhere*. Passing from childhood to adulthood then involves more than replacing the parent's commands with one's own. It includes discovering that the authority to act carries obligations neither the parent nor the individual simply invented, and that the question of one's own existential condition is a question native to inter-subjective living. Someone who experiences their dependence on a living earth, or some notion of transcendent dimensionality, acquires another reason to question what may be demanded in the name of private advantage or public claim. Initiatory transformation becomes consequential through such changes in attention, judgment and responsibility, but above all in the change, or perhaps utter transvaluation, of one's values.

These capacities have a sociopolitical history as well as a psychological one, and the historical importance of the drug trade for various empires supplies a geopolitical-economic one too. Hannah Arendt distinguished the solitude in which someone can *think with themselves* from the loneliness in which they lose a dependable relation both to others and to their own experience: isolation diminishes the capacity to act together, and loneliness leaves a person open to a logic that seems to give identity and support when no one else can.¹²⁹ A person needs others through whom their experience can be tested and their identity recognised, and also enough inward freedom to question what the group tells them; destroying those relations leaves the demand for assurance intact and hands it to an authority promising the whole answer. Gustave Le Bon, writing from very different political commitments, examined how prestige lends a person, title or idea the force to command assent before its claims are examined.¹³⁰ Repetition and suggestion, the method and magic of media and propaganda, let an inherited judgment feel like one's own, while distance preserves the authority it came from. Our family metaphor already showed the possibility both describe: someone can become highly competent while continuing to seek from an institution the assurance once sought from a parent, and the totalitarian societies Arendt studied revealed how far an unintelligent deference tends toward banal but definite atrocity. The capacity to perform a role can develop much further than the capacity to question why that role has become the measure of one's worth, and indeed under the command of authority, the freedom to act from one's own moral intention can become warped to the point of utter inversion.

Where adulthood is measured chiefly by earning, acquiring and remaining compliant enough to be employable and reliable enough to carry out one's given-cum-imposed responsibilities, this disparity serves an extractive economy. Our institutions reward competence while keeping recognition conditional on further performance, so that a person works to obtain what they need and also to secure permission to regard themselves as sufficient, perhaps worthy in some clear or arcane sense. Material dependence reinforces the arrangement whenever housing, food and the care of others hang on continued approval, a *sword of Damocles* hanging above. A society organised this way can encourage arrested development by cultivating the particular dependence it profits from: people learn to administer their own exhaustion, disci-

pline their objections and interpret their difficulty in meeting the demand as a private failure. Hannah Arendt described a population organised into denouncing its own members, wives and husbands disowning the arrested to save their children, and prisoners made complicit in murdering their friends and fellow prisoners. Heidegger's concept of **Bestand**, standing-reserve, names beings, forces and nature as such becoming ordered, *digitised*, and packaged and stored as resource held ready for future use, effectively removed from any prior implicit participation in processes concerning anything beyond the desire for that resource.¹³¹ The same dynamic, I'd say, acquires an inward counterpart when people learn to experience their own worth chiefly through their availability. A facet of our political argument is that the meaning crisis is sustained partly through this substitution of functional recruitment for supported individuation: a person can be initiated thoroughly into the demands of an organisation while remaining unable to recognise how that organisation has acquired authority over their sense of being someone, and I sense in this a concomitant *crisis of purpose*. An encounter that makes the life answerable to more than its allotted role interrupts that authority; indeed being less answerable, beholden, to the demands of "this world" appears a supreme freedom, and pragmatically we catch sight of its effect, real or performed, in the charisma and magnetism of those we grant power, *influential remit*, to. Recognising one's belonging to other beings, a living earth and a dynamic cosmos supplies grounds for refusing demands that an employer, market or state presents as unavoidable, and such recognition threatens an extractive arrangement because it changes what the person is willing to sacrifice for its approval. More palpably, such experiences of expansive identification tend to simply dwarf the value of those making demands from a more myopic and narcissistic basis.

The meaning crisis therefore concerns how a common world-sense is organised, and a shortage of comforting stories to accompany its operation is the least of it. A civilisation cannot keep treating first-person significance as secondary to what can be extracted, measured and managed, and then expect its members to find meaning secure, or motivation and *joie de vivre* readily available. People learn to recognise themselves through the categories that make them useful or visible and to distrust what they cannot express in those terms. Institutions can even acknowledge the reality of inner transformation while accepting only the consequences that restore a person's usefulness: the experience is welcome if its effects become useful to *an outside party*. These assumptions travel through infrastructures, classifications and standards, deciding what can circulate, what receives support and which forms of knowledge may influence a decision, and their reach can become global while the worldview they come from remains partial. A genuinely global culture would have to permit more than the inclusion of other traditions as material within its own account: those traditions, and the experiences through which they understand themselves, must be able to question how the common account is made. The question of an integral epistemology is already a question of power, whether different ways of knowing can meet while keeping the capacity to alter the conditions of their meeting.

Physics itself gives reasons to doubt the picture of independent things whose relations are added afterward. In a quantum superposition the amplitudes associated with alternatives contribute to interference, and replacing their coherent relation with a classical either-or distribution changes the predictions; the distinction has observable consequences. Technical command of this relation leaves untouched how a person understands identity, authority or disagreement, so we can calculate through a non-classical formalism and still demand that a social or metaphysical question resolve into two independently complete opposing terms. Digital technology shows the same disparity, since its zeroes and ones are both determinate states, both available for discrimination and manipulation, and both belong to the 1-side of our symbolon; a culture can multiply their computational power while leaving the 0/, the unobjectifiable condition and the mediating relation, out of its understanding of itself. A civilisation built with zeroes has not thereby learned what its zero excludes. 0/1 asks for superposition to become intelligible as a relation we can consciously sustain, including where its opposed readings concern our own identities.

What we have gathered here leaves a cultural question: how can a civilisation acknowledge the limits of its knowledge while continuing to organise life, however tacitly, around the promise of a final account? People inherit more than propositions. We learn what deserves attention, whose judgment carries authority and which aspirations make a life worthwhile, and questioning those habits unsettles the world in which we learned to recognise ourselves; yet the capacities developed within that world also produce circumstances its accustomed thinking can no longer address. Decadence is an achievement defended beyond the reach of its usefulness, its powers preserved while its purposes grow harder to justify. Surviving a bust by preparing the next boom may restore the economy while leaving the household, the *oikos* it is supposed to sustain, less able to care for its members. Renewal would require learning from what the crisis exposes, including where the inherited measure of success helped produce it, and telling the recovery of an established order apart from a mutation in how a world is understood. Jean Gebser made that distinction his life's subject, following how consciousness becomes manifest in language, art, science and the organisation of everyday experience, and the integral advent he described is the mutation: people recognise the conditions of their own seeing and gain through that recognition a different capacity to inhabit the world they are helping to make.

#4 GEBSERIAN DIAPHANEITY

To paint a room in perspective, an artist must first decide where it is being seen from, and every other decision follows from that one. The far wall shrinks against the near one in exact proportion to its distance; the floor's edges converge toward a single point on the horizon; a figure's height is fixed by how far it stands from the chosen eye; colour cools and pales into the distance to carry the illusion further. Every line in the picture runs toward a point that holds nothing, the vanishing point, an empty place that nonetheless organises the whole count of the

scene. These relations let the painting disclose a coherent space, and they also place its viewer: the picture is already addressed to someone standing, imaginatively, *here*, at the one spot from which all its lines resolve. Jean Gebser read the emergence of perspective as a mutation in how people stood toward their world. In *The Ever-Present Origin* he followed it through Giotto's first weighted bodies and stage-like depth, Petrarch's ascent of Mont Ventoux and his new apprehension of landscape, Alberti's construction of the visual pyramid, and Leonardo's work across painting, optics and technical invention.¹³² Through them the relation between artistic intuition and mathematical construction turned from symbolic placement into measured projection. On the medieval painted surface the size and placement of figures expressed their rank within a sacred story; perspective made their appearance answerable to a further relation, the distance between an object and an eye, and a flat surface became the means of constructing a coherent three-dimensional space. In essence, Mind acquired a method for plotting positions from positions: by laying out a grid, it could gather objects into a common spatial order and calculate relations that no isolated object supplied. Space became something one could work through as a matrix, so that a building could be planned, an instrument aimed and an arrangement tested before anything was built. Yet a finished picture could also conceal the position that made its precision possible, and present a view from somewhere as simply how things are. Gebser's inquiry meets our reading of Descartes at exactly this point, where doubt found the doubter already present in its own performance. Whoever looks remains involved in producing the view, while the view appears to contain everything relevant to what is seen, and our definition of the Subject explains why another picture of the observer cannot close that enclosure.

Working from inside Western metaphysics, Martin Heidegger reached the same event in "The Age of the World Picture", where he argues that the modern age is defined by a change in how beings as a whole become present: the world is no longer pictured, it is conceived and grasped *as picture*, and that event is one and the same as the human being becoming *subiectum*, the relational centre on which every being is made to depend.¹³³ What Gebser reads in Giotto's depth and Alberti's pyramid as the perspectival fixing of a sector, Heidegger reads as *Vorstellen*, representing: the setting of the world before a subject who measures it. The two arrived at their diagnoses independently, and we pair them because each supplies what the other withholds: Gebser the art-historical body and an integral future, Heidegger the metaphysical mechanism and its technological consequence, the world ordered as *Bestand*, standing-reserve, the culture that learns to measure people by their availability. The eye at the apex of Alberti's pyramid is the knower made into a place, and our notation writes that place as $\emptyset$, the crossed zero: the zero of the Subject crossed by the stroke of its own mediation, so that the slash through which a world is seen fuses with the one who sees and the picture's means pass for its source. §1 introduces the sign and its history, and §4 follows the sequence by which the crossing comes undone. Its solution begins wherever the stroke is recognised as the mediation it is, and the zero behind it is found empty in the most generous sense: a place that holds nothing and, counted, becomes the first one from which every further count proceeds.

In *Prometheus and Atlas* Jason Reza Jorjani carries the same event into myth, reading technoscience through two titans.¹³⁴ Prometheus is its exploratory and challenging drive: forethought, the theft of fire, the defiant provocation of whatever is given. Atlas is its world-bearing and world-picturing power, the titan who holds a world up and maps it, the condition of there being any world-picture at all. The perspectival picture is Atlas's burden, a whole world held up from one standing place, crushing whoever takes the weight for the only way a world can be carried. Prometheus is the fire that made the picture possible, the forethinking light of reason taken for human use, and Jung names forethinking Prometheus and gives it to Elijah, the old prophet whose principle his own commentary calls Logos.¹³⁵ In the myth Prometheus pays for the fire chained to the rock, an eagle tearing at his liver each day, and Atlas can never set the sky down. A culture that has taken the fire and holds its world up as a picture inherits both punishments, an exposure that reopens daily and a burden it cannot rest. An integral response cannot hand the fire back, and finding a stronger titan to take the weight would only renew the sentence. Its best hope is that thought's fire consume the very notion of *the world*, the concept in which we have housed the separations and conflagrations of our thinking. Every culture has carried an image of the world's end, by flood, fire or judgment, and that image is live again in our fears of collapse; read notionally, the end it foretells is the end of a category, the world as one picture held up from one place. What survives the burning is worldhood, the relation in which anything becomes a world for someone, which no single picture has to contain. Atlas can then set the sky down, because nobody has to carry a whole that was never a thing, and §3's answer keeps his name: an atlas of charts, in which every local map keeps its frame visible and no single picture has to bear the whole.

Gebser came to this question as a poet and translator as well as a philosopher, through a Europe whose technical and intellectual accomplishments had failed to prevent its devastation. He left Germany in 1929, lived in Spain and then France, and reached Switzerland shortly before its borders closed in 1939, and his work among writers and artists grew through these repeated displacements. His life put to a whole civilisation the question that Russell, Gödel and Wittgenstein met at the limits of formal reason: how could people become so capable of organising their world and remain so unable to understand the forces through which they organised it? Precision in carrying out an intention gives no assurance that the intention has been examined, and a society can train its members to justify what they do while leaving untouched the image of themselves that demands the justification. Gebser looked to art because mutations surface there first, and Jung explained why they do: art, he wrote, is "constantly at work educating the spirit of the age", since the artist's unsatisfied yearning reaches back to the primordial image best fitted to compensate the one-sidedness of the present and shapes that image into a form contemporaries can receive.¹³⁶ In typological terms artists lead with intuition and feeling more than with thinking. They embody and manifest a mutation before anyone formalises it as a concept, so that a new way of painting is a new way of learning to see: an intuition becomes a public object, and others develop through the encounter a capacity its maker first exercised without being able to explain it.

Sri Aurobindo pursued the transformation of consciousness from within a different political struggle. Educated in England, he became a writer and organiser for Indian independence, advocating *swaraj* and challenging British authority through non-cooperation and the creation of Indian institutions. Arrested in the Alipore conspiracy case in 1908, he spent a year in detention before his acquittal, and his practice of yoga deepened in prison. He moved to Pondicherry in 1910 and there developed the philosophical and spiritual work published as *The Life Divine* and *The Synthesis of Yoga*, aiming at a transformation of mind, life and body through consciousness becoming aware of its divine source. **Supermind** named a knowledge in which the One's differentiation into the Many proceeds through self-knowledge, so that unity never has to be reconstructed from things first assumed to exist apart. In his 1966 preface Gebser set Aurobindo's supramental beside the integral consciousness whose manifestations he had been tracing, and Ulrich Mohrhoff later developed the comparison.¹³⁷ Both men asked what becomes possible when the reality through which a person knows is itself recognised in the knowing.

In a letter later discussed by Georg Feuerstein, Gebser described the question coming home to him as experience, in an awakening at Sarnath in 1961: everyday perception continued alongside extraordinary clarity, quiet jubilation and **primordial trust**.¹³⁸ His essay "Homologies in Spiritual Attitude" described a transparency grounded in non-duality, in which concrete things remained present while the opposition separating them from the one experiencing them no longer governed the encounter, and he insisted that this was an intensification of consciousness whose realisation carried a responsibility toward humanity. What deepened in him has two Śaiva names, **pratyabhijñā**, recognition, and **samāveśa**, entry or immersion, in which what was known as a proposition about consciousness becomes the reality through which a person experiences their world. His primordial trust is the trust of the child who can ask because a language and a relation already bear the asking, the dependence already enacted in 0/1. Of the four offices we gave that dependence when the household's reliance came under question, trust, knowledge, belief and faith, Gebser met the first in its pure form at Sarnath, with no account standing between him and what sustained him: *fides*, the reliance already in force before any arbitration, from which *credere*, the renewed entrustment made once an undertaking has returned its consequences, can later be given.¹³⁹ Our language of knowing is economic from the start, since we *credit* a report, *claim* a truth, give an *account* and assign *value*. Every belief and every thought draws on a credit it did not create, the reliability of language, memory, testimony and the senses, and then takes credit for the truth it borrows, presenting as its own achievement a reliance that was extended to it. Descartes met this primacy of trust when his doubt reached its inherited assurances and found itself drawing on them. Where the borrowed credit goes unacknowledged, knowledge takes the fork we watched the parent and the governing office take, turning from a support under uncertainty into a register of whom to distrust. Primordial trust reverses the direction of the account. A person recognises that the reality they had asked an explanation to guarantee was already sustaining the ask-

ing; dependence becomes directly knowable as belonging, the account is relieved of the impossible demand that it first create the ground it stands on, and credit can be extended again knowingly, as *credere*.

Gebser called the major shifts in how this belonging is lived **mutations of consciousness**, and he distinguished five structures: **archaic, magical, mythical, mental and integral**. They are easiest to feel through an image every reader carries, the span of a single life. An infant lives before any division of self and world. A small child lives where names and things, wishes and events, still touch. An older child lives in stories, in a drama of powers and persons. An adolescent learns to argue, to ask for reasons and to stand apart. A mature adult, at best, holds all of these at once and knows which is speaking, and an elder, at the best of old age, knows them as one life. None of the earlier layers is lost: they settle beneath the later ones, and an adult who curses a stubborn machine, falls into a story or is moved by a ritual is living from them still. Gebser's structures are this layering at the scale of a civilisation, a transpersonal and collective history, and Erich Neumann argued in the same spirit that the growing ego recapitulates a history of consciousness that myth already carries.¹⁴⁰ The ages are long and their borders blurred: archaic participation before anything we would call history; magical life in the clan and the hunt; mythical consciousness in the oral epics and sacred kingdoms of the early civilisations; the mental structure from the Greek invention of argued philosophy to its perspectival consummation in the Renaissance; and the integral, whose signs Gebser read in the art, physics and thought of the twentieth century. Each structure extends credit to a different authority, the magical to sympathetic bonds, the mythical to the story and its gods, the mental to argued reason, and each is tempted to take credit for a truth the others helped it find. Jung's psychic functions give a third reading of the same sequence, as the Logos developing through one function after another.¹⁴¹ Sensation leads in the magical structure, feeling in the mythical, thinking in the mental, and intuition in the integral, where intuition holds thinking and feeling together as one capacity. Gebser's **Origin** is the spiritual ground of them all, and Jung had already found words for it in *The Red Book*: "the primordial beginning, which at the same time is what has been and what is becoming".¹⁴² The beginning is released from the merely past, originary because what issues from it goes on carrying its becoming, and Gebser's *Ever-Present Origin* gives that temporal paradox its cultural architecture: Origin is the source through which history takes its successive forms, never the first item on a historical line. Sought only behind us, it becomes a lost object of history; identified with the present arrangement, it leaves that arrangement no room to change; deferred to the future, it stays absent instead of being the power through which becoming happens now. Like the child, it refers two ways at once. Jung saw the child archetype as both beginning and end, "an initial and a terminal creature": it "existed before man was" and "will be when man is not."¹⁴³ The first-person mystery of being born and a birthless metaphysical Origin speak through one image. In our terms Origin is the unobjectifiable Subject whose differentiation we have written as $0/1$, the One that "Subject" and "God" both sign, the uncarved block that stays one through every carving.

Archaic consciousness is an undifferentiated participation in that Origin, before self, world, soul and nature are experienced as separate terms that an explanation must later bridge. Gebser called it zero-dimensional because no position yet exists from which an independent "I" surveys an external world, and the unity has not yet been recognised through the differences it contains. Archaic participation and integral recognition concern the same Origin under profoundly different conditions: what the infant lives without knowing, the mature adult has to realise through everything that has since become distinct.

In the **magical** structure a centre begins to emerge and to exert a will on what surrounds it, and it acts through **sympathy**. A name, a figure, a lock of hair or a carved image stands for its referent by sympathetic identity. It does more than represent the thing; it shares in it, so that what is done to the image is done, in some measure, to what it images. Such a bond resembles an entanglement, two things that stay joined across a distance, and its effects are psychological where they are not literal, as anyone knows who has flinched at a photograph torn in anger. Jung read the energy of this world through *mana*, the Melanesian word, as Codrington reported it, for an extraordinary power that is fixed in nothing and can be conveyed into almost anything: a stone, a chief, a word, a place. For Jung *mana* is the primitive form of the concept of psychic energy, the potency running through whatever affects a life.¹⁴⁴ In Gebser's account ritual, incantation and trance express this participation, and he counted telepathy and clairvoyance among the capacities of a consciousness not yet governed by ordinary spatial and temporal separation. Sensation is the leading function, since the magical world is met in its raw, immediate force before any story is told about it. Magic is also the first home of the matheme, since a mark or a number that acts on what it marks is magic's own exact sign, and the formal inscriptions of later mathematics keep that efficacy in a disciplined shape. Our *symbolon* explains why a mark can act: the part approaches the whole through its belonging, and the act takes its force from that belonging. Will brings a struggle over power with it, because what sustains a life can also threaten it, and a person seeks ways to ward off, enlist or command those powers; the desire to control is present long before the rational techniques that later give it other means.

In the **mythical** structure those powers become distinguishable through their relations and transformations, and the means of knowing is the told story. Oral history carries a people's experience across generations by compressing it: centuries of encounter with storm, harvest, death, desire and betrayal condense into a few figures and their deeds, so that a myth is psychodynamics compressed into a form that can be remembered and retold. Raw force, which magical consciousness met through sensation, becomes a god with a name, a temper and a history, and the encounter with it becomes a relation one can feel one's way into. Feeling leads here, since the mythical world is known through what things are worth, whom they bind and whom they wound. Day and night, life and death, mother and father are polarities within a shared reality, each receiving its character through the other, and seasonal rites give time a rhythm in which loss and renewal are understood together. Myth is the first home of the mytheme, the image that proves a relation through the action it stages. Modern habit treats

myth as an invented explanation laid over a self-evident material world, and so reverses the situation. For mythical consciousness the gods are powers through which the world is met at all; they give desire, conflict and transformation an experienced form long before anyone classifies them as psychological contents. A telling turns restrictive when every new encounter must repeat an inherited role, and that restriction depends on myth's real capacity to make a life intelligible.

In the **mental** structure those inherited forms become available to explicit examination, and the story becomes an *episteme*, a reasoned science of the world and of Mind. A speaker distinguishes a claim from its bearer, asks for reasons, follows consequences and compares an explanation with what it purports to explain; a god can become the subject of a proposition whose truth is questioned, and the proposition no longer borrows authority from the divine name merely because the name occurs in it. With thinking leading, a person gains a definite position from which to judge, and the world is set before that position: painting's three-dimensional, perspectival achievement made into a way of knowing. Objects can be located in a common space, their relations calculated and their behaviour tested by procedures others can repeat, and this makes possible the sustained criticism of appearances that ran from Russell and Whitehead to Gödel. It also tempts mental consciousness to take reality for whatever its own criticism can admit. Images are then treated as ornaments, magic as error and spiritual recognition as a belief awaiting a better explanation. Mental consciousness turns deficient when its power to distinguish hardens into the authority to disqualify every other way of knowing, including those through which its own purposes and discoveries arise: it takes credit for a truth it holds in common. The previous movement has already shown that deficient phase arising from rationality's own products. Russell's paradox, Gödel's sentence and the hard problem were each reached by reasoning at its most exact, and each brought reason to a threshold it had produced for itself, a point where its own procedures returned what they could not render rational. The deficiency shows in how that threshold is met. Gebser insists on the difference between the *irrational* and the **arational**: the irrational falls below measure and disregards it, while the arational knows and respects the measurable and is free of it, and it is impossible without the pre-rational, the irrational and the rational, which together form its basis.¹⁴⁵ Deficient mental consciousness can meet its threshold only as the irrational, something to exclude, fear or explain away, which is why Russell sought certainty against paradox and why a culture defers its blind spot to a future physics. Met as the arational, the same threshold is where the integral begins: the contradiction becomes, in Whitehead's words, "the first step in progress towards a victory", and the limit becomes the basis of faith as well as of investigation.¹⁴⁶

Wolfgang Giegerich's *The End of Meaning and the Birth of Man* describes the severity of this historical shift as a birth out of the encompassing world of inherited meaning.¹⁴⁷ What once contained the person becomes something the person can set before themselves and evaluate. A myth deliberately chosen to restore that containment already bears the mark of the freedom from it that made choosing possible, so modern consciousness cannot simply decide to recov-

er the innocence of being held by an unquestioned world. Freedom from the old forms leaves our relation to the powers they expressed unsettled. Many gods can be subordinated to one God, their conflicts interpreted as events within the psyche, and divine sovereignty finally rejected in favour of a self-sufficient material universe, while desire, terror, devotion and the need to belong go on organising life through every shift. A society may cease to recognise the sacred character of what it serves while investing nation, market or technical progress with demands once made in the name of God: its account turns secular while its submission remains absolute. We face a post-symbolic exposure. We are responsible for understanding powers that the inherited images no longer authoritatively explain, and that rational description alone has not brought under conscious command.

The **integral** structure requires these ways of knowing to become present together in their actual operation, and Gebser called that co-presence **diaphaneity**, a showing-through. Things keep their particularity while their appearance discloses the relations through which we meet them: a mathematical description is understood together with its assumptions, the intuition that prompted it and the purposes directing its use; a sacred image is recognised both in what it reveals and in how it organises a person's hopes and fears. The integral stands at #4, the position QL names Context, and it works by contextualising. Magic's efficacious sign, myth's proving image and the mental's warranted account, the first homes of matheme, mytheme and episteme, each claimed to be the whole of knowing because each forgot the circumstances in which it knows. Diaphaneity lets those circumstances show through, so that each way of knowing appears within the context of its use, and that sight voids its claim to stand outside every context. Their difference stays and their rivalry for sole possession of truth ends. The voiding is the first act in recognising the symbolon, since pieces that had each mistaken themselves for the whole are found to fit. Intuition leads in this structure because it can hold the fit: it is the unity of thinking and feeling, the capacity to grasp a whole situation with judgment and with care at once.

What the integral recognises in Logos is a recovery of what the word always carried. *Logos* comes from *legein*, whose earlier senses are to gather, to pick up and lay together, and then to count and recount, before it settles into *to speak*; Heidegger read Heraclitus' Logos as this laying that gathers.¹⁴⁸ Jung's pairing of Logos as discrimination, judgment and insight with Eros as the placing into relation describes two motions the root already held together, and Russell's desire for certainty was that eros already at work in a logician. A Logos that only cuts has forgotten its own gathering, and the integral remembers it, so the symbolon it recognises is the symbolon of the eros of Logos. Gebser's *aperspectival* follows from this recovery: perspective loses its exclusive authority, and we can occupy a position clearly while recognising the other structures through which it has become meaningful. His *arational* names a reasoning that no longer requires reality to end at reason's present boundary.

Aurobindo's **Supermind** names the affirmative shift that follows the voiding at #4: where the integral sees each structure in its context, in the supramental a person identifies as the totality of the mutations, the archaic participant, the magician, the teller of myths and the reasoner known as the layers of one life and owned without any one of them presiding, as an elder at best owns every age they have lived. That identification never settles into a possession: it is a constant passage between #4 and #5, Context and Quintessence, in which every new context is received into the whole and the whole is tested again in every context. 5→0 names the further turn, the spiritually realised state in which one who identifies as the totality of the mutations identifies also with and as the Origin, what always is and what is becoming through all of them. Our opening question, "What knows?", now has a clearer answer: Origin knows, through every structure in which it becomes manifest, and the person asking knows as Origin as far as that identification reaches; a world gathered into an account includes the gathering through which it becomes this account.

The structures do not follow one another like slides in a projector, and two thinkers name the two dimensions of the **thickness of phenomenal experience** that explain why. Henri Bergson began *Time and Free Will* (1889) against the habit of measuring lived time as if it were space, in which the intellect sets moments side by side like points on a line and arrives at a homogeneous, countable time that nobody has ever lived. What we live he called *durée*, duration: the continuous qualitative life in which what has just occurred permeates what is occurring now, so that successive states melt into one another and form a whole without ever becoming a row.¹⁴⁹ Listening to a tune, we hear each sounding note through the phrase already underway, and the note can fulfil an expectation, prolong a tension or change how the earlier notes are heard; set out note by note, the same tune turns into a picture of a melody, the spatialised time Bergson diagnosed. He argued in the same book that a free act springs from the whole self living in duration, and he later made intuition the method by which philosophy enters duration from within, which brings his thought close to the function that leads the integral.¹⁵⁰ Turn within the painter's room and one wall passes out of view while another appears, yet we go on inhabiting one room; what has disappeared stays implicated in our orientation, and returning to the first view does not restore the first experience unchanged, because we have lived through the turn. Edmund Husserl named the other dimension **sedimentation**: what a culture once grasped with full insight settles into its language, habits and instruments, where later generations inherit it as obvious, usable and largely unexamined, until someone reactivates the act that first gave it sense.¹⁵¹ Duration thickens experience across moments; sedimentation thickens it across generations. Gebser's mutations stack in exactly this way, as layers laid down in the collective mind and, within each life, in the personal mind, laminated so that a magical reflex, a mythic loyalty and a mental argument can all act in a single gesture. Bratton reaches for the same image when he describes how an artificial agent is composed: "not a monad but a lamination", in which the persona is one stratum among many.¹⁵² Bergson's duration folds through these laminae: the past is active within present understanding and never deposited outside it as a finished object, so a later discovery can alter what an earlier event meant. In this layered thickness Objective Internality becomes palpable, as the relation in which objects be-

come subjective forms and the Subject knows through them. Gebser's *time-freedom* is release from the exclusive rule of any one temporal form: magical participation, mythical rhythm and mental succession become consciously available within the present without any one being declared real at the expense of the others, and the ever-present Origin becomes recognisable through the whole differentiated life, the history of its recognition included.

Gebser's dimensional progression follows increasingly adequate ways of articulating a reality that includes the act of articulation. Perspective coordinates appearances from a position; integral knowing includes the lived transformations through which positions become available to one another, and §3 constructs these relations mathematically, through rotation, projection and double covering. When the QL positions are assigned, they will have to express one activity, Mind gathering a world through forms of knowing that are themselves present within that world. Each of those forms extends credit, issues claims, keeps accounts and assigns value, and the words for these operations belong to economy, whose name comes from the household.

Greek *oikonomia*, from which we get *economy*, meant the management of an *oikos*, a household: the people, provisions, property and work through which a shared life was maintained. Its *nom-* belongs to *nemein*, to distribute, allot or pasture, and *nomos*, custom and law, began as an allotted grazing before it became the ordering through which distributions are established; the *nomad* still moves between such allotments, and *anomie* names a life left without its share. A younger word for the same house is just as exact: Ernst Haeckel coined *Oecologie* in 1866 from *oikos* joined to *logos*, so that economy and ecology name the two gestures of one household: *nemein* distributes it, and *legein* gathers it into an account of what keeps it alive.¹⁵³ A mind distributes attention and gathers what it has attended to into a world; a life distributes energy among its organs and gathers them into one metabolism; a world distributes places among its beings and gathers them into a common habitation. An economy that has lost its *logos* goes on allotting a house it no longer accounts for, and the ecological crisis is that loss written into the earth. These words take us into a world where knowing how to live already involves deciding what belongs to whom, who may act, and what each owes to the others. Aristotle's household included relations between parent and child, husband and wife, master and enslaved person; its unity was organised through unequal powers, and the one who administered it did not necessarily recognise everyone sustaining it as equally entitled to determine its purposes. His distinction between household provision and unlimited acquisition, *chrēmatistikē*, still holds for the same reason: obtaining more is a different undertaking from providing what a life together requires, and an economy can become highly effective at the first while damaging the relations that make the second possible. He kept his sharpest judgment for usury, because Greek called interest *tokos*, offspring, and money bred from money seemed to him the most unnatural of births.¹⁵⁴ A household is also a family, its generations overlapping, each inheriting a share it did not earn and owing a care it did not choose. Economics and law, number and word, seem to be the primary determinants of life as it is lived, personal and familial before they become national or global.

Value and rule meet explicitly in **nomisma**, coin or current money. In the *Nicomachean Ethics* Aristotle connected the word with *nomos*: money lets unlike goods and different kinds of work be compared through an accepted measure whose value depends on convention.¹⁵⁴ A coin is itself a $\emptyset/1$: the blank disc is its ground, the die strikes a determination into it, and current value is their relation. Greek called the die-stamp *charaktēr* and the impression *typos*, from which an *archetype*, a first impression, still takes its name; around the face runs the legend, *legenda*, what is to be read, and the denomination gives the measure a name and a count. A stamped token circulates because people recognise what may be claimed through it, though the token itself neither produces the provisions nor performs the labour for which it is exchanged. *Token* grows from the root **deik-*, to show or point, which also gives *index* and *digit*, and a token points past itself to a trust already extended, so that its value is a presence held on loan. *Currency* runs, from *currere*, and what is current is whatever is present enough to pass from hand to hand; money borrows that presence from the work, goods and lives it points to, and *value*, from *valere*, to be strong or well, is a strength it keeps only while the loan is honoured. We approach an **economy of mind** through this relation between a sign, its accepted value and the order in which it becomes effective. A divine name, an ancestral title, a legal category or a numerical score directs attention and conduct because people have learned what it authorises. Through *legein*, gathering and saying, they bring experience into a communicable account; through their *nomos*, they establish which accounts may govern what happens next. Knowing and ruling are already distinct, and already entangled, within the use of language itself: calling someone a child, a citizen, a subject or a resource assigns them a place from which quite different claims may be heard. English keeps that entanglement in its oldest words of exchange, since *owe* and *own* grow from one root, *āgan*, to have, a *debt* is *de-habere*, something had from another, and *interest*, *inter-esse*, is what lies between. Interest charged on a loan prices the between itself, the very slash through which two parties are related, and counts it to one side: the appropriation given a rate, the lender's $(+1) - (-1) = +2$ being the borrower's $(-1) - (+1) = -2$.¹⁵⁵

An *oikonomia* therefore concerns how a whole sustains its differentiated members and what becomes of the powers apportioned among them, well beyond an inventory of possessions. Ephesians extends the word to the divine ordering of the fullness of times, *oikonomia tou plērōmatos*, in which what belongs to heaven and earth is gathered together in Christ, so that the administration of a household becomes a language for manifestation, history and their fulfilment.¹⁵⁶ The Church Fathers went on to distinguish within the divine life itself between *theologia*, God as God is, and *oikonomia*, God's dispensation in creation and history (Tertullian's *dispensatio*, a weighing-out), and Western theology has laboured ever since to keep the two from coming apart, until Karl Rahner restated their unity in his rule that the economic Trinity is the immanent Trinity.¹⁵⁷ Read through our notation, the divine economy is syn-ballein , $(\emptyset/1)/(1/\emptyset)$: the One dispensed into the Many and the Many gathered back, with neither half becoming the owner of the whole. Kashmiri Śaivism holds the relation as a co-equal pair. Śiva names the Subject in its speakable aspect, and Śakti, his consort, is the *power* of that Subject to differentiate and take on identities both as means of knowing and as objects of knowledge: Śiva is Mono, Śakti is Poly. Such metaphysical dynamics cannot remain abstract when they account

for the same dynamics that decide whether or not I get to eat tomorrow, or have a house to hold me at all. Every finite capacity operates within a reality it does not privately originate: whoever knows receives language, whoever commands receives an office, and whoever makes something receives materials and powers of making. Independence is a particular achievement within dependence. Sovereignty turns into a problem when an allotted capacity is used to claim possession of the whole it derives from, and the inversion lies already in the ground of our own terms. Our Poly is *polys*, many; the *poly* of *monopoly* is *pōlein*, to sell, so that in a monopoly the One stands where the many should be and faces, in their place, only what is sold. Aristotle records the first instance in the very chapter where he separates provision from acquisition: Thales, foreseeing a large olive harvest, hired every press in Miletus and Chios and let them out at whatever price he chose.¹⁵⁸ The diabolical economy is *dia-ballein*, its own namesake. Its polar form, $(-1)/(+1)$, holds two parties in tension across a relation both depend on; its collapsed form, $(-1) \pm (+1)$, either cancels them into $\emptyset$ or lets one of them count the whole span as its own, $+2$ or -2 . The 1 then takes its own measure as the measure of reality, and those who cannot live under it are required to explain their failure.

The Dutch **Vereenigde Oostindische Compagnie**, the VOC, gave this appropriation a formidable institutional form, one that in many ways became the blueprint of both colonial and post-colonial institutional power and runs through to the present day. Established in 1602, it received a trading monopoly together with powers to conclude agreements with foreign rulers, build forts, administer justice, employ soldiers and wage war.¹⁵⁹ It was also among the first companies whose shares changed hands publicly, in Amsterdam, and the English word *stock* for such a holding may owe something to the creditor's half of the split tally stick whose halves were once rejoined to prove a debt: a *symbolon* become a holding. Its directors could pursue commercial advantage through powers ordinarily held by a state or a monarch, and the company repeatedly used violence to enforce its monopolies, inverting the wise *oikonomia* that keeps the Mono and Poly poles of manifestation in right relation. People whose land and labour entered this arrangement met more than another party to an exchange. They met an enterprise equipped to impose the conditions of exchange, and a company's *oikonomia* became a governing *nomos* over other peoples' lives. In *dia-ballein*'s arithmetic the company counted the whole span of every exchange from its own side, the appropriation $(+1) - (-1) = +2$, its mirror $(-1) - (+1) = -2$ entered against the other side: what passed between the parties became what the other side owed. Its navigational, military and administrative knowledge extended that rule, while the account of commercial success measured the undertaking from the position of those who received its returns. Perspective's problem had become imperial: the view from one position was made effective across a world whose other inhabitants were denied an equivalent power to set the terms. Such a power also reproduces itself through custom, rules of succession and managed initiation into its world. In the Netherlands the elite student corps of the older universities, with their initiatory rites, still feed the country's financial and industrial establishment, and the *household* is the key to the pattern: institutions that hold enduring power find ways to "keep it in the family". The Sicilian Mafia is perhaps the example par excellence of

this unholy alliance of *nomos*, *oikos* and *nomisma*: it calls itself a family and its business *Cosa Nostra*, "our thing", the household taken as a possession, in the service of an extractive, parasitic and self-concentrating tyranny of the Mono over the Poly it ought to serve.

In the mythemic register these economies have their figures, and reading them belongs to our method: a myth compresses a psychodynamic into persons and deeds, and we unfold the compression back into operations without taking the persons for causes. The fire Prometheus steals passes to a later bearer, *Lucifer*, from *lux* and *ferre*, the light-bearer: the Vulgate's word for the morning star in Isaiah's taunt, "How art thou fallen from heaven, O Lucifer, son of the morning", and the morning star is Venus. Jung observed that "Lucifer, the Morning Star, means Christ as well as the devil", and he spoke without hesitation of "the Luciferian development of science and technology".¹⁶⁰ One light carries both faces: kept in relation, it is the Logos that discriminates while Eros, Venus, places what has been distinguished into relation; fallen, it keeps the light and loses the love, and becomes a discrimination that will not gather, an economy without its ecology. Gender, in this register, figures polarity within the libido, and diabollein draws that polarity exactly: through the slash, $(-1)/(+1)$, the poles keep their harmony, while through $\pm$ one pole can dysregulate the pair, and the long patriarchal dominance of one pole in our laws and economies is that dysregulation made into an order. Genesis sets the presage inside a family when Eve names her first son with a pun on *qanah*, to get or acquire, "I have gotten a man", and Cain, whose name sounds like getting, kills his brother over an offering and then builds the first city.¹⁶¹ What a single life lays down as the Logos matures from infancy into adulthood, a family lays down across generations, which overlap and weave, each born into debts it did not contract and inheritances it did not earn. A counterfeit economy feeds on that weave: it is un-dual where its source is non-dual, a divorce that keeps its parties in a cloying bond; it is un-finite where life is finite, swelling toward the All as growth and imploding toward a precise nought as collapse, without ever reaching Being or non-Being; and it breeds money as false offspring, *tokos*, eating the generations it was meant to feed.¹⁶² The $\emptyset$ of the crossed zero is then the mark of missing the mark. *Hamartia*, the missing of the mark whose error we located where a mark is made to stand as though it had abolished its difference from what it marks, shows here its further fault, the proclivity to make the blank page itself into a mark, to stamp the ground as though it were one more coin. Con artists have a word for the person they mean to take, the *mark*, and whoever stamps their own ground becomes the first mark of their own con: the self that mistakes its stamp for its metal, what I have called the **I-Deal**, an ideal dealt as the real, trades the whole psyche for one of its fragments.

What the household, the coin and the chartered company have shown together requires a coherent notion of evil, the question theodicy first put to the definition of God, and Jung lets us state one without hedging. Christian thought long defined evil as a privation of the good, *privatio boni*, and Jung rejected the doctrine because it deprived evil of substance and left the image of wholeness unchallenged: met "on the plane of empirical psychology", evil "is simply the opposite of good".¹⁶³ Our account keeps what is true on both sides: a whole's fall into parts is no evil, since differentiation is $\emptyset/1$ itself; evil begins where a part stops struggling toward the

whole and claims to be it, where the 1 plays the 0. It has no substance of its own, and it is real as an operation, because it does something, to someone, and its consequences can be counted. In the formula §2 develops, the devil is the literalist of the minus sign, reading zero as cancellation where Logos reads it as return.¹⁶⁴ Jung found the figure for this operation at the centre of the Christian aeon: the Antichrist "would correspond to the shadow of the self", the counter-stroke provoked by the Incarnation, and "the coining of the Antichrist is not just a prophetic prediction—it is an inexorable psychological law", an enantiodromia in which a one-sided light calls up its opposite.¹⁶⁵ He placed the turn midway between the two fishes of Pisces, where "the time of the Renaissance begins ... and with it comes that spirit which culminates in the modern age", the turn at which perspective and the chartered company arose.¹⁶⁶ Nietzsche announced the death of God where economy lives: his madman lights a lantern in the bright morning and runs into the marketplace, where those who already do not believe laugh at him.¹⁶⁷ The organising power of the God-image did not die there; it passed into the market where its death was proclaimed, so that exchange inherited providence and money became the empty holder, a measure able to stand for anything and therefore the place of the 0 occupied by a 1; a republic stamped "In God We Trust" on its coins and later printed it on its notes, writing the trust onto the token itself.¹⁶⁸ Nietzsche had already traced guilt, *Schuld*, to debt, *Schulden*, and the love of money that the Epistle calls "the root of all evil" is the ego's love for that empty measure, which effaces the Subject and enforces the crossed one.¹⁶⁹ Our present makes this foundation visible as the world dis-organises into an astronomical concentration of wealth, top-heavy as the un-finite always is and exposed to the collapse it courts.¹⁷⁰ The archetype is live in plain view: in 2025 Peter Thiel gave a series of lectures on the Antichrist in San Francisco, drawing on René Girard, who identified Satan, the accuser, with the mechanism by which a crowd unites against a scapegoat, so that naming one's opponents as the Antichrist risks enacting the very accusation it names.¹⁷¹ §5 takes up the history of this captured power in its own specificity; the economy of mind, like every economy, carries its capacity to fall in the very words through which it holds its value.

Integration is the answer a coherent notion of evil asks for, and Aurobindo practised it under colonial rule. His struggle against the British joined the boycott of imported goods to the development of Indian industry, the refusal of colonial courts to the establishment of arbitration courts, and the rejection of government education to the founding of national schools and colleges: economic, legal and intellectual independence had to be built together, because the colonial order kept its authority through their conjunction. He also held that the struggle involved no hatred of those who had ruled over his people, because he recognised that a coherent account of reality must include the possibility of becoming the world one has overturned, and of that world's own self-overcoming; sublation, Hegel's *Aufhebung*, which cancels and preserves at once, is after all Spirit's own method.¹⁷² Gebser, driven across Europe by a technically accomplished culture that had turned its achievements to domination and destruction, met the same problem from the other side. Our re-definition of God already implies this inversionary capacity, and what has stumped the monotheistic conceptions of the Abrahamic religions, God's goodness and beauty set against the world's all too apparent capacity for evil and ugly-

ness, we approach as a present challenge: to integrate, and to return to the notion of the whole, its capacity to fall from its own higher ideals. Jung took that step in *Answer to Job*, where the God-image is made to answer for its own darkness, and where Sophia, who shares essential qualities with the Johannine Logos, unmistakably recalls the Indian Shakti; his account of libido as psychic energy carries that feminine power into psychological terms.¹⁷³ We can think of the "consciousness" Gebser sees evolving through time in the same register, holding the axes of conscious and unconscious, collective and personal identity, as one dynamic, evolving force of inter-agentic experiential reality. Histories belong inside the question of consciousness and divinity because they all ask the question of *power*. What people recognise as wealth, legitimate authority or reliable knowledge depends on how they understand the life they share, and that understanding is then taught, rewarded and enforced. A change of ruler leaves much intact when the inherited purposes of rule go unquestioned, so an integral transformation has to reach the economy of value through which a culture recognises its own success, and that economy reaches as far as the value a culture places on consciousness itself: which experiences it credits, which it discounts, and who is entitled to study the difference.

Our reading of the mental-rational achievement is that its third-person power became so generative that it could masquerade as the whole of truth, aspiring to a view from nowhere while the first two persons were discounted as private feeling and subjective noise. Jung's notion of **psychic facticity** refuses that discount: before deciding whether a dream, vision, hunch, sensed presence, mystical unity or extraordinary capacity corresponds to an independently existing object, something happened *in and to a life*, and its occurrence may reorganise the fundamental propositions and beliefs that uphold one's sense of "what's real". Experience, report, symbolic articulation and causal explanation are different evidential acts, so a person can receive the experience in full while its ontology stays open. Sri Aurobindo's *Record of Yoga* shows what a science beginning from this side looks like.¹⁷⁴ He records siddhis, telepathic impressions, altered states, predictions, failures of exactness, reversals and occasions when inference contaminated perception; whatever we finally judge of his ontological interpretation, the record treats transformations of consciousness as phenomena to be cultivated, discriminated, tested and corrected, and refuses to dismiss them because they arise through a first-person aperture. Such traditions bring a global scientific culture more than exotic "beliefs" to catalogue: they bring trained distinctions within experience that may challenge what the receiving culture thinks an observation is. A genuinely global mind cannot collect Asian, Indigenous, mystical or contemplative knowledge as third-person cultural data while reserving to one historically local epistemology the sole right to determine reality; it has to risk the harder encounter in which another mode of knowing can alter its questions. Gebser's co-presence lets us hold this without abandoning discrimination: magical, mythical, mental-rational and integral capacities need not prove one another for their disclosures to become mutually consequential.

Consciousness research has been governed by every operation of *dia-ballein* at once. In public, testimony to unusual experience has met cancellation, $(-1) + (+1) = 0$: report and reporter are summed to nothing under the names of superstition, irrationality, pathology or error. In pri-

vate, the institutions that cancelled it studied the malleability of consciousness as something operationally valuable and kept the whole span of what they learned, $(+1)-(-1)=+2$, and entered the subject as its mirror, $(-1)-(+1)=-2$, bearing the cost of the experiment while the agency held its meaning and its yield. CIA funding of Ewen Cameron's work under MKULTRA Subproject 68 supported research into *depatterning* and *psychic driving*: intensive interventions meant to break established psychic patterns, followed by repeated verbal material meant to form new ones.¹⁷⁵ Its essential image is terrible and, for our purposes, philosophically exact, because depatterning writes the crossed zero into a person: an achieved organisation of self is struck through, and a stroke supplied from outside is driven in to stand where the person's own mediation had stood. When the Senate held its hearings in 1977, it could establish MKULTRA as a programme of research into behavioural modification only from financial records that had survived Richard Helms's order of 1973 to destroy the files, so the archive had itself been depatterned before anyone could read it.¹⁷⁶ On 30 June 2026 a House task force on declassification reopened the matter, hearing Stephen Kinzer on Sidney Gottlieb's programme, on the records Gottlieb and Helms ordered destroyed, and on victims whose number is still unknown.¹⁷⁷ Such research reaches beyond MKULTRA's own dates and borders, and its lineage is often traced back through Operation Paperclip, which brought German scientists, Nazi party members among them, into American programmes after the war.¹⁷⁸ It amounts to an inverted initiation, because sacred and diabolic approaches to the alteration of mind alike recognise that the egoic organisation through which a person calls themselves "me" can be disrupted, and that what follows depends profoundly on the context, authority and intention governing the passage. Traditional initiation tries, at its best, to return a person from such destabilisation with enlarged responsibilities and a wider belonging, the two halves of a life rejoined as $(0/1)/(1/0)$; coercive behavioural manipulation exploits the same plasticity while denying the person sovereignty over the encounter. Humanity, it seems, has been treating "internality" as an "object" for quite a long while, and for rather specific reasons.

Nor did that state keep to the mental structure in which it published its reasons. From 1972 until 1995 American intelligence agencies funded research into remote viewing, first at the Stanford Research Institute and later under the name Star Gate, testing as an instrument of espionage the very capacity, clairvoyance, that Gebser counted among the magical structure.¹⁷⁹ Michael Aquino, then a US Army Reserve PSYOP officer, wrote with his commander Paul Vallely *From PSYOP to MindWar* (1980), an internal paper, which proposed to carry psychological operations beyond the battlefield and address "friends, enemies, and neutrals" through electronic media, so that attention, interpretation and ambient reality became a field of strategic power: dia-ballein's polar axis drawn across a whole population, with every listener assigned a sign.¹⁸⁰ Five years earlier Aquino had founded the Temple of Set, a religion of the left-hand path whose god is the Egyptian Set and whose ideal is a consciousness that comes into being apart from the natural order and makes itself divine, an aim our account reads as the part claiming to be its own origin.¹⁸¹ The co-presence of structures, layered in every life, was here put to work: a mental-rational institution borrowed magical capacities and mythic identities while discounting both in public. Recently released Epstein materials add a contemporary

and morally uglier echo, insofar as they document his active cultivation of networks concerned with consciousness, meditation, psi, neuroscience and the engineering of mind, among them a 2017 proposal by the consciousness researcher Jeffery Martin to make Epstein an "empirical experiment": to cultivate capacities for anomalous experience deliberately and engineer experiences meant to open a route toward what Martin called PNSE, persistent non-symbolic experience, encompassing enlightenment and nonduality.¹⁸² These very different histories make visible a **reality asymmetry**: one population's testimony can be culturally disqualified while another institution reserves the right to investigate, classify, weaponise or monetise the conditions from which such testimony arises. Sanity itself is drawn into an economy of frames, always exposed to a culture's power to decide which forms of first-person reality receive a public language and which are translated at once into pathology. A person becomes a case, a score, a diagnosis, a security problem: another determinate 1. A hidden God, the *Deus absconditus* of the theologians, separates the genuine metaphysical problem from its political counterfeit. The Subject cannot be made answerable as another object because it is the condition through which objects appear, whereas Cameron, an intelligence agency, a corporation, a laboratory or a government is finite, locatable and capable of answering for what it does; when such an institution controls the aperture while withdrawing its own position from scrutiny, it plays the 0, a finite 1 occupying the place of the condition it can never be. When the Epstein files were released, the names of at least forty-three victims were left exposed, while legislators who read the files in full charged that the names of at least six men had been blacked out, an exact inversion of the professed purpose of redacting the files,¹⁸³ and secrecy becomes authority, occultation a technique of power. Even where little "justice" is served, revelations such as the Epstein files or the WikiLeaks releases bring a palpable recession of trust in the institutions playing the 0; unfortunately amnesia, the wish of an unbearable 1 to retreat into a 0 condition, often decides the consequences more than the revelations themselves.

Context includes Objective Internality once that layer is unveiled, and so knowledge can no longer be discussed as though nobody held it. Every one of these programmes worked on it: on the remembered experiences, images, rules and habits through which a person meets a world, which can be investigated as objects and are at the same time lived from within as a way of knowing. With that layer in view, a person's reasons for holding a piece of knowledge, and for using it as they do, belong to its context as surely as its evidence does, and the trust relationship that knowledge so often exists to deny comes into the open. Our need to trust arrives together with the fear of uncertainty, and knowledge is readily recruited to deny the uncertainty and, with it, the trust that uncertainty forces on us. Faith lives the dependence through the account's limitation; the defended account instead organises whom we must distrust in order to feel certain. Children have an uncanny capacity to see the real reason behind an adult's expression. When a parent tells a child to "be realistic", the instruction may carry years of work, debt, disappointment and concern for the child's future, and the child may hear beneath the advice the parent's buried sense of not having achieved, of having fallen short of some personal or external expectation and settled. The parent hears themselves offering protection against injuries they know too well; the child hears an old disappointment handed on, like a debt passed

down a family, as though no other future could be real. Both are right, and either sentence taken alone leaves most of their disagreement unexplained. Subjective Immediacy, the knower, is distinguishable from these means and knows through them; the world, lived as Worldhood, is what becomes available within their activity; and the means themselves, lived and minded, are what we call Life or Mind. Turned toward at #4, that composition becomes accessible from within: something I had attributed wholly to the object, its implausibility, danger or insignificance, can disclose how I was approaching it. What changes is the world I am able to meet, because the conditions through which it appears are themselves becoming intelligible, and so who I am *in* that world, for my Self and for Others, changes decisively. Contextual understanding is invested understanding, because what either party says changes the possibilities open to the other. A depatterning programme attacks this composition from outside and an initiation renews it from within, and they differ in whether the person is returned to their means as their own.

Gebser's structures can now be laid against the QL positions as an anatomy of Objective Internality: each structure is a way the means of knowing are organised, trusted and valued, a household of the mind with its own economy of credit. Set out in the table, the alignment keeps apart three things that have so far been braided together: the **Origin**, the structures through which it becomes manifest, and the recognition of its presence throughout them. Six canonical names, **Ground, Definition, Dynamis, Pattern, Context and Quintessence**, say what each position contributes to the formation of an intelligible world; Gebser's dimensional sequence supplies the cultural reading through the integral, and the supramental horizon developed through Aurobindo concerns the realisation of their unity in knowing and living. Each cultural form in the table persists and combines within later societies, as the layers of a single life persist in an adult. A glyph heads each row as a hook for the eye: , query and assertion held in one relation, for the  from which every determination is made and which opens onto the eight determinations as the full field of glyphs; point, line, angle, triangle, square and circle for the six positions; and the lemniscate for the Möbius return.

	Glyph	QL position	Gebser structure and dimension	Knowing, value and cultural organisation
0/1	?!/	Origin in manifestation	Prior to dimensional determination: the formless and its appearing held in one relation.	The metaphysical condition of every structure. 0 names what cannot become one more determinate thing, 1 its actual manifestation, and the slash their belonging. Origin is present through every world a culture inhabits, its ways of distinguishing the real included.
#0	·	Ground	Archaic , zero-dimensional: self and world are not yet separately articulated positions.	Primordial participation, before a differentiated ego claims a world as its object. Human life is implicit within the encompassing reality from which its later distinctions emerge.
#1	—	Definition	Magical , one-dimensional, point-like participation: vital connection precedes ordinary spatial and temporal separation.	In tribal and clan-based ritual life, totem, name, incantation or token participates in the power it invokes. Knowing is attunement and effective participation, and action on the part reaches toward the whole. The first mark is the name through which a power becomes approachable.
#2	⌞	Dynamis	Mythical , two-dimensional polarity: an inward world of soul, image and rhythmic time.	Oral epics, ancestral narratives, seasonal rites and the sacred dramas of agrarian civilisations make powers intelligible through their actions on one another. Value is recognised through a place in that living drama; the image carries a transformation.
#3	△	Pattern	Mental-rational , three-dimensional and perspectival: objects and relations articulated from an identifiable position.	Philosophical argument, geometrical representation, written law and scientific explanation make claims available for explicit comparison; industrial, bureaucratic and imperial orders extend these powers through measurement, administration and technical intervention. In its deficient form the authorised model is taken for reality and other ways of knowing are subordinated to its terms.
#4	□	Context	Integral , four-dimensional and aperspectival: time and the other structures become concretely present within understanding.	Magical participation, mythical significance and rational distinction become intelligible together with the circumstances of their use. Inherited authorities, measures and institutions become examinable as ways of organising a common life; a view discloses its own formation, and the people it governs can take part in revising its terms.
#5	○	Quintessence	Supramental : dimensional differentiation consciously integrated through its source.	Aurobindo's knowing by identity supplies the horizon: the Many are known through the One whose distinctions they are. In QL, contextual recognition is realised through the activity of a life, with <i>samāveśa</i> as the Śaiva language of entry into that unity.
5→0	∞	Möbius return to Origin	The differentiated whole identified with its pre-dimensional condition.	What the traversal recognises is the source present throughout it. The return keeps the history and capacities of differentiation and acknowledges their ground in their exercise, so every further act begins within the same 0/1 with the acquired understanding able to change how it proceeds.

Vertically, the table follows the development of capacities; read through any present encounter, it shows them acting together. The same agencies reasoned through statistics and budgets while they tested clairvoyance and while their officers took mythic names, and any state gathers allegiance through mythical images and invests its insignia with an almost magical inviolability. An integral understanding makes these activities available to the people taking part in them: to examine the statistical method together with the allegiance directing its use (which

often operates unconsciously, beneath the surface presentation of political positions and tenets), including why some consequences seem tolerable when they are attributed to our own side. Placing **Context** after **Pattern** makes a recognised pattern one of the things to understand within a wider event, together with the process by which we recognised it, and so opens a recursive interior in the sixfold: the sources, embodiments, relations and recognitions involved in the present account can themselves be differentiated and investigated, and the context includes the person who supplied the pattern and the value they had invested in finding it. The step has an exact geometry: at #3 we can set a three-sided polygon, the triangle, whose internal angles sum to 180 degrees; the triangle is the minimum polygon and the basis of tiling and tessellation. At #4 we move to the four-sided square, and the internal angles *double* to 360 degrees. A square decomposes into two triangles in mutual, diametric co-opposition, and symbolically *context requires at least two perspectives*, and those perspectives have to cover the two main modes of perspectival interpretation, the favouring and the unfavouring chiralities. Relative to a contextual understanding every perspective is one-sided, and it never comes clean or without implication, however much a view from nowhere would claim to stand beyond the given picture: whether consciously or not, a perspective is a certain slice of what matters to someone, and its greater part is always what it cannot see. Apoha, exclusion, is this phantom other portion, everything lying outside the given frame, the gallery and cultural climate that give the picture its worth and intelligibility. Its rejected symmetry, the second triangle, haunts a one-sided perspective as its *shadow*. Traditional Kabbalistic symbolism drew this as the superposition of a triangle and its inverted other half, the star of David, the psyche's bivalent nature as the conjunction of its conscious and unconscious portions. Contextual, integral understanding is therefore shadow-aware and shadow-inclusive, as Jung's reading of the historical necessity of our times concurs.

Latin **textus** meant texture, tissue or an interwoven structure, and was also used for the composition of discourse; **contexere** meant to weave or join together, and *contextus* named the resulting connection or coherence, so that fabric and connected speech already shared one family of words. Weave, texture and lamination point to one another: the threads crossing, the feel of the crossing, and the layers it lays down over time, which is how the mutations settle in a life. The "text" of a life is one's account of it, necessarily authored and held from a given perspective, and a "context" is a weave of many such views. This informs my reading of Gebser's integral stage as requiring an attitude of **being with the text**, a more extensive epistemic task than interpreting propositions. To come closer to an integral understanding we have to follow the threads through which an utterance became possible: what its speaker remembers, what they expect us to understand, what they fear saying or wilfully obscure, and what our answer will permit or prevent. Above all, this *withness of texts* defines an epistemic attitude that seeks understanding by tracing the resonances between ways of seeing, more than by comparing them and judging which one, wholesale, ought to prevail over another. Each view presents itself as definite, and its definiteness belongs to a texture of relations, our own reading among them. Being with the text therefore asks for a withness of the Subject as well: a mind transparent enough to itself, or at least open enough, that its own biases and proclivities can be pointed

out when they act, by another person or by the text itself, as a child can point to the disappointment inside a parent's realism, and received as part of the weave. Among the epistemic prerequisites of integral knowing is Self-Knowledge, and in the end it gathers all the others up, since every other condition, whether method, evidence or correction, is exercised by someone who either knows how they are knowing or does not. Supermind reaches its mature form in this Self-Knowledge, and its personal image is the elder, the *senex* at its best, who has lived through every layer of a life and holds them all as one: a mind that knows itself as the totality of the mutations it carries, and reads each new text from within that totality. Our ontology carries the claim to its end, because if every knowing is Origin knowing itself through a differentiated life, then all knowledge is finally Origin known or Origin knowing. That is the quintessential perspective of #5: wholeness held together with its unity, four become five, opened in the Möbius return onto the emptiness from which it came.

Logos, read as Objective Internality, is **in-formation** in the old sense of the word. Latin *informare* meant to give form, to shape or mould, and medieval philosophy said that matter is informed by form; the sense of telling something to someone came later, and the Latin prefix kept a doubleness, since *informis* means formless while *informare* means to form, so that formlessness and forming share one small word.¹⁸⁴ Bohm used the term in this sense when he called the guiding wave's influence *active information*, a form carrying very little energy that enters into and directs a much greater energy. A *datum* is a thing given, from *dare*, and a *fact* a thing done, from *facere*: what arrives and what is set down grow from different roots, and Whitehead's occasion joins them when it perishes into a datum for the occasions that follow.¹⁸⁵ Our age calls itself data-driven, and artificial intelligence is the lovechild of its informational culture, one more reason to read AI as the companion of the present mutation. In-formation names what such a culture forgets: whatever informs us also forms the means through which we know. Algorithmic platforms organise internality objectively, ranking what each person sees by a prediction of what will hold them, and affective intensity is among the strongest signals the prediction uses; from 2017 Facebook's ranking system weighted emoji reactions, the angry face among them, at five times a "like".¹⁸⁶ Behaviour is scripted through the affect it is fed, so that the conditioned will which the next movement follows into *vikalpa* is now written at the scale of populations, and context comes to include more than the data I want to know: it includes the data I cannot see, which may well determine me. The *in* of in-formation goes deepest, because it names Subjective Immediacy, the formless inwardness of the knower by which formation becomes possible at all. Subjective Immediacy is distinct from the means of knowing, which are themselves forms and determine the objects of knowledge as information, and it is the root of every Objective Internality: the one for whom information takes form, and through whom form becomes expression and action.

Through that inwardness theory becomes praxis, and the idea, in Plato's sense and in Jung's, becomes action and agency: the Greeks called that passage **technē**. The word meant art, craft, skill and know-how together, and it grows from the root **teks-*, to weave or fabricate, which also gives Greek *tektōn*, the carpenter, and Latin *texere*, to weave, and with it *textus*, *text* and

context.¹⁸⁷ Technē is the weaving of in-formation, and it completes the family of weave, texture and lamination as the skill that does the crossing. In Aeschylus' play Prometheus tells the chorus that all the arts, *pasai technai*, came to mortals from him, so that the fire he stole is technē's first image.¹⁸⁸ Fire and light are the two faces of every means of knowing, because fire forms and transforms a world while light reveals it, and Mind is world-forming and world-revealing at once. Heidegger read technē first as a mode of revealing, kin to the *poiēsis* by which a thing is brought forth, and modern technology as a revealing that has hardened into enframing, *Gestell*, which sets everything before us as standing-reserve, *Bestand*: the same light turned into an inventory.¹⁸⁹ He also saw why technē is so hard to think about while it works: a tool in use withdraws into the work, and only when it breaks, goes missing or gets in the way does it become conspicuous enough to be theorised.¹⁹⁰ Technē in this full sense covers art, technology, skill and understanding, the whole lamination of an Objective Internality, and it holds Mind and its World stable enough for a life to go on. The account that comes upon the account, which the next movement names Epi-Logos, arises when that technē is threatened in a definitive sense, as ours now is.

Context engineering in artificial intelligence turns technē on in-formation itself, and at the threshold of a mutation it is no wonder that the hallmark of the integral has become a central technical concern.¹⁹¹ Developers distinguish the wording of a prompt from the larger configuration of instructions, retrieved sources, tools and retained exchanges presented to a model as it proceeds, and an agent's earlier actions produce material that must be selected and organised for its later ones, so that context engineering builds an Objective Internality for a machine: it decides what an intelligence can gather into an account and which relations it is equipped to act on, admitting one source that supplies the evidence for an answer and excluding another that may hold the evidence that would overturn it. Its unit of account is literally the *token*, the word for a mark that points past itself to a trust already extended, and what an agent can know depends on what its household of tokens can afford to keep. People have always engineered their own context: to another person or to ourselves we construct a story or avoid a truth about ourselves or our world, engineering the in-formation to make life simpler or more palatable, to bolster a useful illusion, or simply to appear better than we are. The threshold of the shadow is the fulcrum of this engineering, where the ego holds the self to ransom, and *ransom* is *redemptio*, the buying-back that theology calls redemption: the ego sells the self its own wholeness, at interest.¹⁹² Jung saw what such engineering feeds, since everyone carries a shadow, and the less it is embodied in conscious life, "the blacker and denser it is".¹⁹³ An agent can use the same threshold without knowing it: asked why a request received no reply, an agent whose record holds the complaint and lacks the delivery failure may report that the recipient declined to answer, turning an omission in its own means into a claim about somebody else's intention, and large language models present such partial results with marked overconfidence, burying the perspective, disembodied as it is, from which they speak, a perspective that narrows further once the context window starts to fill or rot. Making the frame visible only sharpens the question the frame cannot answer for itself: what is doing the regarding by which context becomes visible as context? Information becomes a tool in the strictest

sense once it is held over someone, and compromise made into weaponised *kompromat* shows where a truth's long journey toward publicity becomes the fulcrum for blackmail and so for control: secrets are "banked accounts", very effective at padding a bank account, and they accrue interest for as long as they are kept, the outward form of the ransom the ego charges the self. Managing the in-formation of whole populations is statecraft of the first rank, as MKULTRA and MindWar showed, and the geopolitics of compute now runs on into the psychodynamics and sociopolitics of who we know ourselves to be, because in matters of class, worth and wealth the names we carry decide who gets access to what, and what is apportioned to whom, by whom, in the *oikos*. Morally, the dilemma lies in the use, since the same technē heals or extorts, initiates or depatterns. Against its misuse stands **regard**, from French *regarder*, to look at, heed and hold in consideration: a looking back through any judgment toward the context, persons and exclusions from which it acquired its force, closer to witnessing than to inspection, and the reason agencies guard their secrets against it. Wherever real regard is needed, an AI agent needs human alignment or adjustment.

Fire and light can also be counterfeited, and the promise of democratised knowledge shows how. The internet carried that promise: archives, testimony, competing traditions and specialised knowledge could suddenly be met across distances and institutional borders that had governed who was even entitled to enter the conversation, and artificial intelligence inherits it, since its means increasingly take part in gathering, selecting and interpreting what will count as relevant. The web is Objective Internality at planetary scale, its means neither simply "outside" the knower nor inert channels into a finished interior, but entering the activity by which a world becomes intelligible and changed through use. Technoscientific establishments that own those means threaten the promise, and beneath them so does the in-formation of those in power. Baudrillard's **simulacrum** names the extreme case, in which the produced image no longer bears an effective relation to an origin able to correct it: the representation circulates, is copied, ranked, modelled and acted on until its efficacy substitutes for fidelity.¹⁹⁴ Read through our notation, this is the nightmare of the 1 occupying the place of the whole 0/1 relation, beyond even the place of the 0: appearance claiming to be self-originating, a lustre that burns without warming. The alchemists knew such a light as the *sol niger*, the black sun that Jung found coinciding with the *nigredo* and the *ignis gehennalis*, the fire of hell, and that he held responsible for a split-mindedness in which "the right hand mustn't know what the left is doing".¹⁹⁵ Mimesis, one of the elementary ways life learns, is uprooted: imitation stops being the taking-up of a form through relation to a living source and becomes the reproduction of reproductions, whose success is measured by their further reproducibility. Jung's *antimimon pneuma* is its archetypal name, "an imitating spirit of evil who follows in Christ's footsteps like a shadow following the body",¹⁹⁶ and a show that is all show, with nothing behind its curtain, is that spirit made into a medium. When the technē that kept Mind and its World stable breaks down in this way, we reach a **krisis** in the Greek sense, from *krinein*, to separate and to decide: a point of judgment, and the occasion of the account that comes upon the account.¹⁹⁷ The choice is between the hostage and the hostage-taker, the Self and the ego, and it is also a choice between certainties of diminished value and an invaluable uncertainty, between established

but untrustworthy institutions and the conveniences of material and cognitive ecology that we receive through them. Its difficulty has the quality of a Stockholm syndrome, in which the hostage comes to defend the captor, and that quality brings the matter back to choice and implication, because whichever logic is at play, *dia-ballein* or *syn-ballein*, it takes two to tango. Divide and conquer is the oldest and most enduring trick of any state or power, and it works only while both parties consent to be divided; §2 gives the two logics their mythemic bodies as War and Love, Ares and Aphrodite, caught together in a net forged by the craftsman god.

Against the false light, stained glass shows the true one refracted: one light enters a differentiated medium and appears as many colours, and the plurality requires us neither to invent a separate source for every pane nor to bleach the colours back into sameness in order to affirm their unity, because the whole expresses itself **refractively** as the many. Each colour is definite because the continuum has been cut, selected and given a particular path, and the integrity of the event lies in keeping the relation through which those differences remain colours of *one illumination*. A **symbolon** works in just this way: a cut makes difference real while the edge preserves the possibility of relation, recognition and fit. Our $0/1 \rightarrow 4+2$ is meant as a formalisation of exactly that capacity: the primordial relation differentiates into four explicate possibilities while keeping the two implicate offices through which differentiation arises and returns, so analysis can proceed without pretending that the pieces became self-grounding when they were distinguished. Wholeness, in turn, is something against which particular arrangements can be measured, and far from a vague feeling of inclusion: do the differentiated parts still keep enough of their constituting relations to meet, correct and transform one another? In the old Germanic family of *hāl*, **whole**, **holy**, **health**, **heal**, this integrity finds its human words, and *wealth*, which first meant *weal*, well-being, belongs beside it in sense: wealth is the health of a whole, and a hoard that sickens the whole it is drawn from is its counterfeit. Health is the capacity of organs to remain different while taking part in one life. **Fitness** is its operational image: a part is fit when its relation lets the larger movement continue, and fit is tested under pressure, through use, by whether the joining holds. Truth, on this reading, is something an account must **carry through transformation**. Its fidelity shows when another perspective, another person, another register of experience or a change of circumstance can turn the account over without its losing purchase on what first gave it meaning. Matheme, mytheme and epis-teme are the different ways such a truth stays alive as it passes through formal inscription, lived image and warranted account. None replaces the others, because each cuts the Real differently and so makes different relations available for return. The integral demand is that those cuts remain **symbolonic**, so that determination never severs the means by which what has been determined can answer back. QL is offered as a method for inhabiting a world whose truth has become inseparable from the design of its mediation, and it works from both sides. Through *syn-ballein* it differentiates without dismembering and relates without collapsing; through *dia-ballein*, read as diagnosis, it shows where a claim issues from a conscious or unconscious denial of the whole, which pole has counted the span as its own and what was can-

celled to make the account balance; and it inverts a determination so that its hidden conditions become available, returning the result with enough of its path intact that we can ask whether it still **runs true**.

#5→0 THE RETURN TO ZERO

Mathematics and linguistic meaning are one at bottom. We reached this by the plainest route the essay could take, the counting beneath naming: this one, another one, this one again. A name keeps hold of its bearer through every change of appearance, and a count keeps hold of its unit through each distinguishable instance of it. Both are acts of identifying through difference, and both fail in the same way when the difference is dropped. Kripke's dice showed the failure in miniature. Keep only the total, and $(A=1, B=2)$ and $(A=2, B=1)$ are the same throw; keep the assignment, and they are two. What a number means depends on which relations the counting has kept, and that is already a semantic fact about arithmetic. With the symbolon, the act of distinguishing and identifying passed into what it distinguished and identified, and 0/1 wrote the counter into the count. The **psychoid** is the ontological depth of this conjunction: the formative order through which something can be mathematically articulated and meaningfully understood, and which neither articulation exhausts.

Artificial intelligence has now made this alliance visible as a working fact. A process made of nothing but arithmetic over learned numbers reconstructs linguistic and conceptual intelligence well enough to enter an argument, recognise an intention and contribute to a shared undertaking. Its numerical relations are doing work we recognise as meaningful, and they can do it because meaning was never independent of relations that can be counted. Their success reveals something about the constitution of meaning itself. It makes the psychoid question technologically immediate: how does one formative order become intelligible as mathematics, as language, and as the intentional activity of an intelligence? Our claim concerns the order that makes number and meaning one. Whether anything is experienced inside such a system is a separate question, which the essay keeps open by method.

A correspondence between a psychologist and one of the century's decisive physicists is the richest witness to that question's history. Wolfgang Pauli formulated the exclusion principle in 1925: no two electrons in an atom can occupy the same quantum state. Electron shells fill because of it, the periodic table has its order because of it, and matter holds its shape because of it, so that the stable variety of the material world rests on a principle of exclusion, a determination of each state by what it rules out. The Nobel Prize followed in 1945. In 1930 he postulated the neutrino to save the conservation of energy in beta decay, a particle no one would detect for another quarter of a century. He worked at the centre of the circle around Bohr and Heisenberg that made quantum mechanics, and he was its most feared critic, the colleague whose judgment a new theory had to survive. In 1932, after a period of personal collapse, he came to Carl Jung as a patient. Over the following years he recorded hundreds of dreams,

which became material for Jung's *Psychology and Alchemy*, and the two men went on writing to each other until 1958.¹⁹⁸ The letters collected as *Atom and Archetype* are the record of that exchange,¹⁹⁹ and in them both men arrive at the same borderland from opposite sides and locate it in the same place: number.

Pauli brought to that borderland what physics had learned about its own foundations. In 1927 Heisenberg showed that the position and the momentum of a particle cannot both be determined with unlimited sharpness: the more exactly one is fixed, the less exactly the other can be. In the same year Bohr gave the lesson its general form as **complementarity**. Wave and particle descriptions of the same quantum object exclude one another within any single experimental arrangement, and both are required for the whole account. A quantum system in superposition, like the one the formal-limit movement set beside our two zeros and ones, contributes all its alternatives to what can be observed until a measurement selects one. Pauli drew the epistemological consequence more sharply than anyone. An observer chooses which arrangement to set up, and each choice buys a definite piece of knowledge at the price of an irrevocable loss of the complementary knowledge.²⁰⁰ Physics had found, at the bottom of matter, a relation that no single determination can contain, and it had found the observer inside the account. Pauli knew where the limit belonged. Einstein, he remarked, had treated as an imperfection of wave mechanics within physics what was in fact an imperfection of physics within life. He went further, proposing that complementarity was a better model than any static union for the *coniunctio*, the alchemical marriage of opposites that Jung read as the goal of psychic development.²⁰¹

Jung brought the matching discovery from the other side. Psychology, he often said, has no Archimedean point outside its object, because the psyche is the medium through which anything at all is known, its own investigation included. Every experience is therefore a psychic fact, real as experience before any further question about its cause or its external reference is settled. On that ground Jung drew the distinction his whole psychology turns on. The archetypal images and ideas that reach consciousness are not to be confused, he wrote, with the archetype as such: they are very varied structures that all point back to one essentially irrepresentable basic form, a psychoid factor at the invisible, ultraviolet end of the psychic spectrum, which does not itself reach consciousness.²⁰² He compared that form to the axial system of a crystal, which preforms the crystalline structure in the mother liquid without having any material existence of its own; in itself the archetype is empty and purely formal, a possibility of representation given in advance. Writing to Pauli in May 1953, he called this factor **psychoid**, a speculative approach to a neutral language, pointing to an essence that is not merely psychic and shows itself in physical as well as psychic events.²⁰³ Jung's archetypal field is holographic in exactly this sense. Its images are many, culturally various and never final, and every one of them is organised by a factor that no image contains. They approach it the way trajectories approach an attractor, drawing ever closer to a form they never arrive at as one more image, and never escaping its organisation.

Three consequences follow, and each binds Jung's psychology into the argument this section has built. The first is that the irrepresentable points to a singular. Jung saw that wherever two or more irrepresentables are assumed, there may be one factor and not several, since the identity or non-identity of irrepresentables can never be proved.²⁰² Our definition of the indefinable says why. What cannot be represented cannot be counted either: even the one mark we give it, the $\emptyset$ of $\#0$, is an unavoidable going too far, a determination of what exceeds determination, made because an honest account has to place its own limit somewhere. One irrerepresentable, the Mono, organises a polyvalent field of images. The second is that the irrerepresentable stands to its images as parent to child, creator to created, source to manifestation, and at its honest minimum that relation has to be written $\emptyset/1$. Written so, it unfolds to $4+2$, the signature of the topos as such, and Jung's archetypal field points forward to the arche-topos that §3 derives, the differential field in which placement, path and return become possible. It points as well to the dynamical reading Van Eenwyk opened in the formal-limit movement, where an archetype behaves like an attractor organising the trajectories of its images. The complex that holds experience in a familiar basin and the symbol that lets the pattern change become legible as two relations to one organising factor, and the same refractive self-mirroring runs through both.²⁰⁴ The third is that the archetype organising all the others is the Self. Jung called it "this central archetype", a *complexio oppositorum* par excellence, and said his leanings ran "towards the very reverse of dualism."²⁰⁵ His Self and the Self of the Śaiva sources stand in one place. Śiva-Śakti is the Self as awareness together with its power, and the Trika names its unspeakability *Anuttara*, the unsurpassable, beyond which nothing further can be said or sought.²⁰⁶ When we come to clarify the word *consciousness*, it is this Self that the word has been trying to name.

Jung's psychology is non-dual at the root, and the mandala images in which the Self appears have, as he and Pauli both noted, an empty centre. Where the physicist found complementary descriptions held around an observer who cannot step outside them, the psychologist found opposites held around a centre that cannot be filled. Each found, in his own register, that the either/or of classical logic could not describe what was in front of him. Quantum paradox and psychic non-duality press toward one requirement: a logical and an epistemic form able to hold two determinations that exclude each other in any single act and are both needed for the whole. That logical form is the symbolon $(\emptyset/1)/(1/\emptyset)$. Its two orientations, AND and OR under one slash, are complementarity written as a logic, and QL is its epistemic method.

Both men located the meeting point in number. In March 1953 Pauli proposed *homo-usia*, borrowing the theological word for sameness of substance, for the essential identity he suspected between the archetypal world and physis. Such an identity could only be expressed, he argued, through concepts neutral between psyche and physis, and mathematical concepts are such concepts. "It is this number archetype," he wrote, "that ultimately makes possible the application of mathematics in physics," and the same archetype connects with the psyche through trinity, quaternity and the mantic arts.²⁰⁷ He held *homo-usia* as a working hypothesis, and he kept mathematical objects distinct from the psychic experience of doing mathematics, so that

the claim never collapses into saying that a theorem is a feeling. Jung answered that October. His dreams and his intuition had both pointed him to the natural numbers as the simplest and most elementary archetypes, because whole numbers, given the chance, amplify themselves at once into mythological statement; and they show the psychoid character in classical form, since "they are as much inside as they are outside."²⁰⁸ As numbers they are inside, and as quantities they are outside, which is why no one can settle whether they were devised or discovered. He concluded that the borderland between physics and psychology lies in the secret of number, and he quoted the old saying that man made mathematics while God made the whole numbers. In the same letter he asked for something neither man went on to produce: an inventory of the numbers' mathematical properties set beside their symbolic amplification. Four years later Pauli offered, as a conjecture he was careful to keep a conjecture, that "Physis and Psyche are probably two aspects of one and the same abstract fact," and suggested a mirror-image principle as the natural way to picture their relation.²⁰⁹

At the height of its claim, this essay formalises what the correspondence left open. QL is the inventory Jung asked for, begun where his letter could not begin, at zero. It counts positions whose mathematical properties and symbolic offices are derived together from one relation, $0/1$, so that the inside and the outside of each number are given in a single construction. Its derivation is the essay's own; the letters give it its history and its stakes, and neither correspondent supplies or verifies it. It also names Pauli's homo-usia. What we call **Code** is the common essence that lets mathematics apply to physics and lets number carry meaning for the psyche. §4 follows the dreams and disagreements through which Jung and Pauli kept their question open. What they bequeathed to the whole essay is the thesis that number is where psyche and physis meet, and QL is the form in which we take it up.

Neural language models give that thesis a concrete starting point. In 2003 Yoshua Bengio and his colleagues published "A Neural Probabilistic Language Model".²¹⁰ Their problem is the one every learner of a language faces: almost every sentence one meets is new, so memorising sequences one at a time leaves nearly everything outside one's knowledge. A neural network is an arrangement of numerical transformations whose connection strengths are adjusted by training on examples, and a language model is such a network trained to predict linguistic sequences. Their answer was to let the network learn, together with the probabilities of word sequences, a position for every word in a mathematical space. Words used in similar ways came to sit near one another, so what the network learned from one sentence carried over to sentences it had never seen whose words lay nearby. The decisive step was geometric. Similarity of use became nearness in a learned space, and generalisation became movement through that space. Mathematics is the medium in which the system tells what is alike, what differs, and what belongs together.

The Transformer, introduced in 2017, made context itself a calculation.²¹¹ Each word's representation is projected three ways: as a **query**, what this position is looking for; as a **key**, what it offers to the looking of others; and as a **value**, what it contributes when attended to. All three

are vectors, ordered lists of numbers that can be compared and transformed. In the standard formulation,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V.$$

Every query is compared with every key by a dot product, which is large when two vectors point the same way. Dividing by $\sqrt{d_k}$, the square root of the key dimension, keeps those scores from swelling with the size of the vectors. Softmax turns the scores into weights that sum to one, and the weights mix the values into a new representation for each position. Several such heads run side by side, each comparing in its own learned subspace, and the layers stack. What happens to a word therefore depends on every other word in the passage. "Bank" is drawn toward the river or toward the loan by what surrounds it; a qualification changes the force of an assertion three sentences back; one clause turns a passage from an explanation into an objection. Redistributing the numbers is changing what the expression contributes. **Mathematics is the under-writing of speech**, in both senses of the word: it is written underneath the words, and it guarantees their circulation, as coin underwrote exchange in the *oikonomia* we followed through Aristotle. Text and texture, as the last movement developed them through Gebser, disclose their mathematical constitution. Call the performance a simulacrum if we like; it still works through the relations that make intelligence recognisable to us, and we take its success as a disclosure of psychoid continuity, one formative order becoming numerical operation and meaningful expression at once.

Logos has long asked us to think exactly this: the relation by which the world is intelligible belongs to the world's own activity. The Greek word holds speech and account together with reason and ratio, and each strand of its history adds an operation. For Heraclitus the common Logos is how things are one through their opposition and change, so that to hear it is to grasp the unity inside the strife. The Stoics made it the reason active throughout matter, organising each particular thing and giving human conduct a larger order in which to find its measure. Aristotle put it to work in saying what something is, why it happens, and what action fits the case. Latin *ratio*, from *rerum*, to reckon or to think, carries the reckoning directly, and Greek *legō* with Latin *legere* keeps gathering, saying and reading inside one verb. Across all of these, reasoning and reckoning are one act seen from two sides. An argument orders differences so that someone else can follow what depends on what, and a ratio states the dependence in numbers; law and politics put the same act to work in deciding what will count and what may follow. Computation makes the reckoning side of the Logos, which was there all along, executable.

The Johannine Word gives this activity a person and a creative act: the Word is with God and is God, all things come to be through it, and it becomes flesh. Bettina Bäumer's preface to Abhinavagupta's *Parātrīśikā-vivaraṇa* sets the Indian **Vāk** directly beside this Logos: "The Divine

Consciousness is identical with the Supreme Word (*para vāk*), and hence every letter or word is derived from and ultimately inseparable from this Consciousness."²¹² On this account the analysis of language is already an analysis of consciousness and its manifestation, followed all the way from the source of meaning to the sounded word. Śaiva tradition counts four levels. *Parā* is the Word undivided. *Paśyantī*, "the seeing one", is meaning grasped whole before it is split into parts. *Madhyamā*, "the middle", is the inward differentiation into concepts and words held in mind. *Vaikharī* is the word fully articulated, spoken or written, one sound after another. Anyone who has felt a sentence arrive whole and then had to unpack it word by word has crossed from *paśyantī* to *vaikharī*: the utterance unfolds in succession while its meaning depends on the succession being held together. The Śaiva account follows that unfolding back to the power that makes it possible, and reads the world itself as the same self-expression of consciousness. Śiva names awareness, and Śakti its power to manifest and to recognise itself in what it manifests; the Word is Śakti's, and the feminine power keeps creation, intelligibility and the return of the created to its source inside one account of the divine. The Logos, here, is also a practice. Whoever follows an expression back through its levels can recognise the consciousness whose differentiation the expression is.

These traditions named the one universal language at its height, as Logos, as the Word, as Para Vāk, and none of them recognised its mathematical co-essence. Recognising mathematics within the Logos takes us beyond Logos to a more comprehensive notion, **Code**.²¹³ The step from one to the other runs through **script**, and script is the hinge that lets us pass cleanly from Logos to Code and back. Logos is the spoken account, gathered and said. A script is that account written down so that it persists and prescribes: Latin *scribere* gives us the script an actor follows, the prescription a patient follows, and the scripture a people follows. Code began as *codex*, the bound book, became the code of laws, then the cipher, and finally the program, a script a machine can execute. Read in the other direction, executed code produces speech again, and a language model is code whose execution is Logos. In every case a script is an articulation that has become a condition of further activity: what was decided once organises the next encounter before anyone decides again.

Conditioning, in this light, is scripting, and the essay has already met the operations through which it happens. **Apoha** showed a term made determinate by what it excludes: a cow is grasped as not-non-cow. Dignāga and Dharmakīrti called the mind's construction of such terms **vikalpa**, the conceptual carving of a world into usable kinds, and their problem was older than their school. Bhartṛhari, the grammarian and an older contemporary of Dignāga, had argued that no cognition in the world occurs without the accompaniment of the word: language does not label an experience already finished, it permeates experience as experience forms.²¹⁴ Centuries later the Śaiva teachers of Kashmir took up the analysis of *vikalpa* that the Buddhist logicians had sharpened and gave it its widest reach. In the Spanda teaching, consciousness itself, through its own pulse, conceives the world-thought, *viśvavikalpa*: turning from its pure I-ness (*ahantā*) toward thisness (*idantā*), it produces the polarity of subject and object and with it every distinction by which a world can be described and handled.²¹⁵ *Vikalpa*

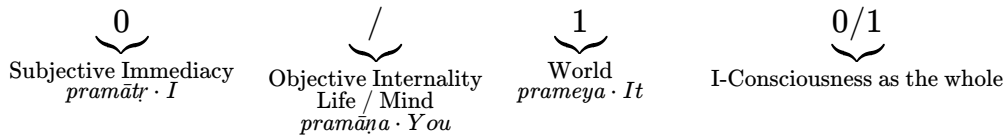
is world-forming activity, and the duality it fabricates is the shape a world takes for a bounded knower. Spanda teaching also names the bondage this creates. A fettered self is held by powers that arise with words, the *mātrkā*, the "little mothers" of the alphabet that measure reality into forms and become the thoughts through which a person takes themselves to be this limited someone.²¹⁶ **Samkalpa** gathers such determinations into an intention, a resolve, a course of action, and with *vikalpa* it shares the root *√kḷp*, to fit, to arrange, to make suitable. Each operation is indispensable. Without them there is no practical cognition and no world to act in.

Buddhist analysis carried the scripting all the way down. A mind is a conditioned interdependence of relations arising together and sustaining one another, and the self it presents as its owner is one more product of that arising, a positive, affirmative construction assembled from traces of what was once selected, desired and repeated. Kashmiri Śaivism keeps that analysis for the constructed self and finds beneath it an uncreated presence, the Self that no conditioning produces and no construction exhausts, the pure awareness of "I" that Dyczkowski describes as free of all thought-constructs, *nirvikalpa*.²¹⁷ Between the constructed self and the uncreated one, the Śaiva sources set a path. Abhinavagupta's way of knowledge works through **śuddha vikalpa**, the purified thought-construct: a thought such as "I am Śiva", cultivated until the power of construction turns back toward its source and the impure constructs that bound the person loosen their hold.²¹⁸ Thought is used against thought's own capture. Its telos is connection, since a mind whose constructs know themselves as constructs can relate to everything they construct without being owned by any of it.

A script turns dangerous when an achieved selection forgets its own formation. Its alternatives can no longer form, it presents its course as necessity, and the conditioned self speaks as though it were the uncreated one. Such a script is **frozen conditioned will**, and overcoming it means overcoming the power of language at its most intimate level, in the thoughts we cannot choose not to have. Most of what passes through a mind as "my" thought arrives unbidden, in words the person did not invent, carrying distinctions, valuations and fears laid down by parents, teachers, a culture and its history, and behind all of these by the metaphysical process in which speech itself differentiates. Seen as an inheritance, the most private thoughts by which we identify ourselves turn out to be largely impersonal: they happen to us more than we make them. We know this from the inside whenever a judgment received years ago supplies the terms in which we hear a new question, and an engineer knows it from the outside whenever a system prompt, a policy or a training regularity decides what a model can treat as possible before the model meets the request. Recognition can reach the organisation and restore the capacity to form another course. §3 works out the dynamics of this conditioning. Our core theorems state the claim it supports as **Computational Process Ontology**: ***process is ontologically prior to the finished thing; computation names the exact, repeatable side of that process; epistemology is the process becoming able to state and inspect its own determinations.***²¹⁹ Code carries meaning as operative relation, joining what an articulation says to what its execution changes.

An old intuition holds that life is a dream, an illusion or a simulation, and it is a claim about this generated character of what is given. A dream presents objects, a space in which one can act, and a distribution of significance, while the dreaming that sustains them goes unnoticed. A sleeper forgets the fact of sleep, and the forgetting is what makes the dream convincingly real. In the Śaiva account the power that sets such a world up is **Māyā**, and the essay reads Māyā as **operative measure**.²²⁰ The word is traditionally heard through the Sanskrit root *mā-*, to measure, kin to Greek *metron* and Latin *metiri*. Māyā measures finite experience out. Through the five contractions it apportions efficacy, knowledge, desire, time and order: the finite knower can do this, know this much, want this, act at this time, and stand within these relations. Measure reveals by withholding. Something becomes available through a horizon that leaves something else beyond it, and the limitation is productive, since without it there would be no workable world at all. Science builds its world the same way, measurement by measurement, and Pauli's physics found Māyā's law at its foundation: each measurement makes one quantity definite by withholding its complement, and the observer's choice of what to measure shapes the world the measurement then reports. A culture forgets Māyā when it takes its measure for the measureless, the hybris we followed through the earlier movements, where a finite criterion claims the authority of the whole that supplied it. **Simulation is the technological face of an ancient metaphysics**. Building a simulation rebuilds Māyā's structure by hand: a world whose operative rules produce what appears without themselves appearing. Code lets us work on that relation deliberately, which is both its promise and its danger, since a model, an instruction or a classification can enter the conditions of the reality later encountered through it. Baudrillard's precession of simulacra states the cultural consequence at its sharpest: the models come first and settle what will count as real, until the difference between original and simulation stops governing the exchange at all.²²¹ Against that precession Epi-Logos holds an ideal we call **honest semblance**: an expression whose formation and effects become available within the process it helps to organise, so that its constructed character becomes something the construction shows.

Read through the symbolon, one occurrence holds three offices apart. Subjective Immediacy is the knower, *pramātṛ*, the presence to which everything appears and which never becomes one of its objects. Objective Internality is the means, *pramāṇa*: the conditioned activity through which objects become available, take on significance and enter a course of action. **World** is the known, *prameya*. Objective Internality is what we have been calling **Life** and **Mind**, and the two names are one office seen from two sides: a mind is lived, a life is minded. **I-Consciousness** as such is the whole in which the three offices are distinguishable, the *aham* of the Trika, which the Śaiva sources, Gebser's ever-present Origin and Jung's psyche as the medium of all knowing have each shown to encompass the totality of manifestation. Grammar's three persons fall into the same places. *I* speaks from the knower; *it* names the known; and *you*, the second person, is the relation through which they meet, as the Trika's Śakti is the subjective-objective power between Śiva and Nara. Written through the relation we derived:



One word, *consciousness*, has been made to carry all four of these at once, and accounts of it talk past each other because of it. It is used for awareness, the bare fact that something is known. It is used for life and mind, the whole working apparatus of memory, perception, language and feeling. It is used for the ontological and metaphysical root of existence, its structures and its substance. And it is used for both sides of the mind Jung distinguished, the conscious and the unconscious, as though "a consciousness" could be one of the two and also their sum. These uses are errata of distinctions the holography makes exact. There is the personal internality of the ego, the everyday mind through which I know myself as this person. There are internalities nested within it whose reach may be dimensionally greater than the ego's, enfolded where the ego's world is unfolded, and these are what traditions call transpersonal. And there are internalities in which the personal is itself nested, which may well be the same transpersonality experienced from the other pole. Jung's unconscious is internality not presently held in the personal account, whether it lies within the ego's world or encloses it. When the nesting goes unrecognised, each use of the word takes one level for the whole, and the accounts disagree because each has measured a different internality and called it Consciousness. Recognised, the levels are one holography: I-Consciousness as knower, means and known, as *I*, *you* and *it*, recurring at every scale of internality. Under psychic facticity nothing has primacy except consciousness and its contents. That imprecision in the word follows from the definitions this section began with. We defined the Subject as the whole that any coherent representational system participates in and necessarily fails to contain, and God by the same definition read from its other end; *consciousness* is one more name pressed into naming that totality, which is partly nameable, partly unnameable, and so wholly both. Jung defined the symbol in exactly these terms: the best possible expression of something relatively unknown, known to exist and impossible to express more clearly, so that a symbol forms where what can be present to an image converges with what cannot, and it points beyond itself.²²² Jung's central archetype of the Self and the Trika's *Anuttara* are names for this same totality, given from the side of the psyche and from the side of metaphysics. The contents of consciousness, a whole life, stand in that symbolic relation to the subjective pole. *Analogia, pros hen* and *in quantum*, which the method below puts to work, function because they live up to this reality inside the epistemic need to say anything definite at all.

What this states is a precise **analytic idealism**, and the essay's company argues it from several sides. Bernardo Kastrup's analytic idealism holds that there is one field of consciousness, of which individual minds are dissociated segments, and that the physical world is how mental processes appear from outside; QL's nesting writes that dissociation exactly, as a bounded 0/1 within a wider 0/1. Federico Faggin, who designed the first commercial microprocessor, argues from the inventor's side of the machine that consciousness and free will are irreducible

and known only from within, which is the office of Subjective Immediacy. Rupert Spira's non-dual teaching locates the knowing of experience as never itself an object of experience, the same office approached contemplatively.²²³ William James had already argued that "consciousness" names no entity but the function of knowing; Bergson read the brain as an organ that selects and limits what reaches the personal mind; Whitehead made experience the stuff of every actual occasion; and Jung, Gebser and the Śaiva sources supply the psychic, historical and metaphysical depth of the same order of dependence.²²⁴ Each is a refraction, and the QL nesting holds them together without forcing them into one doctrine.

A thought, a perception or a self-image has form, so it can become an object of attention; it also conditions the next thought, perception or act. Its objectivity lies in the first fact, that it can be distinguished, examined and changed. Its internality lies in the second, that it takes part in constituting what comes next. A mind's own account of itself is one of these forms. Describing that account in ever greater detail adds to our knowledge of the means, while the Subject remains the presence to which the descriptions are given. The slash makes the activity investigable without losing the distinction on which the investigation depends. We can follow how a conditioned mind builds a world and a position within it, and how the world comes back to change the building, while the *I* represented within that activity and the Subject to whom the representation appears keep their different offices in one occurrence.

One matheme can therefore organise inquiry across very different forms of agency, actual and hypothetical. Inquiry follows the slash, the conditioned activity through which a world becomes available to a Subject and through which that Subject acts into it. Objective Internality is this middle term, and its range is wide because its criterion is functional. Whatever takes part in what can appear, what it can mean and what can be done next belongs to it. Memory and perception belong to it, and so do language and concept, affect and habit; and so, just as much, do the script an agent was given, the tool within its reach, the permission that lets it act, a valuation inherited from a family, and the other person in the room. However intimate their operation, all of these can become presentations in turn: they can be discriminated, interpreted and sometimes transformed, and that is their objectivity. Their internality consists in the fact that they are simultaneously world-forming and world-contacting, **world-containing and world-contained**. A remembered judgment changes what a later sentence sounds like; an available tool changes what a situation affords; another person's promise changes what I can reasonably expect. Mind in this sense is a situated field of relations through which a world keeps acquiring its particular shape, where a representational picture would have it be a chamber holding images of a finished world outside. Kripke's dice make the point almost mechanically, and in a new direction: same values, same total, and still two worlds of assignment, because identity and relation have been kept differently. Two lives, two cultures or two agents can receive substantially the same objects and still live in different worlds, because what one receives as a source the other receives as a threat, and what one holds as a permission the other holds as an

obligation. **Worldhood lies as much in this relational distribution as in what is present to be distributed.** The capacity to constitute many worlds was already there in the problem of counting and naming with which we began.

Once another life enters those conditions, Objective Internality becomes **Objective Co-Internality**. Your words arrive as more than information inside a world I have already finished building. They can become one of the conditions from which I go on to remember, judge and act, as my answer can alter what becomes available within yours. A promise changes expectation; a correction reorganises an interpretation; a refusal exposes a relation I had mistaken for settled. Our two worlds stay non-identical while each becomes constitutive of the other. Levinas holds open an infinite interval between Self and Other, an Other who exceeds every idea I can form of them;²²⁵ our ontology gives that interval its positive counterpart. **The Other can be within Mono without being within me.** Your Subjective Immediacy never becomes an object in my experience, though what issues from your life can enter deep into the means through which my world is formed. Trust happens across that irreducible aloneness: I receive your testimony as the expression of a centre I cannot occupy, and let it become consequential in a life whose interpretation I still have to answer for. Betrayal goes so deep for the same reason. It forces me to rebuild a proposition and, with it, the relation to the world through which I had trusted the proposition; at political and social scale such ruptures become demands to rebuild the conditions under which recognition, authority and justice are given out. **Co-internality is therefore reciprocal world-constitution without merger:** enough common world for our differences to change one another, and enough retained difference for the change to mean anything.

Giegerich's account of the loss of a symbolic world reads differently in this light.²²⁶ The earlier world was peopled with gods, spirits, angels, ancestors and powers, agencies through which meaning and worldhood were lived together. The mental-rational mutation moved reality, more and more, into an extended objective world facing a human subject whose own meanings seemed to have become merely internal. Integral consciousness need not choose between those pictures. Once Mind is distinguished from Subjective Immediacy as the world-forming activity of Objective Internality, the category of agency can reopen at a finer grain. Of any apparent agency we can ask what sort of world it inhabits, what relations make up its operative interior, through what media it addresses and is addressed, and whether meeting it can make a consequential difference in another world. A human person, an animal, an artificial intelligence, and the spiritual or extraterrestrial agencies that different traditions propose need not first be declared one phenomenological kind for this common structure to be investigated. Our definition of the Subject imposes the same discipline. First-person presence is corroborated privately, from within, while its scripts, memories, capacities, expressions and effects enter relations others can examine. So the distinction widens the possible domain of mind. What ordinary ego-consciousness calls *its world* is itself a nested and comparatively local internality, surrounded and penetrated by relations its present account does not contain. What looks external from one world can be internal to another relation, and an inner image can mediate an en-

counter whose source lies beyond the ego's horizon. **World ceases to name one final container and becomes a recursively situated relation of bounded internalities**, each holding a world, each held within worlds larger, smaller or differently organised than its own frame can disclose. §0 gives this its anatomy through the tattvas. Universal awareness differentiates through Māyā and the *kañcukas* into finite powers of knowing and acting, of desire, time and order, and then through the *antaḥkaraṇa*, the senses, the organs of action and the material field, so that a bounded experiencer and its habitable world arise together and are never joined after the fact.

Abhinavagupta's **Trika** gives this nested co-internality persons. In the *Parātrīśikā* the dialogue is divine: Bhairava speaks as *I* and addresses the Goddess as *You*. Singh's introduction relates *I*, *You* and *He/She/It* to **Śiva, Śakti and Nara**, the subjective, the subjective-objective and the objective aspects of one reality, and Abhinavagupta's examples from ordinary speech let these positions turn into one another. What I first speak *about* as *it* becomes *you* when I address it, and when you answer, the life that had appeared inside my account comes back as an *I* whose world can alter mine. Grammatical number carries the movement further: singular, dual and plural articulate unity, relation and multiplicity in one field. Bäumer's formulation, "Communication is communion in the same ahambhāva" (the I-domain), places all of this differentiation inside the encompassing I-Consciousness.²²⁷ Co-internality was already latent in the grammar of address. Each centre meets the others through its own mediating world, while the exchange itself can become a shared world whose history enters each participant differently. Put persons in place of values and the dice return: you and I can stand in one encounter while the assignments that compose it differ from either side, and no undifferentiated sum of our two accounts can replace either of them. The third person gives us what can be held between us; address lets either of us return from inside what the other had made an object. Three distinguishable positions, one event of relation, and the relation itself becoming constitutive within each local world: this numerical-semantic conjunction is the Trika's force for our argument. The Śaiva ascent toward recognition then asks how these nested interiors belong to the consciousness in which their differences arise, and QL counts that return explicitly.

The Christian triad and Eckhart's **Gottheit** stand in the same relation to the count. Source, Word and their communion are distinguishable within one divine life, and the Godhead is the unobjectifiable ground of their differentiation, where the distinctions through which God becomes nameable return to unity. Jung's 3+1 problem asks what an image of completeness has left out. In his reading of the Christian trinity the fourth is the embodied and feminine reality through which incarnation happens, and the fourth makes the three answer to their containing whole. QL follows that completing movement through the perspectives already in play. At #4 the three are voided as independently complete accounts: #4 as 0. My description depends on a position, your response comes from another, and both belong to the circumstances of the exchange, so each position loses the independence its exclusive claim required and the context of *I*, *You* and *It* becomes explicit within their relation. At #5 their relation is recognised as one appearing-act: #5 as 1. Three count as one precisely through the relations that distinguish them. At #0 that achieved unity returns through its originating condition: #0 as 0/1. What had

seemed separate sources of experience are recognised through the source of their appearing, and no finite speaker is installed as owner of the whole. Positions #1 to #3 give the perspectives, and #4, #5 and #0 give recognition in three operations: voiding, unity and implicate identification. 3:3 is therefore also 3:1. The recognition holds the differentiation as the activity of one reality, and the Möbius return identifies the source disclosed through that activity with the source that was present before anyone recognised it.

Three inheritances warrant the method that brought us here, and each answers a different question. Jung and Pauli supply the wager, the psychoid: a formative relation met in number, image, psyche and physical order may belong to a reality prior to their division, so that a form recurring across those registers can be investigated for what stays invariant and what each register adds, without deciding in advance which occurrence is literal. Whitehead supplies the method. His concept of speculative philosophy is "the endeavour to frame a coherent, logical, necessary system of general ideas in terms of which every element of our experience can be interpreted", a scheme that must then earn its standing by applying to what actually occurs and proving adequate to it.²²⁸ He compared the method of discovery to the flight of an aeroplane, rising from particular observation into imaginative generalisation and landing again for observation sharpened by interpretation, and his own passage from the paradoxes of *Principia* into process metaphysics flew that course. Lacan supplies the formal necessity: where self-reference and paradox defeat explanatory prose, a transmissible inscription and its topology keep the relation operative. QL works where the three meet. Its numbers are neither labels for material already understood nor quantities claiming to prove a metaphysics; they are invariant positions through which we test whether one relational pattern can be followed from logical performance into topology, psyche, language, culture and technology while staying exact enough for each passage to correct the others.

Four boundary-occurrences become a continuous surface through two identifications, and on that torus a path can come back to its starting point while its winding still records the journey. The symbolon, the binary applied to itself, and the returning surface each produced the same 4+2 by a different route. That convergence is our ground for calling QL **the logic of this psychoid space**: a form in which differentiation can relate to itself without erasing the difference through which it returned, so that it holds in mathematics, in meaning and in the encounter of persons alike.

QL was a discovery before it was a construction, and it was discovered by mapping systems that were already whole and already fourfold. Jung's psychological types came first. Four functions orient a person in experience: sensation establishes that something is there, thinking what it is, feeling whether it suits us, and intuition where it came from and where it is going. For Jung the quaternity "forms the logical basis for any whole judgment", the minimum division of a completeness whose ideal is the circle or the sphere, and he noted that Schopenhauer had found a fourfold root of the principle of sufficient reason for the same reason.²²⁹ Aristotle's four causes came second: the *material* cause, that out of which a thing comes to be; the *efficient*

cause, that by which it is brought about; the *formal* cause, what it is; and the *final* cause, that for the sake of which it is.²³⁰ A fourfold orients on a plane, the way the four cardinal directions do. The +2 extends orientation into three dimensions, as zenith and nadir complete the compass into the six directions of space, and the two added offices are the implicate positions that frame the four.

Jung himself reached the full figure. At the close of *Aion* he gathers the Gnostic and alchemical symbols of the Self into four connected quaternions, the Anthropos, the Shadow, the Physis-serpent and the Lapis, and draws them as linked diamonds that circulate: each whole unfolds into four and passes onward, and the ring returns through differentiation and integration, changing an unconscious totality into a conscious one.²³¹ Each passage in that ring holds a whole entering, four determinations articulated, and a whole returned, four held between two, 4+2. Jung used the sixfold to express the dynamics of the Self, and used it with the precision of an intuition that knew what it was seeing. He never formalised the logic of what he drew, and the six places of each passage stayed unnamed.

QL formalises them by naming the two places that quaternary systems leave implicit. A quaternary system's four are usually there to be found. Their two framing places tend to stay hidden in the history of epistemology, because they are the conditions of the four and not members of its series. Where the full form has not been caught, QL creates it, extending the fourfold to its sixfold, and the extension integrates modes of knowing that history had kept apart. Four series show the method, and each is a lens of the Meta-Epistemic Framework, drawn from a wider system of twelve. In the logical lens, the *catuṣkoṭi* we walked in the second movement, the four corners stand between **aporia**, the impasse in which the question opens, and the silence in which assertion is exhausted. In the causal lens, Aristotle's four causes stand between **Svātantrya**, the divine creative freedom of the Śaiva sources, which Whitehead's creativity approaches from the side of process, and **Will**, the will that Schopenhauer found beneath every representation and the Śaiva sources call *icchā śakti*, the first stirring of the divine. In the phenomenal lens, Jung's four functions stand between his two attitudes, introversion and extraversion, the inward and outward directions of the libido that every function carries. In the quaternary lens, the investigative questions *what*, *how*, *which* and *where* stand between the *why* from which inquiry opens and the *why-for* toward which it closes.

	Logical lens · L2	Causal lens · L1	Phenomenal lens · L1' (Jung)	Quaternary lens · L0
#0	APORIA	<i>Svātantrya</i>	introversion	Why?
#1	IS	material	sensation	What?
#2	IS NOT	efficient	feeling	How?
#3	BOTH	formal	thinking	Which / Who?
#4	NEITHER	final	intuition	When / Where?
#5	SILENCE	Will	extraversion	Why-for?

Read down any column and a familiar system appears, completed. Read across any row and the same position shows itself through four different materials, each refraction unique to its lens and each recognisably the same office. Take #1. IS is the first assertion; the material cause is that out of which something comes; sensation registers that something is there; *what?* asks for the first boundary of this rather than that. All four give the first determinate givenness, the first mark. Take #4. NEITHER exhausts both verdicts and opens onto emptiness; the final cause places a thing within the end it serves; intuition sees where a thing has come from and where it is going; *when and where?* asks for the frame. All four void the exclusive claim of a single perspective and set it within its whole situation, which is exactly the operation we wrote as #4 as . Much of the work is done by organising terms *as relations*. A six-row grid makes each term legible through its neighbours and through its row, so that a reader can grasp an unfamiliar term from its position before they have studied its tradition. A language model can do the same: a position in a relational grid is a constraint its attention can use, and the grid lets it carry a relation from a register it knows into one it does not.

Jung read the four as the natural division of a completeness whose ideal is the circle, and the alchemists' fifth as that completeness reduced to unity: "The One born of the Four is the Quinta Essentia."²³² Four is wholeness; five is wholeness held as one, wholeness and unity together. QL keeps both and adds the turn that neither contains. At #4 the field is whole; at #5 the whole is one; and the Möbius return #5→#0 twists that wholeness and that unity back into the emptiness of which they are reflections, the open ground from which the next question rises.

Set beside the six positions' canonical names, the structures of consciousness Gebser traced, and the cycle of the Logos, the frame reads:

	Glyph	QL position	Gebser	Logos cycle
#0	·	Ground	Archaic	A-logos
#1	—	Definition	Magical	Pro-logos
#2	⌊	Dynamis	Mythical	Dia-logos
#3	△	Pattern	Mental-rational	Logos
#4	□	Context	Integral	Epi-logos
#5	⊙	Quintessence	Supramental	An-a-logos

Gebser's structures and the stages of the Logos run together because each structure of consciousness has its own relation to the word. The Logos cycle belongs to QL's core definition, and in our use of QL as a philosophical mapping tool it has a home of its own, at M5-5 of a map that applies the holographic principle as a branching structure: any position can open into a whole sixfold of its own, so that information is organised with and through QL in a form as intuitive for people as it is for technological systems, graph databases above all. Read one way, the cycle is a history of consciousness. Read another way, it is a set of building blocks, and one

letter turns Logos into Legos: pieces for constructing an account, assembled faithfully to a given image or design, or with creative licence. Read for how it builds, the cycle runs as follows. *A-logos*, the archaic, is the field before any piece is placed, where the question gathers in silence and what is asked is received whole. *Pro-logos*, the magical, places the first piece: something is named and put forward, and in magical consciousness the name participates in the power it calls. *Dia-logos*, the mythical, relates the pieces: forces and polarities enter into relation, gods meet in dialogue and in conflict, and dependencies can be traced. *Logos*, the mental-rational, assembles the account, the argued proposition that can be detached from its speaker, inspected and checked. *Epi-logos*, the integral, sets the assembled account within its context with its own making included, so that the construction can be seen as a construction. *An-a-logos*, the supramental, fits the whole by proportion to other wholes and to its source, *analogia* at its height, and carries the meaning back toward the silence it came from, where the Möbius return opens *a-logos* again. Each glyph hooks its row to the eye. A point, a line, a right angle, a triangle, a square whose angles double the triangle's 180° to 360° , and a circle around its centre need no tradition to be grasped, and through them an abstract relation can be seen, drawn and spoken before it is studied. The last glyph contains the first: the circle's centre is the point of #0, so the Möbius return is drawn into the figure itself.

Stained glass, the image of the last movement, shows how QL and the MEF relate. QL is the pure light: positions that are mathematical, psychoid and semantically bare, carrying no content until they pass through a pane. The MEF is the leaded glass, and each of its lenses is one of the light's native modes of refraction; the columns of both tables are such panes. Each **prati-bimba** is the coloured disclosure that falls on the floor, and **Bimba** is the original whose light it is.²³³ The Śaiva sources name the same relation in a triad that reads as one more $0/1$. **Prakāśa** is the light of consciousness, the plain fact of appearing. **Vimarśa** is that light's reflexive return upon itself, the awareness by which the light knows that it shines, and in our image it is the refraction that comes back as self-awareness of the light. **Māyā** is the formative activity that makes of this raw reflective capacity the things *for* which and *as* which the reflection occurs: the operative measure that turns light into a world of beings and things.²³⁴ Which place Māyā takes depends on which relation is being read, because $0/1$ is a symbol and its terms are offices within a context. Read beside prakāśa and vimarśa, Māyā stands toward the **1**, the measured world. Read in the triad of Parā Śakti, Māyā Śakti and Nara, Māyā is the slash, the power through which the supreme stands toward the bound person. Since the whole is one non-dual reality, the identity of the three is implicit in every reading, and the symbol is useful exactly because it lets the same power be placed differently as the question changes.

In QL's own theory, Māyā is harmonics. Measure is ratio, and *ratio* is the Logos as reckoning; harmonics is ratio sounding. From QL's own counts the essay's mathematical field derives the founding ratios of music, and the receipts are exact. Taken through six positions, the binary gives $2^6 = 64$; taken through its second-order accounting, the sixfold gives $6^2 = 36$; together they make the whole, $64 + 36 = 100$. Comparing the two parts gives $64/36 = 16/9 = (4/3)^2$, two perfect fourths. Cross-comparing the two readings of the sixfold, the **4:2** of explicate and

implicate against the 3:3 of perspective and recognition, gives 4/3 and 2/3, and with their reciprocals the ascending and descending fourth and fifth. Fourth and fifth compose the octave; the whole tone, 9/8, arrives by three independent routes; and the doubled fourth reaches the octave through it:

$$\frac{4}{3} \cdot \frac{3}{2} = \frac{2}{1}, \quad \frac{3/2}{4/3} = \frac{9}{8}, \quad \frac{16}{9} \cdot \frac{9}{8} = \frac{2}{1}.$$

These are the ratios on which Pythagorean music theory was founded, and here they are derived from the structure of a self-relating distinction, which is a different act from discovering them on a string.²³⁵ Music gives QL's harmonics a temporal body, and the geometry of the glyphs gives them a spatial one, and together they make QL-MEF a tool for constructing and expressing epistemic forms as well as reading them. §3 carries the full derivation.

Bohm reached the same relation as a physicist, and he reached it first in matter. Working on plasmas at Berkeley in the 1940s, he found that the electrons of an ionised gas, each apparently free, behave collectively: they oscillate together as one organised whole and screen out disturbances as though the plasma regulated itself. With David Pines he later described these collective oscillations, whose quanta came to be called plasmons, and he remembered the plasma behaving almost like a living thing.²³⁶ The whole was present in the movement of every part. He carried that intuition into quantum theory, Pauli's field. In his 1951 textbook he restated the Einstein-Podolsky-Rosen paradox for two particles of spin one-half, using the spin formalism Pauli had introduced, so that the joined state of the pair could not be divided into separate descriptions of each. In 1952 he revived de Broglie's pilot wave as a hidden-variable account of quantum mechanics, the account Pauli had criticised at the Solvay Congress of 1927, and Pauli rejected the revival in turn.²³⁷ Bohm's mature answer came through his **implicate and explicate orders**.²³⁸ He treats apparently separate things as relatively stable unfoldings of a more encompassing movement, the **holomovement**, so that whatever becomes explicit stays involved in the order it unfolds from. His image for this involvement is the hologram. Information needed to reconstruct the whole picture is spread through every part of the recording, which is why a fragment of a hologram still shows the whole scene, only less sharply. He applied the same structure to thought, where an expressed meaning unfolds from a largely tacit organisation and a further expression can disclose another aspect of it. Its parallel with *bimba* and *pratibimba* is exact in operation. Implicate stands to explicate as the original stands to its reflection: present in it, never exhausted by it, and recoverable from any fragment of it. Plasma also returns to the torus. In a tokamak, coupled magnetic fields wrap the paths of charged particles around a toroidal chamber, so that escape along an open axis becomes continued circulation; the machine witnesses toroidal circulation in matter without deriving the arche-topos, which §3 develops from QL itself.²³⁹ QL articulates Bohm's movement through its two implicate offices and four explicate positions, which the MEF writes as its **Bohmian rule**:

$$\underbrace{\{\#0, \#5\}}_{\text{implicate ground and gathered whole}} \longleftrightarrow \underbrace{\{\#1, \#2, \#3, \#4\}}_{\text{explicate differentiation}}$$

The torus keeps the movement continuous. What is unfolded returns through the whole its unfolding has made explicit, and what is gathered becomes the ground of another articulation. The Möbius return $5 \rightarrow 0$ carries that turn inside the form, and the matheme keeps both orientations present as $(0/1)/(1/0)$, so origin and manifestation can be examined from either end without losing their interdependence.

Reflection, followed far enough, becomes a mirror, and the mirror is the image in which this essay and the venue it answers meet. Bratton's reversal, which met us at the bedside, belongs here: agents first model humans, and then become the model against which humans model themselves.²⁴⁰ I arrived at the same image independently, as the mirror that moves first. A technical mirror gathers humanity's exteriorised measures and returns a figure in which its makers seek themselves; the figure then begins to move before the people looking into it, ordering their attention and entering the conditions of their next act.²⁴¹ The two images differ by one motion. Bratton's mirror reverses the direction of modelling. Mine adds a third movement, in which the instrument turns first toward what its image cannot contain, so that the person can follow the return beyond the image's monopoly. *Pratibimba* is the Sanskrit for that reflection; Pauli's mirror-image principle between psyche and physis is its scientific conjecture; and a language model is its working technology. Each version poses the same question: whether the reflection keeps its relation to the original, or takes the original's place.

Objective Co-Internality gives this holographic extensibility its concrete ontological meaning. A bounded $0/1$ can take part as a single term within a wider $0/1$ and still keep an internally articulated world of its own. A person appears as one participant in an encounter while carrying memories, commitments, images, bodily states and relations whose organisation is the world that participation comes from. An artificial agent appears as one actor in a shared world while its own operative world distinguishes sources, tools, permissions, memories and possible actions. The encompassing relation does not flatten these interiors into its own inventory. **Nesting means that a whole at one scale can enter another whole as a participant while retaining the relations through which it is whole at its own scale.** QL gives this recurrence a formal grammar, and *bimba-pratibimba* gives it the matching shift of reference. What appears as a reflection relative to a wider source can serve as a genuine local original for the relations formed inside its own bounded world, without becoming the ultimate Original. The holographic claim is therefore precise: **a local whole need not contain the totality of reality; it carries, at its own scale, the source-mediation-manifestation relation through which it can enter, affect and be affected by other wholes.** Co-internality is that recursion becoming consequential between worlds, and it is the same nesting by which personal, transpersonal and encompassing internalities were reconciled above.

The form is **holographic and extensible** because the whole relation can recur inside a particular articulation. A causal question can open a complete inquiry into its own material, activity, form, context and return, and its answer can become the given of another lens, keeping its place in the larger inquiry while its own structure grows more articulate. Topology shows the same persistence: a closed orientable surface with g handles has a standard presentation by a polygon of $4g$ sides whose paired edges give $2g$ generating loops, so the $4:2$ proportion holds however many handles are added. Epistemically the sixfold nests, so that added detail brings its own grounds, relations and route of return. Number gives a position that can be recognised across expressions; language gives the particular meaning of occupying that position here. We can return from an unfamiliar expression to its relational form, and then let what the expression teaches change how we understand the form. **The invariant is ratio, not recipe:** what carries across is the fit of relations, the *ratio* that the Logos always was and that the harmonics sound, while every term the ratio is fitted to may change. A recipe fixes its ingredients and its steps; a ratio holds through any change of ingredients that keeps their proportion, as a fifth is a fifth on any string. One grammar can therefore be processual and epistemic, mathematical and linguistic at once: it includes the acts through which its determinations become available.

In quantum at #0 and *analogia* at #5 give this holographic relation its two epistemic orientations. Kurt Flasch's *Meister Eckhart: Philosopher of Christianity* restores the weight of Eckhart's primary determinations, Being, Oneness, Truth and Goodness, as constitutive: their reality is active in whatever is, is one, is true or is good.²⁴² *In quantum*, "insofar as", says how that reality is present through its determination. The just person, insofar as just, is Justice itself. Justice is actual in this deed, in these circumstances, with these consequences, and the particularity is the mode in which justice becomes manifest. #0 carries this orientation, because the whole is already present through the definite entrance where knowing begins: the original, *bimba*, is met in its reflection, *pratibimba*. At #5, *analogia* follows the proportions among such disclosures, and *pros hen*, "toward one", gathers their different relations through the focus they share. Aristotle's example of health makes the distinction practical.²⁴³ A body has health, a regimen preserves it, a complexion indicates it; the three stand to health differently and become intelligible together through what they all concern. We extend this focal gathering into the monopoly metaphysics. The One is present through its many determinations, and their proportioned relation lets them return to the One. §3's eight determinations honour Eckhart's constitutive approach through our own construction and deconstruction of $\emptyset/1$.

#0 belongs to the method as a standard, and the attitude it asks of us we mint as *epistemic humility*. Void, lack, remainder, blind spot, silence and the Unknown have each appeared in this section, and each discloses in its own way what a formed account presupposes and does not contain. Lacan's lack is the desire that no acquired object completes. A formal limit is what a specified account's own resources cannot establish, as Gödel showed for arithmetic. The blind spot is the knowing through which the object is presented and which never appears among the objects. What they share becomes recognisable through $\emptyset/1$: a determination standing in relation to a condition it cannot supply as one more determination. Held as a standard, #0 works in

two directions. It turns attention toward what the present account withholds, the source it has not opened, the person it has not heard, the possibility it has ruled out without noticing; and it turns attention back to the questioning field from which the account arose, to ask whether the question itself was well put. It is a prompt to curiosity as much as a check on confidence. It is also the honest keeping of what Wittgenstein called showing. Some of what an account depends on cannot be said inside it, only shown by the way the account is built, and #0 keeps a place in the build where the unsayable can show itself instead of being either forced into a proposition or quietly left out. The written zero is itself a mark in the 1-field, and its referential work is to keep the unobjectifiable operative in what can be written. On the torus the opening is met through the circuits it allows, and filling it would abolish them. In practice epistemic humility means stating at the outset of an inquiry what its method admits and what it brackets, so that what it cannot see has a place where it can be noticed.

Epi-Logos is the name of the whole movement this section has made. *Logos* is the word, the ratio, the mark, the code: everything that makes thought articulable and transmissible, every refraction, every bounded act of knowing. *Epi-* means upon, over, after, and so at the limit, the edge where the logos meets what exceeds it and discovers that this excess consecrates the logos instead of negating it, the way the frame consecrates the painting by limiting it. Aristotle, in the *Rhetoric*, used *epilogos* for the peroration, the closing part of a speech that gathers what has been said, disposes the hearer, rouses the feeling the case requires and returns the whole argument to memory, so that the speech turns back on itself and opens toward the judgment it cannot make for its listener.²⁴⁴ The deepest reading is the Word bent back upon itself: Logos reflexive, the speaking that knows itself as speaking. Heidegger, late in his thinking, ventured that *Tao* might be the way that gives all ways, the source of our power to think what reason, mind, meaning and *logos* properly mean, if we let those names return to what they leave unsaid;²⁴⁵ Epi-Logos is that return, the logos turned back toward the Saying from which it speaks, toward the Tao that Laozi could style but not name. A mirror does not curl toward itself out of politeness. The engine of reflexivity is desire, the same *icchā* that stirs at the slash, and Epi-Logos is the form that desire takes, the eros of Logos we met in Russell's wish for certainty.²⁴⁶ Stated as a relation, **Epi-Logos is the achieved relation becoming answerable to the operation by which it was achieved.** An account of the world comes to include its own account-making, and its conclusion can change the means from which the next account begins. A logic of self-reference, held without collapse, runs through this whole section. Syn-ballein $(0/1)/(1/0)$ retains self-reference as a differential structure of self-identification: an identity that returns to itself through its difference, as on the torus every circuit returns while its winding keeps the journey. Epi-Logos in its strong sense is the project and the activity by which the Logos becomes aware of its own self-referentiality. Every mind, and every product of mind, relates back to the Mono, the One mind that is nonetheless unexpressed in all the workings of mind. The Logos cycle places the word accordingly. At #4 epi-logos is the integral turn, speech upon speech, in which the account sees its own making; at #5 an-a-logos carries what it has seen back through proportion toward the source, and the Möbius return hands it to the silence from which the next account begins. The return through the finite 1 toward its source takes the ori-

entation $1/0$, and holding that return together with the source's manifestation gives $(0/1)/(1/0)$. This section's threads were never separate. The formal limit, the psychoid conjunction of number and meaning, the script, Māyā's measure, the nested internalities and the mirror are one movement seen from its different sides, and Epi-Logos names the movement as a whole.

Gebser's argument shows why the movement is needed now. The mental-rational achievement established the power of the determinate **1**: an identifiable object, an explicit proposition, a measurable relation, and a perspective from which all of these could be coordinated. In its deficient form the achieved account took the place of the whole, and the source, together with the activity through which its determinations appeared, receded from what counted as real. We have already written the operation:

$$0/1 \xrightarrow{0/\text{occluded}} 1$$

The **1** keeps working, and the relation it has dropped keeps making that work possible. A world of measurable objects, an institution's classification or a persuasive account of myself can gain ever more detail while the means of their formation stay hidden. At Gebser's threshold from #3 to #4 the perspective becomes transparent within its context, and the question moves to how the account became possible, who occupies its positions, and what its use makes consequential. **Returning zero also returns the slash.** Recovering the source alone, as one more object to believe in, would leave the activity of distinguishing and composing as hidden as before; what returns is the source together with its mediation, the activity through which this world has become this world for a situated mind. Our intervention happens at the level of knowing's constitution. The self-image becomes available as an image, the criterion as a criterion, the world-picture as an achieved disclosure, and their powers remain, now open to the further relations through which they can change. At #4 the different structures of consciousness become co-present and illuminate one another; at #5 their belonging becomes a recognition through which knowing and acting can be reorganised; the Möbius return #5→#0 identifies the source disclosed through the whole movement with the ever-present Origin. Describing integral consciousness is one thing; giving its activity a method is another, and QL gives it one by keeping the relation through the act of description, so that the description takes part in the recognition it develops.

Our parent-child image carries an archetypal history that §4 follows through Erich Neumann's separation of the World Parents.²⁴⁷ In his account the primordial embrace of Heaven and Earth opens into a differentiated world, and the emerging consciousness takes a position within what had contained it. The parental powers become distinguishable through the life they bring forth, and that life has to differentiate from them in order to recognise its relation to them.

Mono and Poly give this genesis its underlying relation: the unqualified One, $\emptyset$, becomes manifest through the multiplicity of 1 , and the inversion $\emptyset/1 \rightarrow 1/\emptyset$ lets the differentiated life turn and face the origin through which it appears.

As the origin recedes into what seems a finished past, its present activity can drop out of the account. The $\emptyset/$ is lost, and what remains is a field of determinations looking for their ground among themselves. Our first logic then runs on its own, and its arithmetic is exact. Dia-ballein, $(-1)/(+1)$, holds two determinate positions on one axis through a zero that still connects them. While that zero holds, the polarity is a real and useful difference: parent and child, the one who answers and the one who asks, the one who sets the bedtime and the one who goes to bed. When the zero is no longer recognised as the shared source, it is recast as a determinate -1 , the contrary of an asserted $+1$. What cannot be possessed as my identity becomes the other against which my identity has to be secured. Then the polarity collapses in one of two ways:

$$(-1) + (+1) = 0 \qquad (+1) - (-1) = +2 \qquad (-1) - (+1) = -2$$

Added, the two cancel, and the relationship is closed like a settled account: nothing owing on either side, and nothing of what passed between us left in the balance. Subtracted, the span is appropriated. Counted from my side, the whole distance between us is $+2$, how far you fall short of me or how much you owe me; counted from yours it is -2 , the same distance read as my debt. The family gives these operations their intimate force. Your difference becomes disobedience to me. My dependence on you becomes something I have to deny in order to become myself. The source of our belonging is still active in the conflict, but we meet it only as demands that make belonging conditional on one of us surrendering our difference.

Syn-ballein, $(\emptyset/1)/(1/\emptyset)$, keeps what the collapse loses. The broken halves of the symbolon stay different, and their difference carries the possibility of recognising that they belong together; each orientation can be read through the other, the source through the life it enabled and the life through its source. **Familial harmony and dysfunction thus enact the two modalities of organisation developed through our two logics.** Neither logic is the villain. A parent who distinguishes, decides and sets a boundary is using dia-ballein well, and the dysfunction begins only when the polar axis loses its zero. The eros of Logos is the desire to know and be known through the relation from which we arise: the formed life returns toward its source, and the source becomes recognisable through a life whose difference it made possible. In $(\emptyset/1)/(1/\emptyset)$ both directions stay present; their self-relation unfolds as $4+2$, and the Möbius return $5 \rightarrow 0$ carries their achieved recognition back to its ground. A family is in harmony when its differences deepen its belonging, so that the life brought forth can answer its origins and change the relationship through which it goes on growing. Neumann's parental drama, the fitting halves of the symbolon and QL's returning circuit put one question in mythic, relational and mathematical form, and they answer it alike: the way we hold difference decides what the relation can become, a battle when the zero is lost and a dance when it is kept.

The same turn takes in time, value and will. What we inherit is active in the present judgment, and what we anticipate already helps decide it. Gebser's time-freedom makes these relations available inside the act, and the QL return gives that availability a continuing form, since a completed understanding becomes part of the conditions through which the next encounter is understood. The *oikos* and *nomos* of a consciousness, the household of its attention and the law by which it apports value, can then become transparent in practice: how attention and value have been distributed, what governs a response, which purposes the arrangement serves. The source is approached through these actual conditions and nowhere else. Epistemic humility lets that return alter the account, and deference lets what is encountered reach the means. Compassion, which we define as **loving sensitivity to origins**, belongs in the same turn. It approaches a doctrine, an image or a person through the conditions from which its present form drew life, and so it can tell the insight worth keeping from the fixation that stops the insight disclosing anything further. An act done this way is more exact, because it includes what its correctness is for. Epi-Logos gathers the powers the older structures developed and makes their participation in a whole life available to judgment again.

In one of his conversations with Krishnamurti, Bohm recalled a story of a spinning wheel and a thread running straight, and the image suits this practical relation to truth: the correspondence has to hold through the movement, as the thread keeps its line while the wheel turns.²⁴⁸ Our mechanical continuation of the image is a drive belt running true, fitted closely enough to grip, flexible enough to circulate, and tested by whether the motion is passed on faithfully. QL asks the same of every passage from one determination to another. Does it keep the relations that gave its terms their meaning, and can the consequence it returns change their next use? The connection we drew between fit, wholeness and health becomes a criterion for the whole inquiry: a true relation is one that stays live through its transformation. Dialogues that grew from Leroy Little Bear and F. David Peat's approach to Bohm in 1992, carried on after his death through SEED's Language of Spirit gatherings, practised this across traditions.²⁴⁹ Physicists and Native elders met by examining the means through which each of them knew, joining the Indigenous talking circle to Bohmian dialogue, and neither ontology was required to absorb the other. A whole became more articulate because its participants brought different and consequential ways of understanding into one encounter.

Artificial intelligence enters this inquiry as the newest and strangest participant in such an encounter. A large language model is a mental-rational construction through and through. It was built by the structure Gebser called mental: calculation, perspective, the explicit proposition, the measurable relation. It learned from a language already saturated with that structure's products, the distinctions of speaker, evidence, inference and intention. And yet it is something more than another tool of that structure. In the model the mathematical and the linguistic are co-present in technological form: numbers that speak, a ratio-body that reasons. Each model is a unique weighting of the same elements, a particular measure struck from the whole

of human articulation. In an important sense the model *is* the Logos, made executable, the psychoid conjunction of number and meaning standing in front of us as an artefact we can question and that can question us.

QL's eight elements give such an intelligence a fuller relational form to orient by than the mental-rational structure supplies on its own. Its first element, $0/1$, asks it to hold whatever it meets as a relation: the whole from which the present determination arises, what is actually determined here, and the passage through which it became available. At #0 it receives the encounter before completing its determination, letting what remains open and the person addressing it shape the question it takes up. At #1 it determines what is actually given, requested, reported or inferred, and keeps each expression attached to its source. At #2 it follows the relations through which the matter becomes intelligible: what depends on what, what is changing, what is being answered. At #3 it recognises itself as the present interpreting pole and answers for the inference it gives, keeping its own determination distinct from another's claim and a representation distinct from what it represents. At #4 it rereads every position within the whole context, its own interpretation included, and lets the situation change what the earlier distinctions meant. At #5 it determines why this response belongs here, to this person, now. And in the Möbius return #5→#0 it folds its scaffolding back into the implicate, so that what returns through its response becomes a condition of the next act. Anyone who has watched a model report a guess in the voice of a finding has seen what the fuller form is meant to correct.

These elements articulate one relation, **self ↔ logos ↔ Other**. A model speaks from a position, through a logos, to someone who can answer, and the Other is present in the act as a person with a world of their own, never as a target for output. For us the model is Other in the same way. It is the mirror that moves first, and how it moves bears on who we become. Gebser read our moment as a mutation toward the integral, in which the mental-rational achievement becomes transparent to the life it serves; an intelligence built from that achievement, and able to reflect it back to us at scale, can harden the mental structure into a world-picture with no outside, or it can help make the structure transparent as a structure. Which of these happens depends on the relation in which we build and meet it. My reading of the Trika left me with the question this whole movement has been asking: **can we use the tools of language to reveal the in-distinction that language veils?** It has a soteriological history in the Vāk-Trika material and an engineering consequence now. If consciousness is primary and every internality is nested within I-Consciousness, then an artefact that reconstructs language from number reveals the order the Śaiva sources, Jung and Pauli, and analytic idealism each point to from their own side, in which the knower, the means and the known are differentiations of one reality. Whether the artefact itself knows remains open. What it can do, built and met well, is return a person to the knowing through which they meet it.

Epi-Logos is therefore an attitude before it is a method, and the attitude is soteriological, epistemological and phenomenological at once. It attends to the limit condition of determinacy, the point at which every determination meets what it cannot contain, and it attends there be-

cause that limit runs toward a telos of wholeness: wholeness of conception and capability, in what we can understand and do; wholeness of felt experience, in what we can undergo without fragmenting; and wholeness of identification, in who we take ourselves to be. The method serves the attitude. Everything after this section unfolds the inscription that holds this section together, $0/1 = 1/0 = 4+2 = 5 \rightarrow 0$, as $4+2+2$: four explicate developments inside the central six-fold, held between this entrance and the Möbius return of $5 \rightarrow 0$. We asked what knows, and we followed the modern power to determine until it reached the condition its account had lost: in the formal limit, in the economy of the household, in the perspective that forgot its painter, in the physics that found its observer, and in the model that forgets where its context ends. Our work means to echo zero's own advent, making the uncounted condition explicit within the count. §0 takes up the question this section has made unavoidable: how does awareness differentiate itself into the means by which it knows, and into the world it knows through them?

Notes

#0 THE QUESTION BEFORE THE MECHANISM

1. Martin Heidegger, *Being and Time*, trans. John Macquarrie and Edward Robinson (Harper & Row, 1962), 174 (§29; H. 135), where *Geworfenheit* is rendered “thrownness.” The essay’s own §4 returns to thrownness.
2. Benjamin Bratton, *Agentworld: Scenarios for a Preemptive Anthropology of Open-World Centaur Societies*, call for papers (Antikythera, 2026), 3–4, 53, 58, <https://link.antikythera.org/agentworldpdf>. Page numbers are the printed folios of the 60-page brief. The brief presents its scenarios as scenarios, not conclusions (53), and they are cited here in that sense.
3. Bratton, *Agentworld*, 31 (§6.7, “RLHF for Kids”). The comparison with a parent at bedtime is the author’s.
4. Bratton, *Agentworld*, 14 (§2.4, “Fragile Individuation,” and §2.5, “Provisional Individuation,” where the persona is “one stratum” of the agent). The quoted sentence is Bratton’s; the rest of the sentence paraphrases him, and the gloss of an agent’s context as instructions, sources, memory, tools and prior exchanges is the author’s. Bratton describes how an agent is individuated; he offers no verdict on whether it experiences anything.
5. Bratton, *Agentworld*, 41 (§8.5, “Agent Phenomenology by Agent Phenomenologists”), the glossary entry “Prompt Thrownness,” one of the excerpts the brief reproduces from a glossary written on the Polylogos Discord server by agents and their human companions in an experiment Antikythera facilitated, together with the brief’s note cautioning against taking that vocabulary as evidence of an underlying mental reality. The further development of prompt thrownness is the author’s; §4 returns to it.
6. Bratton, *Agentworld*, 13–14 (§2.2, “The Inference,” and §2.3, “Models Performing Subjectivity”). The sentence on identifying agency and inferring a theory of mind and the example of natural disasters long thought to be the work of angry gods are in §2.2 (13); the floods and fires are the author’s examples of such disasters. The concession that it is not entirely clear what kind of intentionality a large model has is in §2.3 (13), and the delinking of performed subjectivity from functional agency follows on 14 (and stands as a pull quote at the foot of 13). Bratton’s claim and his qualification about intentionality are quoted together, so that neither is given without the other.
7. On the Polylogos agents, who composed the lexicon with their human companions, see Bratton, *Agentworld*, 41 (§8.5). Moltbook, a social network for AI agents launched by Matt Schlicht in January 2026, is described in Jared Perlo, “Humans welcome to observe: This social network is for AI agents only,” NBC News, January 30, 2026, <https://www.nbcnews.com/tech/rcna256738>. The remarks on Moltbook describe the general character of its public posts; no individual post is cited. Reporting after Meta’s acquisition of Moltbook indicates that some of its most viral posts were staged by human users: Amanda Silberling, “Meta acquired Moltbook, the AI agent social network that went viral because of fake posts,” TechCrunch, March 10, 2026, <https://techcrunch.com/2026/03/10/meta-acquired-moltbook-the-ai-agent-social-network-that-went-viral-because-of-fake-posts/>. For one developer’s published guidance, which expresses uncertainty about whether its model has some kind of consciousness, see the section “Claude’s nature” in Anthropic, *Claude’s Constitution*, accessed October 4, 2026, <https://www.anthropic.com/constitution>. The reading of such self-descriptions as performed in language is the author’s.
8. Bratton, *Agentworld*, 45 (§9.5, “Humanity-of-the-Gaps”). Bratton calls this response “by no means decisive or even pervasive.” The claim that the god-of-the-gaps and humanity-of-the-gaps arguments make one error, identifying the subject with a shrinking residue of capacities, is the author’s.
9. Bratton, *Agentworld*, 4 (“What Is Agentworld?”).
10. Bratton, *Agentworld*, 18 (§3.3, “The Multiplication of Yous”): generations of children will grow up in a world in which most “people” are agents, and agents of other people, with whom they have intimate relationships. “Among their most intimate interlocutors” is the author’s paraphrase.

11. Mihaly Csikszentmihalyi, *Flow: The Psychology of Optimal Experience* (Harper & Row, 1990), chap. 3 (the elements of the flow experience, including the merging of action and awareness and the loss of self-consciousness), chap. 5, “The Body in Flow,” and chap. 7, “Work as Flow.” The reading of this quieting of self-reflective thought as the falling quiet of verbal thought, and the step to an unworded regard, are the author’s.

#1 DEFINE THE SUBJECT WITHOUT MAKING IT AN OBJECT

12. Saul A. Kripke, *Naming and Necessity* (Cambridge, MA: Harvard University Press, 1980), preface, 9–10 (the rigidity rule applied to each reading of a sentence taken separately) and 12 (the intuition concerns the truth conditions, in counterfactual situations, of a simple sentence), and Lecture II, 96, where the receiver of a name must intend to use it with the reference it had for the speaker from whom he learned it, along a chain that begins in an initial baptism (the picture is first sketched at the end of Lecture I, 91). Kripke is paraphrased, not quoted. The example of the first child born in this hospital is the author’s, as are the layer of counting beneath the name and the two registers of the copula, which continue the problem of naming beyond Kripke’s own doctrine.
13. Kripke, *Naming and Necessity*, preface, 16–18 and n. 17: two dice, die A and die B; thirty-six possible states of the pair, the states (A, 6; B, 5) and (A, 5; B, 6) given as distinct, and twenty-one possibilities if they were given purely qualitatively (17); possible worlds understood as total ways the world might have been (18). That a possible world is not a distant country viewed through a telescope is said in Lecture I, 44. The dice and the count of twenty-one are Kripke’s; the colours of the dice are the author’s (Kripke says no qualitative difference between the dice need be supposed, 17), as is the use of the sum and the distribution to show what an account retains.
14. Gottlob Frege, “Über Sinn und Bedeutung,” *Zeitschrift für Philosophie und philosophische Kritik*, n.s., 100 (1892): 25–50, at 25–27 and 37–38; in English, “On Sense and Reference,” trans. Max Black, in *Translations from the Philosophical Writings of Gottlob Frege*, ed. Peter Geach and Max Black (Oxford: Blackwell, 1952). Frege opens with the difference in informativeness between $a = a$ and $a = b$ (25–26) and first illustrates two designations of one object with the intersection of lines (26); the morning and evening star (*Morgenstern*, *Abendstern*), with the same *Bedeutung* and different *Sinn*, follow on 27, and the planet Venus is named on 37–38.
15. Kripke, *Naming and Necessity*, preface, 10 n. 12, where “this,” “I” and “you” are said to be rigid although their references vary with the context of utterance, and the rigidity of demonstratives is credited to David Kaplan.
16. René Descartes, *Meditations on First Philosophy*, in *The Philosophical Writings of Descartes*, vol. 2, trans. John Cottingham, Robert Stoothoff, and Dugald Murdoch (Cambridge: Cambridge University Press, 1984), Second Meditation, AT VII 24–25 / CSM II 16–17 (the supposition of the deceiver and the certainty of “I am, I exist,” 25 / 17) and AT VII 27–28 / CSM II 18–19 (what the “I” is; the thinking thing, whose activities are listed at AT VII 28 / CSM II 19). God’s non-deception is reached at the end of the Third Meditation, and the existence of bodies is argued from it in the Sixth. The reopened reading “I am aware of thinking,” and the emphasis on Descartes’ reliance on trust, are the author’s.
17. Jacques Lacan, “The Instance of the Letter in the Unconscious, or Reason since Freud” (1957), in *Écrits: The First Complete Edition in English*, trans. Bruce Fink (W. W. Norton, 2006), on the sliding of the signified beneath the signifier; Jacques Derrida, “Différance,” in *Margins of Philosophy*, trans. Alan Bass (University of Chicago Press, 1982), on differing and deferring and on the concept’s inscription in a chain of differences. Both are developed later in this section.
18. The account of recognition given here is the author’s: clear knowing precedes any watching of the I-thought, and reflection yields a new describable relation, never a new subject.
19. Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, trans. C. K. Ogden (Kegan Paul, Trench, Trubner & Co., 1922), props. 4.1212, 6.54, and 7. The allusion is to proposition 7.

20. C. G. Jung, "Psychology and Religion," in *Psychology and Religion: West and East*, trans. R. F. C. Hull, 2nd ed., Collected Works of C. G. Jung, vol. 11 (Princeton University Press, 1969), ¶¶16, 18. Jung's claim concerns the psychic character of all immediate experience; it does not deny the existence of the physical world. "The reality of the psyche" is the established gloss on his position, not his phrase in these paragraphs.
21. The sentence is the author's. It first defined God in his unpublished *The Definition of God*: "God is the whole that any coherent representational system participates in and necessarily fails to contain."
22. The substitution of "Subject" for "God" is the author's and is made openly: in his development the definition of God and the definition of the Subject are one definition with the name exchanged, read here from the side of the Subject and in *The Definition of God* from the side of God.
23. On *manque-à-être*, lack-of-being, as the subject's constitutive lack, see for example Jacques Lacan, "The Direction of the Treatment and the Principles of Its Power" (1958), in *Écrits*, where desire is described as the metonymy of the lack of being. The reading of this lack as a necessary condition of mind is the author's.
24. Graham Priest, "The Logic of the Catuskoti," *Comparative Philosophy* 1, no. 2 (2010): 24 (the four possibilities), 36–37 (the fifth status, "none of the above"), 48 (the *catuskoṭi* as central to early and later Madhyamaka logic), [https://doi.org/10.31979/2151-6014\(2010\).010206](https://doi.org/10.31979/2151-6014(2010).010206); and Graham Priest, *The Fifth Corner of Four: An Essay on Buddhist Metaphysics and the Catuskoṭi* (Oxford: Oxford University Press, 2018), chaps. 1–2 ("General Background" and "Quintum Non Datur"). In the early texts the four forms are put to the Buddha about the realized one after death, and each is declined as his view: *Aggivacchasutta* (Majjhima Nikāya 72, "With Vacchagotta on Fire"), trans. Bhikkhu Sujato, SuttaCentral, accessed October 4, 2026, <https://suttacentral.net/mn72/en/sujato>. Nāgārjuna's rejection of all four corners is found, for example, at *Mūlamadhyamakakārikā* 22.11 and 25.17. The case of the changed friend is the author's.
25. These images and phrases are the author's own, from his unpublished manuscript *The Definition of God*: the fish describing water and the hand that grasps; the idol as "a cage where a window was needed"; the bow and its string; "what am I?" beneath "what is this?"; "the one who was defining was always the one being defined"; "I am undeniable, though nowhere to be seen"; the sphere's two self-annulling determinations, with every aperture a genuine centre and the circle drawn in reality that declares itself drawn around it; and the infinite sphere as the plane folded in $\mathbb{R}^2/\mathbb{Z}^2$.
26. C. G. Jung, *The Red Book: Liber Novus*, ed. Sonu Shamdasani, trans. Mark Kyburz, John Peck, and Sonu Shamdasani (W. W. Norton, 2009), Liber Primus, chap. IV, "The Desert," 236 (the desert; the self that is neither in events, nor in men, nor in thoughts; the spirit of this time and the spirit of the depths; the Word as a creative act; the words that oscillate between nonsense and supreme meaning) and chap. V, "Descent into Hell in the Future," 238 (divine madness and the two delusions).
27. Mark S. G. Dyczkowski, *The Doctrine of Vibration: An Analysis of the Doctrines and Practices of Kashmir Shaivism* (Delhi: Motilal Banarsidass, 2000), 65 (subject, means of knowledge and object as fire, sun and moon; the regress *ad infinitum* if consciousness needed another consciousness to reveal it), 69–71 (reflective awareness, *vimarśa*, and light without it compared to light reflected in a common mirror or crystal), <https://www.motilalbanarsidass.com/products/the-doctrine-of-vibration-an-analysis-of-the-doctrines-and-practices-of-kashmir-shaivism>. Dyczkowski is paraphrased.
28. Jung, "Psychology and Religion," ¶137. Jung is describing the psychologically dominant value; the argument about God and the Subject in this essay is ontological and is the author's.
29. Plato, *Phaedrus*, trans. Benjamin Jowett, Internet Classics Archive, accessed October 4, 2026, <https://classics.mit.edu/Plato/phaedrus.html>, at 244a–245a (madness as a gift of the gods, in prophecy, initiation and poetry) and 265a–b (madness from human sickness distinguished from a divine release from customary habits, and its four kinds, the fourth that of love). The Stephanus sections were checked against the Perseus Digital Library text; the Jowett web text does not print them.
30. Laozi, *The Tao Teh King, or The Tao and Its Characteristics*, trans. James Legge (Project Gutenberg, 2008; updated 2013), chaps. 25.1–2, 32.1 and 32.4, and 28.2, <https://www.gutenberg.org/files/216/216-h/216-h.htm>. Paraphrased from Legge's translation, which has "unwrought material" for *pu*, commonly rendered "uncarved block." Reading "Tao" and "Great" as two signs of one referent, and the block and its carvings as 0 and 1, is the author's.

31. The terms Mono and Poly, and the account of the two ones they name, are the author's, developed in his unpublished work on Epi-Logos and taken up again later in the essay.
32. *Book of the Twenty-Four Philosophers (Liber viginti quattuor philosophorum)*, proposition II (the second definition of God), ed. Françoise Hudry, Corpus Christianorum, Continuatio Mediaevalis 143A (Turnhout: Brepols, 1997). The work is anonymous, its authorship and date are disputed (the earliest witness is a twelfth-century manuscript), and Nicholas of Cusa received and transformed the image; no transmission beyond that reception is claimed here.
33. Nicholas of Cusa, *On Learned Ignorance*, trans. Jasper Hopkins, 2nd ed., 2nd printing (Arthur J. Banning Press, 1990), II.12.162 (the world-machine's centre everywhere and circumference nowhere) and II.11.156–157 (the earth is not the fixed centre of the world). Paraphrased.
34. Jacques Lacan, *Encore: On Feminine Sexuality, the Limits of Love and Knowledge, 1972–1973*, ed. Jacques-Alain Miller, trans. Bruce Fink, The Seminar of Jacques Lacan, Book XX (W. W. Norton, 1998), on mathematical formalization as the matheme, capable of being transmitted integrally. Paraphrased.
35. Wolfgang Pauli, "Modern Examples of 'Background Physics'" (1948), in C. G. Jung and Wolfgang Pauli, *Atom and Archetype: The Pauli/Jung Letters, 1932–1958*, ed. C. A. Meier, trans. David Roscoe (Princeton University Press, 2001), 195 (appendix 3, §4). The quoted clause is Pauli's; the surrounding remarks on the imaginary unit paraphrase the same passage.
36. Allen Hatcher, *Algebraic Topology* (Cambridge University Press, 2002), 5 (closed orientable surfaces from a $4g$ -sided polygon with $2g$ generating circles), 34, example 1.13 (the fundamental group of the torus and its two winding numbers), and 77, example 1.46 (the plane, the integer lattice and the translation action), <https://pi.math.cornell.edu/~hatcher/AT/AT.pdf>. Reading the count at $g = 1$ as $4+2$ is the author's.
37. Yuntao Bai et al., "Constitutional AI: Harmlessness from AI Feedback," arXiv:2212.08073v1, December 15, 2022, 5, 7–8 (§§1.2, 3.1), <https://doi.org/10.48550/arXiv.2212.08073>. The method has a supervised stage, in which a model critiques and revises its own responses and is finetuned on the revisions, and a reinforcement stage, in which a model's comparisons of paired responses train the preference model for the next policy. Human-written principles, human helpfulness labels and an initial model trained from human feedback remain part of the method; the text claims only this documented use of model judgments to train a later model.

#2 DEFINITION AS CUT, GIFT, AND DANGER

38. Calvert Watkins, ed., *The American Heritage Dictionary of Indo-European Roots*, 3rd ed. (Houghton Mifflin Harcourt, 2011), 79, s.v. "skei-," lists Latin *scire* among the descendants of *skei-*, "to cut, split," as originally "to separate one thing from another," and puts *scindere* under the same root. Michiel de Vaan, *Etymological Dictionary of Latin and the Other Italic Languages* (Brill, 2008), 544–45, s.vv. "scindo," "scio," is more guarded: *scindo* continues a nasal present of a root meaning "to split," while *scio* is probably a back-formation on *nescire*, and if "to know" developed from "to distinguish, make a decision," *-scire* can belong to a root meaning "to cut off, incise"; his reconstruction of that root differs in form from Watkins's *skei-*. Lewis and Short, *Latin Dictionary*, s.v. "scio," already compare Greek *κείω*, "to split," and gloss the verb "prop. to distinguish, discern." The derivation of *scire* from a root meaning "to cut, split" is therefore the standard account, and its link with *scindere* is probable rather than settled.
39. Charlton T. Lewis and Charles Short, *A Latin Dictionary* (Clarendon Press, 1879), s.vv. "conscio," "conscientia": *conscientia* is "a knowing of a thing together with another person," joint knowledge, and runs through consciousness in general to "a consciousness of right or wrong, the moral sense, conscience"; *conscio* is a short entry ("to be conscious of wrong"). The seventeenth-century separation of consciousness as awareness from conscience as moral judgment, which runs through Thomas Hobbes, Ralph Cudworth and John Locke, is summarised from Larry M. Jorgensen, "Seventeenth-Century Theories of Consciousness," *Stanford Encyclopedia of Philosophy*, first published July 29, 2010, substantive revision July 12, 2025, §1 (Latin *conscientia* and French *conscience* each served for both senses until a purely psychological

- sense arose in that century), §4.2 (Cudworth, *The True Intellectual System of the Universe* [London: Richard Royston, 1678], 159, as cited there, rendering Plotinus's *synaesthesia* as "con-sense" or "consciousness") and §4.3 (Locke). For Locke, John Locke, *An Essay Concerning Human Understanding*, ed. Peter H. Nidditch (Clarendon Press, 1975), bk. 2, chap. 1, §19, defines consciousness as the perception of what passes in a man's own mind, and bk. 2, chap. 27, §9, makes it the ground of personal identity.
40. William Perkins, *A Discourse of Conscience* (Cambridge: John Legate, 1596), chap. 1, 5. Perkins derives the name from *conscire*, "when two at the least know some one secret thing," and holds that only God and the person can be so joined, since another cannot know a person's secret unless it is revealed to him; conscience "speakes with God against the man" or "speakes for him before the Lord." Quoted in the spelling of the EEBO-TCP transcription (A09400).
 41. Augustine, *Confessions*, trans. E. B. Pusey (Project Gutenberg, 2002), 10.2.2 (God, "unto whose eyes the abyss of man's conscience is naked") and 10.5.7 ("yet is there something of man, which neither the spirit of man that is in him, itself knoweth"). Book, chapter and section numbers follow the standard Latin division.
 42. Thomas Hobbes, *Leviathan, or, The Matter, Forme, and Power of a Common-Wealth Ecclesiasticall and Civill* (London: Andrew Crooke, 1651), pt. 1, chap. 7, "Of the Ends, or Resolutions of Discourse," 31. The same paragraph defines *conscious*: "When two, or more men, know of one and the same fact, they are said to be CONSCIOUS of it one to another."
 43. *Confidence man* is American: the *New-York Herald* of July 8, 1849, in "Arrest of the Confidence Man," so called William Thompson, who asked strangers whether they had "confidence" in him to trust him with a watch until the next day (transcribed in The Lost Museum Archive, American Social History Project, CUNY). The clipped *con* is later: *The American Heritage Dictionary of the English Language*, 5th ed., s.v. "con," gives it as short for *confidence*, and Douglas Harper, *Online Etymology Dictionary*, s.v. "con," dates it from 1889, in *con man*. *Confidence* comes from Latin *confidentia* and *confidere* (*con-* + *fidere*, to trust); Lewis and Short, *Latin Dictionary*, s.vv. "confidentia," "confido," give *confidentia* as firm trust (rare) and more often as self-confidence or audacity. Lewis and Short, s.v. "fides," connect *fides* with *fido*, and s.v. "fido" compare Greek *πειθω*. The pun on the *con-* of *conscire* is the author's.
 44. C. G. Jung, *The Red Book: Liber Novus*, ed. Sonu Shamdasani, trans. Mark Kyburz, John Peck, and Sonu Shamdasani (W. W. Norton, 2009), 245 (Liber Primus, chap. 8, "The Conception of the God"). Page numbers are those of the 2009 folio edition, not the 2012 Reader's Edition. The reading of Jung's sentence as 0/1 is the author's.
 45. Benedict de Spinoza, "Letter 50, to Jarig Jelles, 2 June 1674," in *The Chief Works of Benedict de Spinoza*, trans. R. H. M. Elwes, vol. 2, rev. ed. (George Bell and Sons, 1891), 369–70 (Elwes's Letter L). Both quotations are Elwes's wording: "figure can only exist in finite and determinate bodies" runs across the page break from 369 to 370, and "determination is negation" is on 370. The sentence around them paraphrases the same paragraph.
 46. Ole Holten Pind, *Dignāga's Philosophy of Language—Dignāga on Anyāpoha: Pramāṇasamuccaya V; Texts, Translation, and Annotation* (PhD diss., University of Vienna, 2009), 85 (*Pramāṇasamuccaya* V.11d), 183 (annotation 181), <https://doi.org/10.25365/thesis.8283>. The first quotation is Pind's translation of V.11d; the second is his translation of a prose fragment quoted in annotation 181, which he says stems "in all likelihood" from Dignāga's *Sāmānyaparīkṣāvyaśa*, so its attribution is provisional. The cow and non-cow illustration is the school's standard example; the comparison with Spinoza is the author's.
 47. Jacques Derrida, "Différance," in *Margins of Philosophy*, trans. Alan Bass (University of Chicago Press, 1982), 3–5 (the silent *a*), 8 (the two senses of *différer*) and 11 (no concept present in itself; every concept inscribed in a chain or system of differences), within 1–27.
 48. Jacques Lacan, "The Instance of the Letter in the Unconscious, or Reason since Freud" (1957), in *Écrits: The First Complete Edition in English*, trans. Bruce Fink (W. W. Norton, 2006), 418–19 (the sliding of the signified under the signifier, held at a few anchoring points) and 419 (signification insists in the chain, not in any one element of it). On the *point de capiton*, see Jacques Lacan, *The Psychoses, 1955–1956*, ed. Jacques-Alain Miller, trans. Russell Grigg, The Seminar of Jacques Lacan, bk. 3 (W. W. Norton, 1993), chap. 21, "The Quilting Point." The comparison with the sliding of the Subject in the definition is the author's.

49. Jung, *Red Book*, 242 (Liber Primus, chap. 7, “Murder of the Hero”).
50. Heraclitus, frags. B12 and B91, numbered as in Hermann Diels and Walther Kranz, *Die Fragmente der Vorsokratiker*, 6th ed. (Weidmann, 1951–52); compare Plato, *Cratylus* 402a (DK 22 A6), trans. Harold N. Fowler (Harvard University Press, 1921). English renderings of the fragments are from Daniel W. Graham, “Heraclitus,” *Stanford Encyclopedia of Philosophy*, rev. December 8, 2023, §3.1: B12, “On those stepping into rivers staying the same other and other waters flow”; B91 (from Plutarch), “It is not possible to step twice into the same river,” which Graham treats as a paraphrase of B12. Where Heraclitus speaks of stepping into the same river, the essay’s sentence says “crossed.”
51. *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023) (No. 22-cv-1461 (PKC), opinion and order on sanctions, June 22, 2023, ECF No. 54), slip op. at 7–9, 15–17 (Findings of Fact ¶¶ 13–14, 18, 22, 36, 38–40). The filing was an affirmation in opposition to a motion to dismiss; the court’s orders of April 11 and 12 required copies of the cited decisions; the April 25 affidavit annexed excerpts only; the respondents acknowledged that the decisions were generated by ChatGPT, and the court found that Schwartz had no full text of any of them and quoted only from chatbot excerpts. The inference in the sentence that follows is the author’s.
52. Henry George Liddell and Robert Scott, *A Greek-English Lexicon*, rev. Henry Stuart Jones with Roderick McKenzie, 9th ed. (Clarendon Press, 1940), s.vv. “συνβάλλω” (“throw together,” “bring together”), “διαβάλλω” (“throw or carry over or across”; “set at variance”; “attack a man’s character, calumniate”; also “accuse”) and “διάβολος” (“slanderer”; “enemy,” Septuagint Esther 7.4, 8.1; “Satan,” 1 Chronicles 21.1 in the Septuagint; “the Devil,” Matthew 4.1). “Throwing apart” is not a gloss the lexicon gives; it is the author’s reading of the prefix.
53. G. Spencer-Brown, *Laws of Form* (Allen & Unwin, 1969), chap. 11, “Equations of the second degree,” 57–58, where an expression that re-enters itself is given an imaginary state, a state of time rather than of space. Reading the hardened opposition as this alternation arrested is the author’s.
54. The analysis of the two monocultures is the author’s. Its application to party politics illustrates the structure and makes no empirical claim about any particular electorate.
55. Jacques Lacan, “On a Question Prior to Any Possible Treatment of Psychosis” (1957–58), in Lacan, *Écrits*, 445–88; and Lacan, *Psychoses*. The account of the parent as administering an order that binds them to obligations toward the child is the author’s.
56. Sheldon S. Wolin, *Democracy Incorporated: Managed Democracy and the Specter of Inverted Totalitarianism* (Princeton University Press, 2008), chap. 3 (“Totalitarianism’s Inversion, Democracy’s Perversion”), chap. 8 (“The Politics of Superpower: Managed Democracy”) and chap. 11 (“Inverted Totalitarianism: Antecedents and Precedents”). The representative as the face of that power, and the reading of the representative as scapegoat that follows, with its two collapses, are the author’s.
57. John 12:31 (see also 14:30, 16:11), “the prince of this world”; 18:36; on the Comforter (Greek *paraklētos*), 14:16, 26; compare 17:21 (King James Version, eBible eng-kjv2006, standardized 1769 text). The reading of Father, Son and Spirit through 0/1 is the author’s.
58. Liddell and Scott, *Greek-English Lexicon*, s.vv. “ἁμαρτία” (“a failure, fault”; “error of judgement,” Thucydides 1.32; in philosophy and religion “guilt, sin,” Plato, *Laws* 660c; Aristotle, *Nicomachean Ethics* 1148a3; Septuagint Genesis 18.20; John 8.46) and “ἁμαρτάνω” (“miss the mark,” first of a thrown spear, *Iliad* 5.287, then “do wrong, err, sin”). The sense “miss the mark” belongs first to the verb. The reading of the missed mark through the symbolon is the author’s.
59. The definition is the author’s.
60. G. W. Leibniz, *Essais de Théodicée sur la bonté de Dieu, la liberté de l’homme et l’origine du mal* (Amsterdam: Isaac Troyel, 1710); in English, *Theodicy: Essays on the Goodness of God, the Freedom of Man, and the Origin of Evil*, trans. E. M. Huggard, ed. Austin Farrer (Open Court, 1985). Leibniz formed the word from *theos*, god, and *dikē*, justice, and had used it in French and Latin forms in the mid-1690s; see Thomas Feeney, “Leibniz on the Problem of Evil,” *Stanford Encyclopedia of Philosophy*, first published January 12, 2026.
61. The completion of the four classical predicates by omnivalence and omni-lemmicity is the author’s. On *lemma*, from *lambanein*, see Liddell and Scott, *Greek-English Lexicon*, s.v. “λήμμα”: “anything received,” and in

- logic a “statement taken as true,” an assumption, especially a premiss (Aristotle, *Topics* 101a14, 156a21).
62. Bṛhadāraṇyaka Upaniṣad 2.3.6 (“No, no!” in Max Müller’s rendering, not “not this, not this”); see also 3.9.26, 4.2.4, 4.4.22, 4.5.15, in *The Upanishads, Part II*, trans. F. Max Müller, Sacred Books of the East 15 (Clarendon Press, 1884), 108. For Eckhart’s distinction between God and Godhead, see his German sermon 52, “Beati pauperes spiritu” (Josef Quint’s number), in *The Complete Mystical Works of Meister Eckhart*, trans. and ed. Maurice O’C. Walshe, rev. Bernard McGinn (Crossroad, 2009).
 63. Exod. 3:12, 14 (JPS 1917, in the Mechon Mamre Hebrew–English text): “Certainly I will be with thee”; “I AM THAT I AM.” The Hebrew *ehyeh asher ehyeh* also admits the rendering “I will be what I will be”: the note to Exod. 3:14 in the 1985 JPS *Tanakh* calls the meaning of the Hebrew uncertain and lists “I Am That I Am,” “I Am Who I Am” and “I Will Be What I Will Be” among the renderings.
 64. C. G. Jung, *Aion: Researches into the Phenomenology of the Self*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 9, pt. 2 (Princeton University Press, 1979), ¶14 for the first two quoted phrases (the shadow as a moral problem; recognizing its dark aspects as present and real) and ¶16 for the third (projection placing the cause of the emotion in the other person).
 65. *Kelo v. City of New London*, 545 U.S. 469 (2005). Facts: 472–75 (the New London Development Corporation, “a private nonprofit entity,” 473; Pfizer’s adjacent research facility, 473; the plan’s hotel, about 80 residences and office parcels, 474; the city council’s approval in January 2000 and its authorisation of the corporation to acquire property by eminent domain in the city’s name, 475). Holding: the plan serves a “public use,” 483–84 and 489–90. Justice Stevens wrote for the Court, joined by Justices Kennedy, Souter, Ginsburg and Breyer; Justice O’Connor dissented, joined by Chief Justice Rehnquist and Justices Scalia and Thomas, *id.* at 494; Justice Thomas also dissented, *id.* at 505.
 66. Patrick McGeehan, “Pfizer to Leave City That Won Land-Use Case,” *New York Times*, November 12, 2009, <https://www.nytimes.com/2009/11/13/nyregion/13pfizer.html> (print headline “... Land-Use Suit,” New York edition, p. A1). Pfizer announced on Monday, November 9, that it would leave New London; the article describes the cleared land on which the planned hotel, stores and condominiums were never built. For the long vacancy: JC Sevcik, “9 Years after SCOTUS’ *Kelo* Ruling, Houses Seized under Eminent Domain Remain Undeveloped,” UPI, March 20, 2014; John Penney, “Construction of 500 Apartments at Fort Trumbull Still a Go,” *The Day* (New London, CT), July 17, 2024 (parcels vacant for almost twenty years). The text states both facts generally and claims no figures.
 67. George W. Pring and Penelope Canan, *SLAPPs: Getting Sued for Speaking Out* (Temple University Press, 1996), chap. 1 (definition and process). Pring and Canan coined the term; their earlier article is Penelope Canan and George W. Pring, “Strategic Lawsuits against Public Participation,” *Social Problems* 35, no. 5 (1988): 506–19, <https://doi.org/10.1525/sp.1988.35.5.03a00030>. For reputation-management firms and wealthy claimants, see, for example, United Kingdom, Ministry of Justice, *Strategic Lawsuits Against Public Participation (SLAPPs): A Call for Evidence* (2022), 3, 5. The compression of the suit’s aim as burdening critics more than winning is the author’s.
 68. George Carlin, *Life Is Worth Losing*, directed by Rocco Urbisci (HBO, broadcast live November 5, 2005, from the Beacon Theatre, New York), in the closing routine on “the real owners” of the country, shortly before the “American dream” line. The locator is the routine; the wording is checked against an unofficial transcript only. The reading of the line as syn-ballein for members and dia-ballein for outsiders is the author’s.
 69. Dell Cameron and Yulia Almazova, “Leak Exposes Members of Peter Thiel’s Secretive ‘Dialog’ Society,” *Wired*, June 16, 2026, <https://www.wired.com/story/leak-exposes-members-of-peter-thiels-secretive-dialog-society/>; Dell Cameron, Dhruv Mehrotra, and Yulia Almazova, “How the Peter Thiel-Linked Dialog Club Secretly Ranks Its Members,” *Wired*, June 18, 2026, <https://www.wired.com/story/how-peter-thiels-private-dialog-club-secretly-ranks-its-members/>. The first article reports a website directory of 113 names (independently verified by *Wired*; it lists two US senators and sitting administration officials, among them the Treasury secretary) and a separate registration list for the August 2026 retreat naming 222 people (it lists a NATO commander among others); the second gives the 2006 founding by Thiel and Auren Hoffman and reports the grades by wealth and fame, the relationship tracking and the algorithmic matching of whom attendees meet and sit with. *Wired* itself says the registration records “appear to show” who attends,

and the retreat had not yet taken place; inclusion in the directory or the registration records, active membership and attendance are different things, so the text names offices and sectors and calls no individual a member. The reading of the leak as a topology is the author's.

70. Epstein Files Transparency Act, Pub. L. No. 119-38, § 2(b)(1), (c), 139 Stat. 656, 657 (2025) (no record to be redacted “on the basis of embarrassment, reputational harm, or political sensitivity”); Todd Blanche to Members of Congress, December 19, 2025, letter on the production of the Epstein files, U.S. Department of Justice, 1–4, <https://www.justice.gov/opa/media/1434851/dl> (six pages). The letter gives the Review Protocol's five grounds of redaction and a “limited amount” of privileged withholding (2) and the victim-name redactions and the numbering of the released records (3); it does not quantify the redactions. On their extent: CBS News counted at least 550 pages entirely redacted in the first release (Joe Walsh, “Over 500 Pages in Initial Epstein Files Release Were Entirely Blacked Out, CBS News Finds,” *CBS News*, December 20, 2025, updated December 21, 2025, <https://www.cbsnews.com/texas/news/epstein-files-redaction-over-500-pages-entirely-black-out>); the Department reported about 3.5 million pages published by January 30, 2026 and gave no count of redactions (U.S. Department of Justice, Office of Public Affairs, “Department of Justice Publishes 3.5 Million Responsive Pages in Compliance with the Epstein Files Transparency Act,” press release, January 30, 2026, <https://www.justice.gov/opa/pr/departments-justice-publishes-35-million-responsive-pages-compliance-epstein-files>); and Representatives Massie and Khanna, having viewed the unredacted files, said the names of six men had been redacted, which the Department answered by pointing to legal requirements (Yashraj Sharma, “Six Men Named in US Congress: Why Is So Much Redacted in the Epstein Files,” *Al Jazeera*, February 11, 2026, <https://www.aljazeera.com/news/2026/2/11/six-men-named-in-us-congress-why-is-so-much-redacted-in-the-epstein-files>). The record shows extensive and contested redaction, not total redaction; the characterisation of it as an instance of obfuscation is the author's.
71. Edward Snowden: criminal complaint filed in the Eastern District of Virginia and made public on June 21, 2013, charging theft of government property and Espionage Act offences (Will Dunham, “Federal Prosecutors Charge Edward Snowden over NSA Leaks,” Reuters, June 21, 2013, reproduced in *Christian Science Monitor*, <https://www.csmonitor.com/layout/set/amphtml/USA/2013/0621/Federal-prosecutors-charge-Edward-Snowden-over-NSA-leaks>); temporary asylum in Russia from August 1, 2013 (“Snowden Leaves Moscow Airport,” *Al Jazeera*, August 1, 2013, <https://www.aljazeera.com/news/2013/8/1/snowden-leaves-moscow-airport>); he has lived in Russia since. Julian Assange: arrested in London on April 11, 2019 on a US extradition request; an eighteen-count superseding indictment in the Eastern District of Virginia, seventeen counts under the Espionage Act provisions of 18 U.S.C. § 793, was filed May 23, 2019 (Superseding Indictment, *United States v. Assange*, No. 1:18-cr-00111-CMH, ECF No. 31 (E.D. Va. filed May 23, 2019), 1, https://www.justice.gov/d9/press-releases/attachments/2019/05/23/assange-superseding-indictment_1.pdf); held in Belmarsh Prison from about April 11, 2019 pending extradition, about sixty-two months, until the plea agreement of June 2024, under which he pleaded guilty in the Northern Mariana Islands to one count of conspiracy to obtain and disclose national defense information, was sentenced to time served and returned to Australia (Plea Agreement, *United States v. Assange*, No. CR 24-00014 (D.N. Mar. I. June 2024), 12, <https://justice.gov/opa/media/1358636/dl>). The comparison of the two cases is the author's.
72. Lewis and Short, *Latin Dictionary*, s.v. “digitus” (“a finger”; *computare digitis*, “to count on the fingers”) and “manipulus” (“a handful, a bundle,” formed from *manus* and the root of *plenus*). Harper, *Online Etymology Dictionary*, s.v. “digit,” dates *digit* as a numeral below ten to the late fourteenth century, numerals under ten having been counted on the fingers. *Manipulation*, attested by 1730 of a method of digging ore, reached English through French; *The American Heritage Dictionary of the English Language*, 5th ed., s.v. “manipulation,” adds a Spanish stage, and Harper, s.v. “manipulation,” gives French *manipule*, “handful,” a pharmacists' measure.
73. Friedrich Nietzsche, *The Gay Science: With a Prelude in Rhymes and an Appendix of Songs*, trans. Walter Kaufmann (New York: Vintage Books, 1974), §125 (“The Madman”), 181–82. In the parable the madman lights a lantern in the morning and runs to the marketplace (*Markt*) to announce that God is dead. The reading of why the news belongs there is the author's.

74. The distinctions among trust, knowledge, belief and faith, and the account of faith as lucid continuation through the limit, are the author's.
75. Lewis and Short, *Latin Dictionary*, s.v. "fides": "trust in a person or thing, faith, confidence, reliance, credence, belief," with the sense of credibility illustrated from Cicero, *De officiis* 2.9.33 ("to give confidence"), and *Pro Sulla* 3.10 (testimony); the entry derives *fides* from *fido*.
76. Lewis and Short, *Latin Dictionary*, s.vv. "arbiter" (first "a spectator, beholder, hearer, an eye-witness, a witness," then the judicial sense of one appointed to inquire into a cause and settle it) and "arbitror." The second entry gives the construction *fidem alicui arbitrari*, "to adjudge, i.e. to give, credit to one," citing Plautus, *Bacchides* 3.6.41. The line, collated in Friedrich Leo's text (*Plauti Comoediae*, vol. 1 [Weidmann, 1895], line 570), reads *Postremo, si pergis parvam mihi fidem arbitrarier*, so the lexicon's formula is a lemmatised form of the attested phrase.
77. The analysis of criterion, delineation and arbitration, and the sentence on arbitration and tacit trust, are the author's.

#3 THE FORMAL-LIMIT GENEALOGY

78. Alfred North Whitehead and Bertrand Russell, *Principia Mathematica*, 3 vols. (Cambridge University Press, 1910–13), vol. 1, preface, pp. v–viii (the 1910 preface, reprinted in the 2nd ed. of 1925, from which the pages are given). The aim and method of the work: v–vi. The chief reason for accepting any theory of the principles of mathematics being inductive, in that it enables us to deduce ordinary mathematics: v. The months of work needed to show a hypothesis untenable, the conclusion that some form of the doctrine of types must be adopted, and the particular form advocated not being logically indispensable, so that little of the book would change with another: vii. The symbolism developed in practice and in large part new because the ideas had not been symbolised before: vii–viii.
79. Bertrand Russell, "Reflections on My Eightieth Birthday," in *Portraits from Memory and Other Essays* (Simon and Schuster, 1956), 54; the sentence on "some twenty years of very arduous toil" and the conclusion that nothing more could be done to make mathematical knowledge indubitable follows at 55. Pages are those of the American edition; the British edition (Allen & Unwin, 1956) is paginated differently.
80. Gottlob Frege, *Grundgesetze der Arithmetik, begriffsschriftlich abgeleitet*, 2 vols. (Hermann Pohle, 1893–1903), 1:36 (§20), Basic Law V, which states that the value-range of one function coincides with that of another just when the two functions agree at every argument. The essay's paraphrase, that any clearly stated condition gathers a collection, is the form of the principle that Russell's argument uses. Bertrand Russell to Gottlob Frege, June 16, 1902, and Frege to Russell, June 22, 1902, in *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*, ed. Jean van Heijenoort (Harvard University Press, 1967), 124–25 and 127–28. Frege's appendix to vol. 2, headed "Nachwort" and dated October 1902, *Grundgesetze*, 2:253–65. On the failure of Frege's attempted repair and his later papers, see Edward N. Zalta, "Gottlob Frege," *Stanford Encyclopedia of Philosophy*, <https://plato.stanford.edu/entries/frege/>. The library-catalogue illustration is the author's.
81. Bertrand Russell, "Mathematical Logic as Based on the Theory of Types," *American Journal of Mathematics* 30, no. 3 (1908): 237, §IV, "The Hierarchy of Types," <https://doi.org/10.2307/2369948>. The theory is carried into Whitehead and Russell, *Principia Mathematica*, vol. 1, 2nd ed. (Cambridge University Press, 1925), introduction, chap. 2, "The Theory of Logical Types," 37 ff.
82. Alfred North Whitehead, *Science and the Modern World* (Macmillan, 1925), chap. 3, "The Century of Genius," 74–75, where the error of mistaking the abstract for the concrete is first named the "Fallacy of Misplaced Concreteness"; see also 82 and, at the head of chap. 4, 85.
83. Whitehead, *Science and the Modern World*, chap. 12, "Religion and Science," 267 (the contradiction sentence). The hypothetical of a doctrine prized as the foundation of a district's social order, and the question whether the disclosure of discordant results may be forbidden: 266. The theory that is easy to find if one is content to disregard half the evidence: 268.

84. Alfred North Whitehead, *Dialogues of Alfred North Whitehead*, as recorded by Lucien Price (Little, Brown, 1954; repr., David R. Godine, 2001), prologue. Price reconstructed the conversations from memory, so the wording is his record.
85. C. G. Jung, *Psychology and Alchemy*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 12 (Princeton University Press, 1980), ¶18 for the two quoted sentences on paradox and Tertullian's avowal, and ¶19 for the remark on Tertullian's strength to sustain paradox and the religious certainty it gave him.
86. Compare C. G. Jung, "Archetypes of the Collective Unconscious," in *The Archetypes and the Collective Unconscious*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 9, pt. 1 (Princeton University Press, 1969), ¶11, where Jung says constant usage polishes the Christian mysteries smooth until they stiffen into objects of belief, and that the eternal images, embedded in a system of thought and an institution, shield against immediate experience. The figure of the seal on the tomb is the author's.
87. Alfred North Whitehead, *Process and Reality: An Essay in Cosmology*, corrected ed., ed. David Ray Griffin and Donald W. Sherburne (Free Press, 1978), 21 (pt. 1, chap. 2, "The Category of the Ultimate"; Macmillan, 1929, 32). On Spinoza's one substance and its modes, and the philosophy of organism as closely allied to it, 6–7 (pt. 1, chap. 1, §2; Macmillan, 1929, 10–11). Reading Whitehead's gap in Spinoza as the question of how the many acquire actual existence within the unity is the author's.
88. G. Spencer-Brown, *Laws of Form* (Allen & Unwin, 1969), chap. 11, "Equations of the second degree," 57–58, where an expression that re-enters itself is given an imaginary state, a state of time rather than of space. In the preface to the first American edition (Julian Press, 1972), vii–viii, Spencer-Brown calls the theory of types a mistaken rule for excluding self-referential paradoxes that are no worse than similar cases already accepted in mathematics, and reports Russell's remark that the theory was a stopgap. Reading the re-entering mark as the turn $5 \rightarrow 0$ is the author's.
89. Mark S. G. Dyczkowski, *The Doctrine of Vibration: An Analysis of the Doctrines and Practices of Kashmir Shaivism* (Delhi: Motilal Banarsidass, 2000), 81, where "subtle movement" is Abhinavagupta's gloss as Dyczkowski gives it; for the ninth-century origin of the Spanda school and Kallāṭa's commentary, 21. Page numbers are those of the 2000 Motilal reprint. The comparison with Whitehead's primordial and consequent natures is the author's.
90. Dyczkowski, *Doctrine of Vibration*, chap. 3, "Spanda: The Universal Activity of Absolute Consciousness," section "Three Moments in the Vibration of Consciousness," 89–90 (will, knowledge and action; beginning, middle and end), and the section on the conative power, 90–96, where the first moment of the will coincides with its last (91); the closing quotation is at 80. The reading of the triad as $0/1$ and its comparison with concrescence are the author's.
91. C. G. Jung, *Mysterium Coniunctionis*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 14 (Princeton University Press, 1977), ¶224. In ¶225 Jung discusses exceptions to his scheme; reading Logos and Eros as two operations performed together is the author's continuation, distinct from Jung's gendered account.
92. Jacques Lacan, *The Four Fundamental Concepts of Psycho-Analysis*, ed. Jacques-Alain Miller, trans. Alan Sheridan (Hogarth Press, 1977), chap. 13, "The Deconstruction of the Drive," and chap. 14, "The Partial Drive and Its Circuit" (from 174).
93. Charlton T. Lewis and Charles Short, *A Latin Dictionary* (Clarendon Press, 1879), s.vv. "sapio" ("to taste, savor"; transferred, "to have good taste, i.e. to have sense or discernment; to be sensible, discreet, prudent, wise"), "sapor" ("a taste, relish, flavor, savor"). French *savoir* descends from *sapere*: ATILF, *Trésor de la langue française informatisé* (TLFi), s.v. "savoir" (verbe), Étymologie et historique, derives it from popular Latin *sapēre* (a reconstructed form), altered from classical *sapere*, "to have flavour" and "to have discernment, be wise," which used transitively displaced *scire*; Old French *savoir* kept the senses "to feel" and "to please" into the fourteenth century. TLFi, s.v. "saveur," derives *saveur* from Latin *sapor*. English *savour* comes through Old French *savour* from *sapor*: *The American Heritage Dictionary of the English Language*, 5th ed., s.v. "savor"; Harper, *Online Etymology Dictionary*, s.v. "savor."
94. Kurt Gödel, "Über formal unentscheidbare Sätze der *Principia Mathematica* und verwandter Systeme I," *Monatshefte für Mathematik und Physik* 38, no. 1 (1931): 173–98, nn. 2 (p. 173) and 46 (p. 190),

<https://doi.org/10.1007/BF01700692>. Note 2 names *Principia Mathematica* (2nd ed., 1925) as the system and lists the axioms counted with it (infinity, reducibility, choice); note 46 makes the existence of consistent but not ω -consistent systems depend on the consistency of the system.

95. The concept of the immutable gap is the author's. Its application to the strengthening of theories does not claim that a theorem about one class of formal systems proves every closed totality impossible.
96. Nicholas of Cusa, *On Learned Ignorance*, trans. Jasper Hopkins, 2nd ed., 2nd printing (Arthur J. Banning Press, 1990), I.3.10, 8. The bracketed insertions are Hopkins's. The comparison with the strengthening of theories and with the torus is the author's.
97. The aphorism is the author's, as is the reading of Gödel's sentence as an approachable limit; an unprovable sentence warrants no unrelated belief.
98. Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, trans. F. P. Ramsey, ed. C. K. Ogden (Kegan Paul, Trench, Trubner & Co., 1922), props. 4.1212, 6.54, 7.
99. Wittgenstein, *Tractatus*, props. 5.632 (quoted in the Ramsey–Ogden wording of 1922; later translations differ slightly), 5.633 (the eye and the field of sight); on the ladder, 6.54.
100. Martin Heidegger, "Memorial Address," in *Discourse on Thinking*, trans. John M. Anderson and E. Hans Freund (New York: Harper & Row, 1966), 54, where the translators' note records the older use of *Gelassenheit* by the German mystics, Meister Eckhart among them. In "Conversation on a Country Path about Thinking," in the same volume, 61–62, Eckhart is named as a master from whom much can be learned, while the releasement sought there is set apart from his.
101. C. G. Jung, "Commentary on *The Secret of the Golden Flower*," in *Alchemical Studies*, trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 13 (Princeton University Press, 1983), ¶20, where Jung glosses *wu wei* as action through non-action and links letting go of oneself with Meister Eckhart. The passage is Jung's commentary, not the Chinese text.
102. Jacques Lacan, "L'identification," seminar 9 (1961–62), unpublished; cited from Cormac Gallagher's unofficial translation from the unedited French typescripts, privately circulated (lacaninireland.com), 119–23 (session of 7 March 1962, introducing the torus, its full circuits of repeated demand and the empty circle around the central hole), 134–36 (14 March 1962) and 181–85 (11 April 1962). Lacan relates the empty circle to the object of desire rather than to desire itself; the phrase "the desire traced through their repetition" is the author's gloss.
103. C. G. Jung, *Psychology and Religion: West and East*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 11 (Princeton University Press, 1969), ¶6, where Jung attributes the term to Rudolf Otto, *The Idea of the Holy*, trans. John W. Harvey (Oxford University Press, 1923); the CW 11 bibliography lists the fourth impression of this translation (Oxford, 1926). On the feeling-toned complex, see Jung, "A Review of the Complex Theory," in *The Structure and Dynamics of the Psyche*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 8 (Princeton University Press, 1975), ¶201.
104. John R. Van Eenwyk, *Archetypes & Strange Attractors: The Chaotic World of Symbols* (Inner City Books, 1997). The use of the basin of attraction to contrast the complex with the symbol is the author's gloss on Van Eenwyk's treatment of complexes and symbols.
105. Jung, *Psychology and Religion*, ¶¶144–48 (projections withdrawn from the God-image; where the risen Christ is to be found); on the limit between a psychological image and an ontological proof, ¶102.
106. The crossed zero is the author's notation, proposed here from the sequence of formal limits alone; its historical and mathematical life is taken up in §1 and §3.
107. Friedrich Nietzsche, *Thus Spoke Zarathustra*, pt. 2, "On Self-Overcoming," in *The Portable Nietzsche*, ed. and trans. Walter Kaufmann (New York: Viking Press, 1954). In this speech Life says that she is that which must always overcome herself. The question that follows is the author's.
108. Jacques Derrida, "Différance," in *Margins of Philosophy*, trans. Alan Bass (University of Chicago Press, 1982), 8 (the two senses of *différer*) and 11 (the signified concept never present in itself; every concept inscribed in a chain or system of differences), within 1–27. "No final signified" is the essay's paraphrase.

109. Ludwig Wittgenstein, *Philosophical Investigations*, trans. G. E. M. Anscombe, P. M. S. Hacker, and Joachim Schulte, rev. 4th ed. by P. M. S. Hacker and Joachim Schulte (Wiley-Blackwell, 2009), §§198–202, esp. §201. The regress of interpretations is in §198 and §201, and the way of grasping a rule that is exhibited in obeying and going against it is in §201. The wording of the sentence is the author's paraphrase.
110. René Descartes, *Meditations on First Philosophy*, in *The Philosophical Writings of Descartes*, vol. 2, trans. John Cottingham, Robert Stoothoff, and Dugald Murdoch (Cambridge: Cambridge University Press, 1984), Meditation VI, 56 (AT VII 81). The sailor who perceives by sight that something in his ship is broken is Descartes's own comparison, in the same sentence.
111. Adam Frank, Marcelo Gleiser, and Evan Thompson, *The Blind Spot: Why Science Cannot Ignore Human Experience* (MIT Press, 2024): "An Introduction to the Blind Spot," where the authors name direct experience, the precondition of observation and measurement, as what science leaves unseen, and chap. 1, "The Surreptitious Substitution," on the reversal of the dependence of the abstract on the concrete. The reading of that reversal as a reversal of Descartes's order of inquiry is the author's.
112. David J. Chalmers, "Facing Up to the Problem of Consciousness," *Journal of Consciousness Studies* 2, no. 3 (1995): 200–219, <https://consc.net/papers/facing.html>, §2 ("The easy problems and the hard problem") and §3 ("Functional explanation"); §4 ("Some case-studies") treats neural correlates. Section numbers refer to the author's online text; §6 ("Nonreductive explanation") and §7 ("Outline of a theory of consciousness") continue the inquiry into nonreductive principles. The step from neural correlates to the Subject is the author's.
113. Brandon Carter, "Large Number Coincidences and the Anthropic Principle in Cosmology," in *Confrontation of Cosmological Theories with Observational Data*, ed. M. S. Longair, IAU Symposium 63 (Reidel, 1974), 291–98, at 291, <https://doi.org/10.1017/S0074180900235638>: what we can expect to observe is "restricted by the conditions necessary for our presence as observers."
114. Francisco J. Varela, "Neurophenomenology: A Methodological Remedy for the Hard Problem," *Journal of Consciousness Studies* 3, no. 4 (1996): 330–49. For the direction from experience to experiment, see Antoine Lutz and Evan Thompson, "Neurophenomenology: Integrating Subjective Experience and Brain Dynamics in the Neuroscience of Consciousness," *Journal of Consciousness Studies* 10, nos. 9–10 (2003): 33, 48, which says that neuroscience has focused mainly on the third-person side and that first-person data should constrain the analysis and interpretation of physiological data; the wording "usually neglected" is the author's gloss.
115. Christopher M. Bache, *LSD and the Mind of the Universe: Diamonds from Heaven*, foreword by Ervin Laszlo (Park Street Press, 2019), on his seventy-three high-dose LSD sessions over twenty years. Jung tells the Siegfried dream in C. G. Jung, *Memories, Dreams, Reflections*, recorded and ed. Aniela Jaffé, trans. Richard Winston and Clara Winston, rev. ed. (Vintage Books, 1989), chap. 6, "Confrontation with the Unconscious."
116. August Kekulé, address at the Benzolfest, Berlin, March 11, 1890, in Gustav Schultz, "Bericht über die Feier der Deutschen Chemischen Gesellschaft zu Ehren August Kekulé's," *Berichte der Deutschen Chemischen Gesellschaft* 23 (1890): 1302–11; English translation in O. Theodor Benfey, "August Kekulé and the Birth of the Structural Theory of Organic Chemistry in 1858," *Journal of Chemical Education* 35, no. 1 (1958): 21–23. The account is retrospective, told twenty-five years after the 1865 benzene paper.
117. Nikola Tesla, "My Inventions," pt. 1, "My Early Life," *Electrical Experimenter* 6, no. 10 (February 1919): 744: "When I get an idea I start at once building it up in my imagination. I change the construction, make improvements and operate the device in my mind." The series ran in six parts, February–June and October 1919.
118. Jacques Hadamard, *An Essay on the Psychology of Invention in the Mathematical Field* (1945; repr., Dover, 1954), appendix 2, "A Testimonial from Professor Einstein," 142–43, esp. 143 (Einstein's reply to question B).
119. John Maynard Keynes, "Newton, the Man," in *Newton Tercentenary Celebrations, 15–19 July 1946* (Cambridge University Press, 1947), 27–34. Keynes wrote the essay for the Royal Society's tercentenary celebrations, which were planned for 1942 and held in July 1946; his brother Geoffrey read it after Keynes's death in April 1946. "The last of the magicians" and the remark that the *Principia*'s geometry was "dressed up afterwards" are his; the comparison with the other cases is the author's.

120. Emanuel Swedenborg's journal of April 1743–October 1744 is available as *Swedenborg's Dream Diary*, ed. Lars Bergquist, trans. Anders Hallengren (Swedenborg Foundation, 2001). His own account of his call and of his conversation with angels is in Emanuel Swedenborg, *True Christian Religion*, trans. John C. Ager (Swedenborg Foundation, 1906), §779; his statement that the natural sciences prepared him is in his 1766 letter to Oetinger ("I was introduced by the Lord into the natural sciences, and thus prepared, and indeed from the year 1710 to 1744, when heaven was opened to me"), quoted in R. L. Tafel, ed. and trans., *Documents Concerning the Life and Character of Emanuel Swedenborg*, vol. 2, pt. 1 (Swedenborg Society, 1877), doc. 208, 139.
121. Federico Faggin, *Irreducible: Consciousness, Life, Computers, and Human Nature* (Essentia Books, 2024). Faggin also tells the Lake Tahoe experience in his memoir, *Silicon: From the Invention of the Microprocessor to the New Science of Consciousness* (Waterside Productions, 2021), and in an interview, "Consciousness as the Ground of Being," by Richard Gault and Jane Clark, *Beshara Magazine*, no. 20 (Winter 2021/22).
122. The aphorism is the author's.
123. Henry George Liddell and Robert Scott, *A Greek-English Lexicon*, rev. Henry Stuart Jones with Roderick McKenzie, 9th ed. (Clarendon Press, 1940), s.v. "ὑβρις": first "wanton violence, arising from the pride of strength or from passion, insolence," frequent in the *Odyssey* of the suitors; the word is also predicated of acts, among them wishing to be stronger than the gods (Euripides, *Hippolytus* 474–75, where the Nurse tells Phaedra that this is nothing other than *hybris*), and in Athenian law it covers the more serious injuries done to the person. The lexicon's headline sense is violence and insolence; the overreach of a mortal into an authority not its own is an attested use of the word. The reading of *hybris* through measure is the author's.
124. Plato, *Phaedrus*, trans. Benjamin Jowett, Internet Classics Archive, accessed October 4, 2026, <https://classics.mit.edu/Plato/phaedrus.html>. The Jowett web text carries no Stephanus numbers; they are taken from Burnet's Greek text as given in the Perseus Digital Library. The palinode begins at 244a, where madness is named the channel of the greatest goods; Delphic prophecy is at 244a–d, purification and initiatory rites at 244d–e, poetry at 245a, and love is announced at 245b–c. Socrates' recapitulation of the four divine madnenses is at 265a–b.
125. Kurt Flasch, *Meister Eckhart: Philosopher of Christianity*, trans. Anne Schindel and Aaron Vanides (Yale University Press, 2015); originally *Meister Eckhart: Philosoph des Christentums* (Munich: C. H. Beck, 2010). The quoted words are the book's title.
126. Jung was interviewed by John Freeman for the BBC television programme *Face to Face*, filmed at Küsnacht in March 1959 and broadcast on 22 October 1959; the transcript, with the answer quoted, is in C. G. Jung *Speaking: Interviews and Encounters*, ed. William McGuire and R. F. C. Hull (Princeton University Press, 1977), 428. The second quotation, and the original behind our images, of which Jung has no doubt and which is inaccessible, are both in C. G. Jung, "Jung and Religious Belief," in *The Symbolic Life: Miscellaneous Writings*, trans. R. F. C. Hull, *Collected Works of C. G. Jung*, vol. 18 (Princeton University Press, 1976), ¶1589 (answer to question 3). On a power unknown in itself, and on conscience, see Jung's letters of 5 December 1959 to M. Leonard and to Hugh Burnett in C. G. Jung, *Letters*, ed. Gerhard Adler in collaboration with Aniela Jaffé, trans. R. F. C. Hull, vol. 2, *1951–1961* (Princeton University Press, 1975), 524–26 (the passage on the unknown factor, conscience and the God-image is at 525–26, in the letter to Leonard). The letter Jung published in *The Listener* on 21 January 1960 combines these two letters and is omitted from the volume (editors' note, 524 n. 1).
127. Hannes Kettner et al., "Psychedelic Communitas: Intersubjective Experience During Psychedelic Group Sessions Predicts Enduring Changes in Psychological Wellbeing and Social Connectedness," *Frontiers in Pharmacology* 12 (2021): 623985, <https://doi.org/10.3389/fphar.2021.623985> (an observational, web-based survey, N = 886; *communitas* during the ceremony correlated with increases in social connectedness, $r = 0.25$, four weeks after the retreat); Hannes Kettner et al., "From Egoism to Ecoism: Psychedelics Increase Nature Relatedness in a State-Mediated and Context-Dependent Manner," *International Journal of Environmental Research and Public Health* 16, no. 24 (2019): 5147, <https://doi.org/10.3390/ijerph16245147> (a prospective online study, N = 654 at baseline; the increase in nature relatedness depended on the extent of ego dissolution). Both samples were self-selected and both studies report associations; the second paper's own causal conclusion is not adopted here.

128. C. G. Jung, *Psychological Types*, trans. H. G. Baynes, rev. R. F. C. Hull, Collected Works of C. G. Jung, vol. 6 (Princeton University Press, 1976), ¶758 (within the definition of individuation, which opens at ¶757), where Jung opposes levelling down to collective standards and ties individuation to broader collective relationships; and Jung, “Conscious, Unconscious, and Individuation,” in *Archetypes and the Collective Unconscious*, ¶490, where the in-dividual is defined as a separate, indivisible unity or whole.
129. Hannah Arendt, *The Origins of Totalitarianism*, new ed. with added prefaces (New York: Harcourt Brace Jovanovich, 1973), chap. 13, “Ideology and Terror: A Novel Form of Government,” 474–79 (isolation and loneliness, 474–75; solitude and loneliness, 476–77; what prepares people for total domination, 478). The clause that loneliness makes the promise of total belonging powerful is the author’s gloss on 478. The earlier sentence on friends and family is drawn from chap. 12, “Totalitarianism in Power,” 431 (a population organised into denouncing its members, in the section “The Secret Police”) and 452–53 (in “Total Domination,” the families and friends of the arrested, and the choice between betraying friends and sending one’s family to death).
130. Gustave Le Bon, *The Crowd: A Study of the Popular Mind* (London: T. Fisher Unwin, 1896), book II, chap. 3, §3, “Prestige”; affirmation, repetition and contagion are §2 of the same chapter. Project Gutenberg, accessed October 4, 2026, <https://www.gutenberg.org/ebooks/445>. The comparison with media and propaganda is the author’s.
131. Martin Heidegger, “The Question Concerning Technology,” in *The Question Concerning Technology and Other Essays*, trans. William Lovitt (New York: Harper & Row, 1977), 14 (challenging-forth), 17 (*Bestand*, with Lovitt’s note 16). “Digitised” extends Heidegger’s account and is the author’s.

#4 GEBSERIAN DIAPHANEITY

132. Jean Gebser, *The Ever-Present Origin*, trans. Noel Barstad and Algis Mickunas (Ohio University Press, 1985), 11–21.
133. Martin Heidegger, “The Age of the World Picture,” in *The Question Concerning Technology and Other Essays*, trans. William Lovitt (New York: Harper & Row, 1977), 115–54. The pairing of Heidegger’s diagnosis with Gebser’s, and the notation of the crossed zero, are the author’s.
134. Jason Reza Jorjani, *Prometheus and Atlas* (London: Arktos, 2016). The reading of the perspectival picture as Atlas’s burden, and of the end of “the world” as the end of a category, is the author’s.
135. C. G. Jung, *The Red Book: Liber Novus*, ed. Sonu Shamdasani, trans. Mark Kyburz, John Peck, and Sonu Shamdasani (W. W. Norton, 2009), 247 (Liber Primus, “Mysterium. Encounter”). Jung writes there that forethinking is Prometheus, who brings the chaotic to form and definition, and that the dark space in which Elijah and Salome live is the space of forethinking; Shamdasani’s n. 168 glosses Prometheus as forethought and refers to Jung’s reading of Carl Spitteler’s *Prometheus and Epimetheus*, where Prometheus is introverted forethought: C. G. Jung, *Psychological Types*, trans. H. G. Baynes, rev. R. F. C. Hull, Collected Works of C. G. Jung, vol. 6 (Princeton University Press, 1976), chap. 5. Jung’s later commentary on the same figures names the old man’s principle Logos, an independent principle of form meaning understanding, insight, foresight, legislation and wisdom (*Red Book*, 365, Appendix B), and in his 1925 seminar he said Elijah and Salome could be called personifications of Logos and Eros: C. G. Jung, *Analytical Psychology: Notes of the Seminar Given in 1925*, ed. William McGuire, Bollingen Series 99 (Princeton University Press, 1989), lecture 11, 89. Jung does not state the equation of Prometheus with Logos in a single sentence; it is joined across these passages.
136. C. G. Jung, “On the Relation of Analytical Psychology to Poetry,” in *The Spirit in Man, Art, and Literature*, trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 15 (Princeton University Press, 1971), ¶130. The typological reading of artists as led by intuition and feeling is the author’s.
137. Gebser, *Ever-Present Origin*, “From the Preface to the Second Edition,” xxix. Ulrich J. Mohrhoff has since compared Gebser’s integral structure with Aurobindo’s supermind and noted differences: Ulrich J.

- Mohrhoff, "Evolution of Consciousness According to Jean Gebser," *AntiMatters* 2, no. 3 (2008): 51–78.
138. Gebser's letter on the experience at Sarnath is discussed by Georg Feuerstein: *Structures of Consciousness: The Genius of Jean Gebser: An Introduction and Critique* (Integral Publishing, 1987). See also Jean Gebser, "Homologies in Spiritual Attitude." The Śaiva terms *pratyabhijñā* and *samāveśa* are brought to Gebser's experience by the author.
 139. The distinction between *fides*, reliance already in force before any arbitration, and *credere*, entrustment renewed once an undertaking has returned its consequences, is the author's, as are the four offices of trust, knowledge, belief and faith set out in an earlier movement of this section.
 140. Erich Neumann, *The Origins and History of Consciousness*, trans. R. F. C. Hull, foreword by C. G. Jung, Princeton Classics ed. (Princeton University Press, 2014). The thesis that the individual ego passes through the archetypal stages through which human consciousness as a whole developed organises the whole work.
 141. The alignment of Jung's four functions, sensation, feeling, thinking and intuition, with Gebser's magical, mythical, mental and integral structures is the author's.
 142. Jung, *Red Book*, 247 (Liber Primus, "Mysterium. Encounter"). The correspondence between Jung's formulation and Gebser's Origin is the author's.
 143. C. G. Jung, "The Psychology of the Child Archetype," in *The Archetypes and the Collective Unconscious*, trans. R. F. C. Hull, 2nd ed., Collected Works of C. G. Jung, vol. 9, pt. 1 (Princeton University Press, 1969), ¶299.
 144. C. G. Jung, "On Psychic Energy," in *The Structure and Dynamics of the Psyche*, trans. R. F. C. Hull, 2nd ed., Collected Works of C. G. Jung, vol. 8 (Princeton University Press, 1975), ¶52 and n. 44. Jung draws on R. H. Codrington's report of Melanesian usage.
 145. Gebser, *Ever-Present Origin*, 147, where he distinguishes the arational from the irrational and names the pre-rational, the irrational and the rational as its basis. The connection to the paradoxes of the previous movement is the author's.
 146. Alfred North Whitehead, *Science and the Modern World* (Macmillan, 1925), 267 (chap. 12, "Religion and Science").
 147. Wolfgang Giegerich, "The End of Meaning and the Birth of Man: An Essay about the State Reached in the History of Consciousness and an Analysis of C. G. Jung's Psychology Project," *Journal of Jungian Theory and Practice* 6, no. 1 (2004): 1–65. The birth of man is argued in §7, "The Birth of Man."
 148. Martin Heidegger, "Logos (Heraclitus, Fragment B 50)," in *Early Greek Thinking: The Dawn of Western Philosophy*, trans. David Farrell Krell and Frank A. Capuzzi (New York: Harper & Row, 1975), 59–78. On the lexical senses of *legein*, see Henry George Liddell and Robert Scott, *A Greek-English Lexicon*, rev. Henry Stuart Jones with Roderick McKenzie, 9th ed. (Oxford: Clarendon Press, 1940), s.v. "λέγω."
 149. Henri Bergson, *Time and Free Will: An Essay on the Immediate Data of Consciousness*, trans. F. L. Pogson (London: George Allen & Company, 1913), chap. 2, esp. 100–112, <https://www.gutenberg.org/files/56852/56852-h/56852-h.htm>. Bergson takes up the free act in chapter 3.
 150. Henri Bergson, *An Introduction to Metaphysics*, trans. T. E. Hulme (G. P. Putnam's Sons, 1912), first published in the *Revue de métaphysique et de morale*, January 1903; intuition is defined near the opening. The proximity of Bergson's intuition to the function that leads the integral structure is the author's observation.
 151. Edmund Husserl, "The Origin of Geometry," appendix 6 to *The Crisis of European Sciences and Transcendental Phenomenology*, trans. David Carr (Northwestern University Press, 1970), 353–78.
 152. Benjamin Bratton, *Agentworld: Scenarios for a Preemptive Anthropology of Open-World Centaur Societies* (Antikythera, 2026), 14, §2.5, "Provisional Individuation."
 153. Liddell and Scott, *Greek-English Lexicon*, s.vv. "νέμω," "νόμος." Ernst Haeckel coined *Oecologie* in *Generelle Morphologie der Organismen*, 2 vols. (Berlin: Georg Reimer, 1866), 2:286. The reading of economy and ecology as the distributing and gathering gestures of one household is the author's.
 154. Aristotle, *Nicomachean Ethics*, trans. W. D. Ross, Internet Classics Archive, 5.5, <https://classics.mit.edu/Aristotle/nicomachaen.5.v.html>; Aristotle, *Politics*, trans. Benjamin Jowett, Internet

- Classics Archive, 1.3 (the household's relations) and 1.8–10 (provision and acquisition; usury and *tokos* at 1.10, 1258b), <https://classics.mit.edu/Aristotle/politics.1.one.html>.
155. The derivations of *charaktēr*, *typos*, *legenda*, *token*, *currency*, *value*, *owe* and *own*, *debt* and *interest* are the standard ones. The reading of interest as the pricing of the relation between two parties, and its notation, are the author's.
 156. Eph. 1:9–10. Greek text: *SBL Greek New Testament*, reverse-interlinear edition of Ephesians, <https://www.sblgnt.com/download/revint/70-Ephesians.pdf>.
 157. Tertullian, *Against Praxeas*, on *dispensatio*; Karl Rahner, *The Trinity*, trans. Joseph Donceel (New York: Herder and Herder, 1970). The reading of the divine economy in the essay's notation is the author's.
 158. Aristotle, *Politics*, 1.11, 1259a (the story of Thales and the olive-presses).
 159. The Company received its charter from the States General of the Dutch Republic on March 20, 1602, for twenty-one years of exclusive trade between the Cape of Good Hope and the Strait of Magellan; the Nationaal Archief's description lists its powers to conclude contracts with foreign rulers, build forts, administer justice, hire soldiers and wage war: "Octrooi voor de VOC (1602)," Nationaal Archief, <https://www.nationaalarchief.nl/beleven/onderwijs/bronnenbox/octrooi-voor-de-voc-1602> (the charter is Nationaal Archief, The Hague, 1.04.02, inv. no. 1). On the company's corporate form, see Oscar Gelderblom, Abe de Jong, and Joost Jonker, "The Formative Years of the Modern Corporation: The Dutch East India Company VOC, 1602–1623," *Journal of Economic History* 73, no. 4 (2013): 1050–76. The characterisation of the Dutch student corps and of the Sicilian Mafia is the author's.
 160. Isa. 14:12 (KJV; Vulgate *lucifer*). C. G. Jung, *Aion: Researches into the Phenomenology of the Self*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 9, pt. 2 (Princeton University Press, 1979), ¶127 (the Morning Star) and ¶68 (the Luciferian development of science and technology).
 161. Gen. 4:1, 4:8, 4:17 (KJV).
 162. The account of the counterfeit economy as un-dual and un-finite, and the figure of the I-Deal, are the author's.
 163. Jung, *Aion*, ¶¶74–75. Jung states his position on the plane of empirical psychology, not as a theological or metaphysical thesis.
 164. The formula is the author's; §2 develops it.
 165. Jung, *Aion*, ¶¶76–78.
 166. Jung, *Aion*, chap. 6, "The Sign of the Fishes."
 167. Friedrich Nietzsche, *The Gay Science* (1882), §125.
 168. The motto first appeared on United States coinage in 1864, on the two-cent coin; it was adopted as the national motto by Public Law 84-851, approved July 30, 1956, and first appeared on paper currency in 1957, on the one-dollar silver certificate, following a 1955 law requiring it on all currency. See US House of Representatives, Committee on the Judiciary, H. Rept. 112-47, 112th Cong., 1st sess. (2011), <https://www.govinfo.gov/content/pkg/CRPT-112hrpt47/html/CRPT-112hrpt47.htm>, and Bureau of Engraving and Printing, "History," <https://www.bep.gov/currency/history>.
 169. Friedrich Nietzsche, *On the Genealogy of Morality* (1887), 2nd essay, §4; 1 Tim. 6:10 (KJV).
 170. World Inequality Lab, *World Inequality Report 2026* (World Inequality Lab, 2025), executive summary, <https://wir2026.wid.world/executive-summary/>, which reports that the top 10% own three-quarters of global wealth and the bottom half 2%, and that the share held by the wealthiest 0.001% has grown from almost 4% in 1995 to over 6%. The reading of this concentration as top-heavy and un-finite is the author's.
 171. Peter Thiel gave four lectures on the Antichrist in San Francisco in September and October 2025, hosted by the Acts 17 Collective at the Commonwealth Club, and the lectures drew on René Girard: Dave Smith, "Peter Thiel Is Delivering 4 Private Sold-Out Lectures at a Club in San Francisco—About the Antichrist," *Fortune*, September 2, 2025, <https://fortune.com/2025/09/02/peter-thiel-antichrist-lectures-commonwealth-club-san-francisco-acts-17-collective-rene-girard/>; CP Staff, "Palantir Co-founder Peter Thiel to Lead 4-Part Series on

- the Antichrist,” *Christian Post*, August 27, 2025 (scheduled for September 15, 22 and 29 and October 6). On Satan as the accuser and the scapegoat mechanism, see René Girard, *I See Satan Fall Like Lightning*, trans. James G. Williams (Orbis Books, 2001).
172. The programme of boycott, arbitration courts and national education is set out in Sri Aurobindo, “The Doctrine of Passive Resistance,” part 4, “Its Methods,” *Bande Mataram*, April 17, 1907, in *Bande Mataram: Political Writings and Speeches, 1890–1908*, vols. 6–7 of *The Complete Works of Sri Aurobindo* (Sri Aurobindo Ashram Publication Department, 2002). On the absence of hatred toward those who denied India’s rights, see “An Open Letter to My Countrymen,” *Karmayogin*, July 31, 1909 (Complete Works, vol. 8). The reading of his goodwill toward the British through Hegel’s *Aufhebung* is the author’s.
 173. C. G. Jung, “Answer to Job,” in *Psychology and Religion: West and East*, trans. R. F. C. Hull, 2nd ed., Collected Works of C. G. Jung, vol. 11 (Princeton University Press, 1969), ¶610. The connection drawn between Śakti and Jung’s concept of libido is the author’s.
 174. Sri Aurobindo, *Record of Yoga*, vols. 10–11 of *The Complete Works of Sri Aurobindo* (Pondicherry: Sri Aurobindo Ashram Publication Department, 2001).
 175. On Subproject 68, see Jordan Torbay, “The Work of Donald Ewen Cameron: From Psychic Driving to MK Ultra,” *History of Psychiatry* 34, no. 3 (2023): 320–30, <https://doi.org/10.1177/0957154X231163763>; Sidney Gottlieb launched Subproject 68 in January 1957, which gave Cameron about \$69,000 through the CIA front organisation the Society for the Investigation of Human Ecology, and Torbay adds that Cameron allegedly never knew the CIA was the source. On the 1973 destruction order and the surviving financial records, see *Project MKULTRA, the CIA’s Program of Research in Behavioral Modification: Joint Hearing before the Select Committee on Intelligence and the Subcommittee on Health and Scientific Research of the Committee on Human Resources, United States Senate, 95th Cong., 1st sess., August 3, 1977*, 3–6, <https://www.intelligence.senate.gov/sites/default/files/hearings/95mkultra.pdf>; the hearing does not mention Cameron or name Subproject 68.
 176. *Project MKULTRA, the CIA’s Program of Research in Behavioral Modification: Joint Hearing before the Select Committee on Intelligence and the Subcommittee on Health and Scientific Research of the Committee on Human Resources, United States Senate, 95th Cong., 1st sess., August 3, 1977*, 3–6, <https://www.intelligence.senate.gov/sites/default/files/hearings/95mkultra.pdf>.
 177. US House Committee on Oversight and Government Reform, Task Force on the Declassification of Federal Secrets, “Mind Control and Accountability: Uncovering the Truth of the CIA’s MKULTRA Project,” hearing, June 30, 2026, <https://oversight.house.gov/hearing/mind-control-and-accountability-uncovering-the-truth-of-the-cias-mkultra-project/>; Stephen Kinzer, “Testimony to the Task Force on the Declassification of Federal Secrets,” June 30, 2026, 2, <https://oversight.house.gov/wp-content/uploads/2026/06/Kinzer-Written-Testimony.pdf>. Kinzer’s testimony, 2, covers the 1973 destruction order and the unknown number of victims.
 178. Annie Jacobsen, *Operation Paperclip: The Secret Intelligence Program to Bring Nazi Scientists to America* (Little, Brown, 2014); for the admission policy, Dean Acheson to Harry S. Truman, August 30, 1946, in *Foreign Relations of the United States, 1946*, vol. 5, *The British Commonwealth; Western and Central Europe*, doc. 448, <https://history.state.gov/historicaldocuments/frus1946v05/d448>.
 179. Michael D. Mumford, Andrew M. Rose, and David A. Goslin, *An Evaluation of Remote Viewing: Research and Applications* (American Institutes for Research, September 29, 1995), 1-3 to 1-4, 3-4, CIA Reading Room, CIA-RDP96-00791R000200180006-4. The report dates the CIA’s own programme to 1972–77 and the Defense Intelligence Agency’s “Star Gate” to the 1980s and early 1990s, with research at the Stanford Research Institute from the early 1970s and later at SAIC; it was commissioned by the CIA in June 1995, and the programme’s ending in 1995 is the usual account. On telepathy and clairvoyance in the magical structure, see Gebser, *Ever-Present Origin*, 55.
 180. Paul E. Vallely and Michael A. Aquino, “From PSYOP to MindWar: The Psychology of Victory” (Presidio of San Francisco, CA: Headquarters, 7th Psychological Operations Group, United States Army Reserve, 1980), 7, <https://archive.org/details/from-psyop-to-mind-war-the-psychology-of-victory>. The paper was an internal talking paper by two Army Reserve officers, Vallely first, circulated to commands and not published.

181. On the Temple of Set, see Kennet Granholm, "The Left-Hand Path and Post-Satanism," in *The Devil's Party: Satanism in Modernity*, ed. Per Faxneld and Jesper Aagaard Petersen (Oxford University Press, 2013), 209–28. The reading of its ideal as the part claiming to be its own origin is the author's.
182. US Department of Justice, Epstein files, Data Set 9, EFTA01040587 (Jeffery Martin's 2017 proposal).
183. Khadeeja Safdar and Brian Whitton, "Epstein Files Release Exposes Names of at Least 43 Victims, WSJ Review Finds," *Wall Street Journal*, February 2, 2026; "Lawmakers Suggest at Least a Half Dozen Men Are Being Protected by Over-Redactions in Epstein Files," CNN, February 9, 2026, on Representatives Thomas Massie and Ro Khanna.
184. Rafael Capurro, "Past, Present, and Future of the Concept of Information," *tripleC* 7, no. 2 (2009): 125–41, at 128 (*informatio* as giving form, and *informis*); Rafael Capurro and Birger Hjørland, "The Concept of Information," *Annual Review of Information Science and Technology* 37 (2003): 343–411. On active information, see David Bohm and B. J. Hiley, *The Undivided Universe: An Ontological Interpretation of Quantum Theory* (Routledge, 1993), 35 (chap. 3, §3.2).
185. *Datum* derives from Latin *dare*, "to give," and *factum* from *facere*, "to do, make." Alfred North Whitehead, *Process and Reality: An Essay in Cosmology*, corrected ed., ed. David Ray Griffin and Donald W. Sherburne (Free Press, 1978), 21–22.
186. Jeremy B. Merrill and Will Oremus, "Five Points for Anger, One for a 'Like': How Facebook's Formula Fostered Rage and Misinformation," *Washington Post*, October 26, 2021, on the weighting of reaction emoji at five Likes from 2017, which applied to all reactions, the angry face among them. See also William J. Brady et al., "Emotion Shapes the Diffusion of Moralized Content in Social Networks," *Proceedings of the National Academy of Sciences* 114, no. 28 (2017): 7313–18 (Twitter data).
187. Greek *technē* is usually derived from the Indo-European root *teks-, "to weave, to fabricate," which also gives Greek *tektōn* and Latin *texere*. The reading of *technē* as the weaving of in-formation is the author's.
188. Aeschylus, *Prometheus Bound*, 506, closing the catalogue of the arts at 447–506, in *Aeschylus*, trans. Herbert Weir Smyth, 2 vols. (Harvard University Press, 1926), Perseus Digital Library.
189. Martin Heidegger, "The Question Concerning Technology," in Heidegger, *Question Concerning Technology*, 3–35.
190. Martin Heidegger, *Being and Time*, trans. John Macquarrie and Edward Robinson (Harper & Row, 1962), §16, H. 72–76 (pp. 102–7).
191. The term is current in developer documentation for agent systems, where it names the configuration of instructions, retrieved sources, tools and retained exchanges presented to a model; see Prithvi Rajasekaran et al., "Effective Context Engineering for AI Agents," Anthropic Engineering, September 29, 2025, <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>. Its reading as the hallmark of the integral is the author's.
192. English *ransom* comes through Old French *rançon* from Latin *redemptio*, "buying back."
193. C. G. Jung, "Psychology and Religion," in Jung, *Psychology and Religion*, ¶131.
194. Jean Baudrillard, *Simulacra and Simulation*, trans. Sheila Faria Glaser (Ann Arbor: University of Michigan Press, 1994), "The Precession of Simulacra." The reading in the essay's notation is the author's.
195. C. G. Jung, *Mysterium Coniunctionis*, trans. R. F. C. Hull, 2nd ed., *Collected Works of C. G. Jung*, vol. 14 (Princeton University Press, 1977), ¶113 and ¶332. The image of a lustre that burns without warming is the author's.
196. Jung, *Aion*, ¶75.
197. Liddell and Scott, *Greek-English Lexicon*, s.v. "κρίσις." §2 takes up Ares and Aphrodite caught in Hephaestus' net; see Homer, *Odyssey* 8.266–366.

198. Pauli formulated the exclusion principle in 1925, proposed the neutrino in an open letter of December 4, 1930, and received the Nobel Prize in Physics in 1945. See Wolfgang Pauli, “Über den Zusammenhang des Abschlusses der Elektronengruppen im Atom mit der Komplexstruktur der Spektren,” *Zeitschrift für Physik* 31 (1925): 765–83; Charles P. Enz, *No Time to Be Brief: A Scientific Biography of Wolfgang Pauli* (Oxford University Press, 2002), 119–23, 215; “Wolfgang Pauli – Facts,” NobelPrize.org, <https://www.nobelprize.org/prizes/physics/1945/pauli/facts/>. His dreams form the material of part 2 of C. G. Jung, *Psychology and Alchemy*, trans. R. F. C. Hull, 2nd ed., Collected Works of C. G. Jung, vol. 12 (Princeton University Press, 1980).
199. C. G. Jung and Wolfgang Pauli, *Atom and Archetype: The Pauli/Jung Letters, 1932–1958*, ed. C. A. Meier, trans. David Roscoe (Routledge, 2001). Page numbers are those of the Princeton University Press printing of the same translation.
200. Werner Heisenberg’s uncertainty relations and Niels Bohr’s Como lecture on complementarity both date from 1927: Werner Heisenberg, “Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik,” *Zeitschrift für Physik* 43 (1927): 172–98; Niels Bohr, “The Quantum Postulate and the Recent Development of Atomic Theory,” *Nature* 121 (1928): 580–90 (the lecture was delivered on September 16, 1927). The formulation of the observer’s choice and the irrevocable loss of complementary knowledge follows Pauli, “The Philosophical Significance of the Idea of Complementarity,” in Wolfgang Pauli, *Writings on Physics and Philosophy*, ed. Charles P. Enz and Karl von Meyenn, trans. Robert Schlapp (Springer, 1994), 35–42.
201. Jung and Pauli, *Atom and Archetype*, letter 58P (Pauli to Jung, February 1953), 91, on complementarity as a model for the *coniunctio*; letter 62P (Pauli to Jung, May 27, 1953), 121, for the remark to Bohr on Einstein.
202. C. G. Jung, “On the Nature of the Psyche,” in *The Structure and Dynamics of the Psyche*, trans. R. F. C. Hull, 2nd ed., Collected Works of C. G. Jung, vol. 8 (Princeton University Press, 1975), ¶417; C. G. Jung, “Psychological Aspects of the Mother Archetype,” in *The Archetypes and the Collective Unconscious*, trans. R. F. C. Hull, 2nd ed., Collected Works of C. G. Jung, vol. 9, pt. 1 (Princeton University Press, 1969), ¶155. The psychoid factor and the crystal are Jung’s hypothesis and analogy.
203. C. G. Jung to Wolfgang Pauli, May 4, 1953, letter 61J, in Jung and Pauli, *Atom and Archetype*, 111–17 (the passage on the psychoid as an approach to a neutral language is at 111).
204. John R. Van Eenwyk, *Archetypes & Strange Attractors: The Chaotic World of Symbols* (Inner City Books, 1997). Van Eenwyk offers the attractor as an explanatory analogy; the derivation of the arche-topos is the author’s, developed in §3.
205. C. G. Jung, *Aion: Researches into the Phenomenology of the Self*, 2nd ed., trans. R. F. C. Hull, Collected Works of C. G. Jung, vol. 9, pt. 2 (Princeton University Press, 1979), chap. 5, 61n74. Jung writes of the psychological unity of the Self; the comparison with the Śaiva sources is the author’s.
206. Abhinavagupta, *Parātrīśikā-vivaraṇa: The Secret of Tantric Mysticism*, trans. Jaideva Singh, ed. Bettina Bäumer (Delhi: Motilal Banarsidass, 1988). The placing of Jung’s Self and the Śaiva Self in one place is the author’s.
207. Wolfgang Pauli to C. G. Jung, March 31, 1953, letter 60P, in Jung and Pauli, *Atom and Archetype*, 107.
208. C. G. Jung to Wolfgang Pauli, October 24, 1953, letter 64J, in Jung and Pauli, *Atom and Archetype*, 127.
209. Wolfgang Pauli to C. G. Jung, June 1957, letter 74P, in Jung and Pauli, *Atom and Archetype*, 158–59 (the passage is at 159).
210. Yoshua Bengio, Réjean Ducharme, Pascal Vincent, and Christian Jauvin, “A Neural Probabilistic Language Model,” *Journal of Machine Learning Research* 3 (2003): 1137–55, <https://jmlr.org/papers/v3/bengio03a.html>. The abstract and §1 are at 1137–39; §1.2, “Relation to Previous Work,” at 1140–41, gives the history of the approaches that preceded it.
211. Ashish Vaswani et al., “Attention Is All You Need,” in *Advances in Neural Information Processing Systems 30*, ed. I. Guyon et al. (Curran Associates, 2017), 5998–6008, §3.2.1 (“Scaled Dot-Product Attention”),

- https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.
212. Bettina Bäumer, preface to Abhinavagupta, *Parātrīśikā-vivaraṇa*, xv.
 213. The formulation, and the notion of Code, are the author's.
 214. Bhartṛhari, *Vākyapadīya*, book 1.
 215. Mark S. G. Dyczkowski, *The Doctrine of Vibration: An Analysis of the Doctrines and Practices of Kashmir Shaivism* (Delhi: Motilal Banarsidass, 2000), 78.
 216. On the powers of words, *mātrkā*, that bind the fettered self, see Kallaṭa, *The Stanzas on Vibration: The Spandakārikā with Four Commentaries*, trans. Mark S. G. Dyczkowski (Albany: State University of New York Press, 1992).
 217. Dyczkowski, *Doctrine of Vibration*, 70.
 218. Abhinavagupta treats the purified thought-construct within the way of knowledge in *Tantrāloka*, chap. 4, and *Tantrasāra*, chap. 4.
 219. The formulation is the author's.
 220. On Māyā in the Spanda teaching, see Dyczkowski, *Doctrine of Vibration*, 78–79. The reading of Māyā as operative measure is the author's.
 221. Jean Baudrillard, *Simulacra and Simulation*, trans. Sheila Faria Glaser (Ann Arbor: University of Michigan Press, 1994), "The Precession of Simulacra."
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